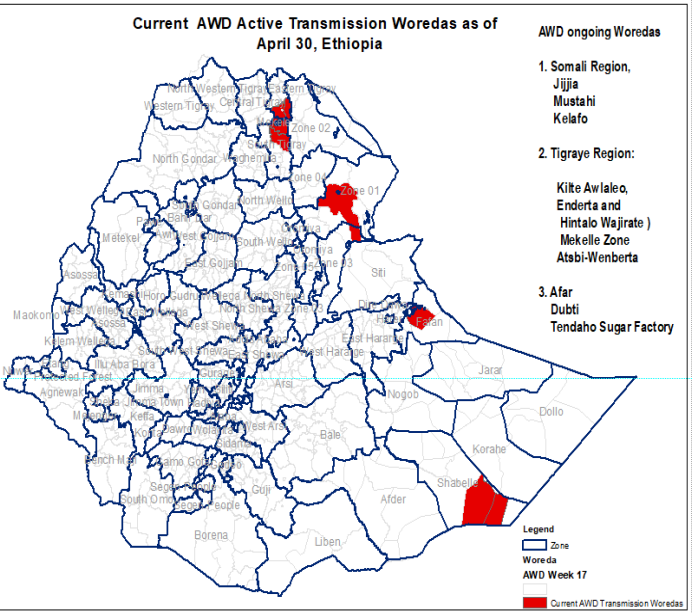
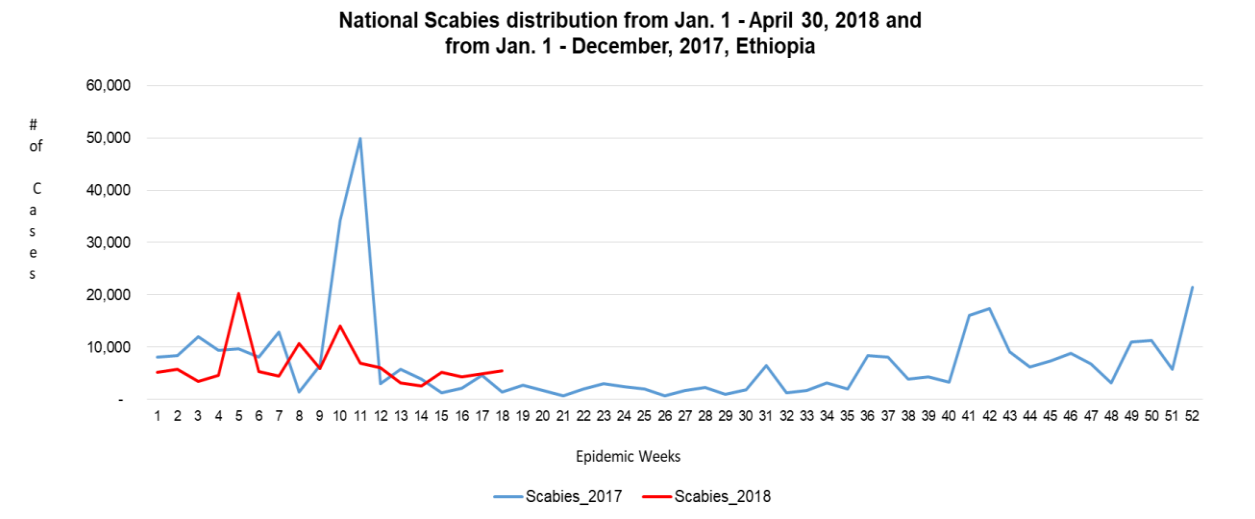
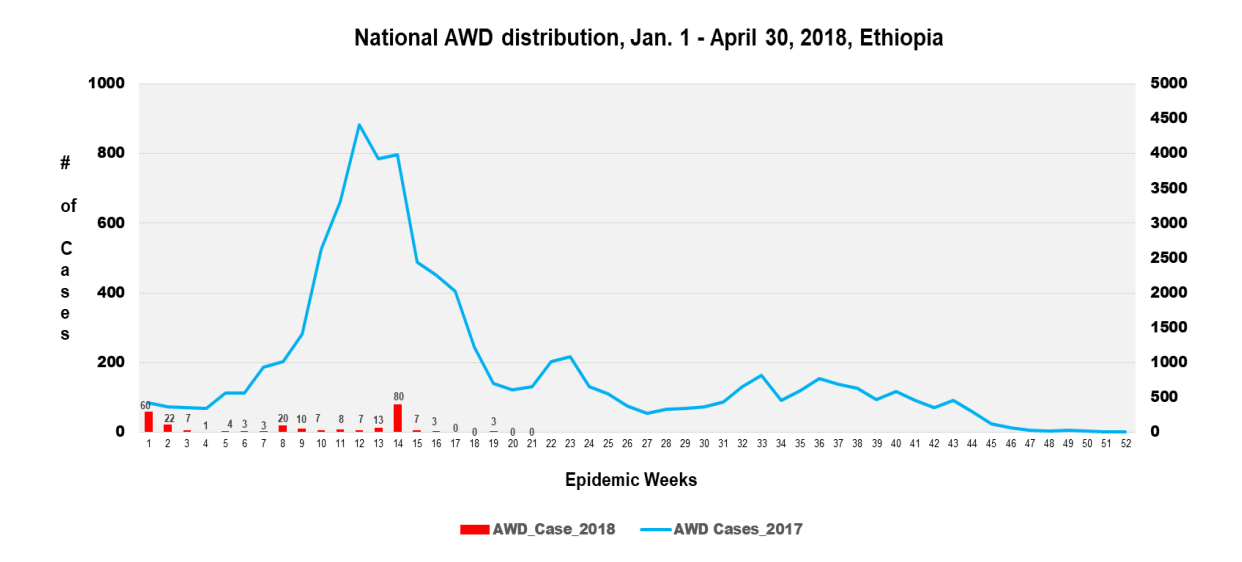




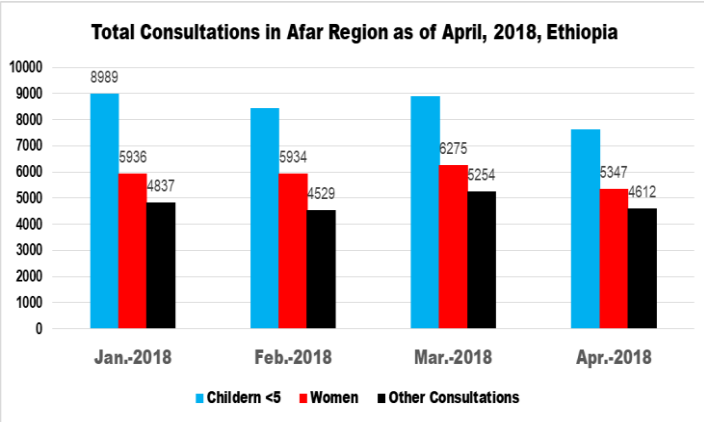
From April 1 – 30, 2018, a total number of 91 AWD Cases with zero death were reported from Somali (26), Tigray (1) and Afar (64) regions. This is significantly lower compared to the same period in 2017. Total Cumulative AWD cases from January 1 to April 30, 2018 is 255.

On April 04, 2018, a new AWD outbreak occurred in Afar region. On epidemiological Week 17, 2018 Zero AWD cases were reported from the affected regions of Afar, Tigray and Ethio-Somali regions.

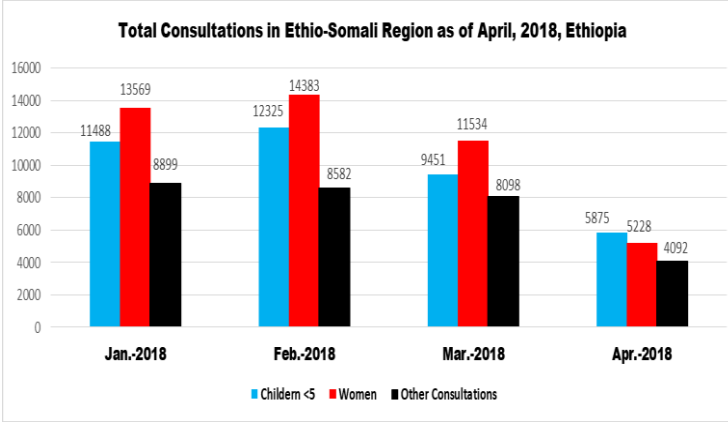
Scabies affected districts from Jan. 1 – April 30, 2018

From April 1 – 30, 2018, a total number of 17,070 new Scabies Cases were reported from Seven regions. SNNP and Oromia regions reported highest number of cases. Please find case distribution in the table below. Comparing to the same period in 2017, Scabies cases were reduced by 65%.

Regions	April, 2018	Jan. 1 - April 30, 2018	2017
Oromia	5,326	28,075	29,876
Amhara	4,689	61,128	100,301
SNNP	5,528	17,076	225,150
Tigray	666	3,384	12,962
B. Gumuz	4	1,528	3445
Addis Ababa	390	1,009	0
Dire Dawa	467	467	0
Total	17,070	112,667	371,734

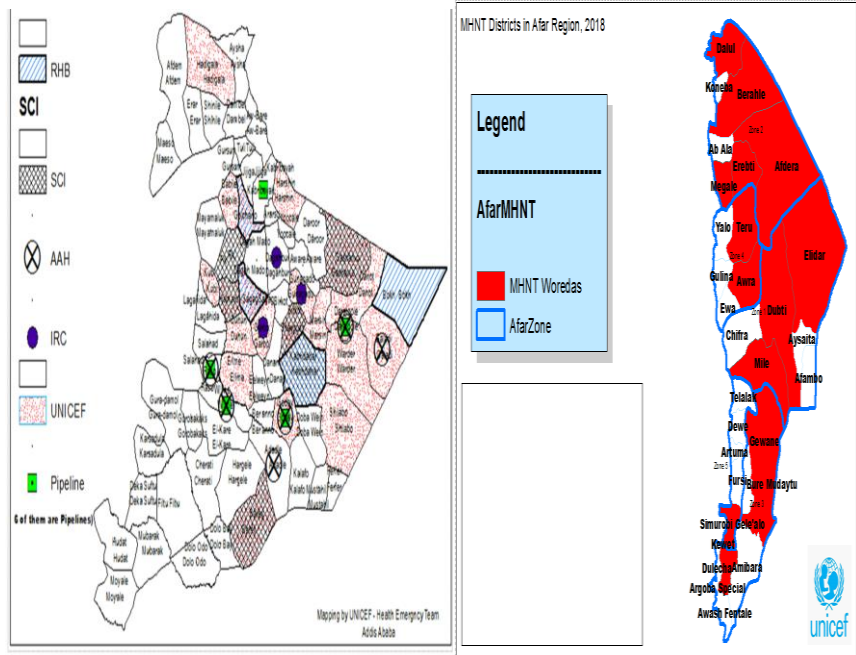


Source: Afar Regional Health Bureau



Source: Ethio-Somali Regional Health Bureau

Location of MHNT team, Afar and Ethio-Somali Region, Ethiopia



From April 1–30, 2018 a cumulative total of 32,787 medical consultations conducted in hard to reach areas of Afar and Somali regions, of which, 13,508 (41%) under five children and 10,575 (32%) were Women.

From January 1–April 30, 2018, a total cumulative 190,196 MHNT consultations were made 76,672 from Afar and 113,524 from Ethio-Somali region. Of which 73,087 (38%) Under five children, 68,206 (36%) Women and 48,903 (26%) were from Other adults.

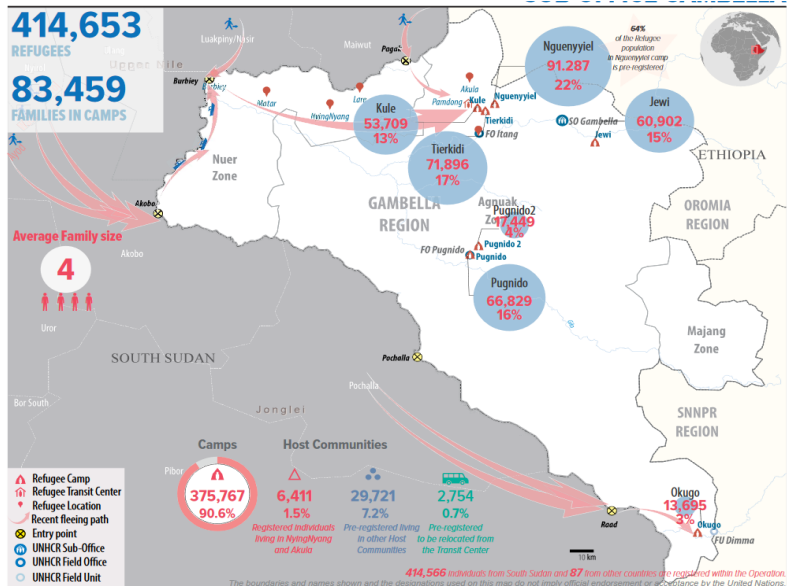
The medical Consultation has been performed by 49 UNICEF supported Mobile Health and Nutrition Teams plus 11 International NGOS and 6 RHBs in Somali in region.

Moreover, total MHNT beneficiaries from both regions became nearly 2.8 million. Of which 1,689,651 from Ethio-Somali and the rest 1,083,044 from Afar region.

South Sudanese Refugees Response, Jan. 1 – April, 2018, Ethiopia

Indicator	2017	April 2018	Cummulative 2018
# of New South Sudanese arrival	87,678	2777	19,178
# of <15 Childern	55,265	1742	10,333
# of 6mth. - <15Yrs Childern	50,152	1498	9,629
# of <15 Childern Obtained arrival Polio Vaccination	59,121	1711	10,344
# of 6mth to 15Yrs childern Obtained Measles Vaccination	53,231	1521	9,684

South Sudanese Refugees in Gambella Sites, April, 2018



Community-level Risk Mapping and Vulnerability Assessments (VRAM) Presenting to WHO - Team

By

Negusu Worku GEBRMICHAEL

June 17, 2022

Presentation Outlines

- 1. Key considerations
- 2. Main activities
- 3. Potential challenges.

- I. Key Consideration
 - Population distribution and Population density
 - U5, 5-14, 15-45, 45+
 - The concentration of population vs. chemical industry vs. IDSR data systematically reported using
 - the political administrative level of region(provinces), zones, districts and villages, health facilities (based on the population catchment area),
 - Data of IDSR (Surveillance): at least a retrospective data 5-10 years (Systematically collected based on a weekly based)
 - Data of GIS Shape files
 - Location of Health facilities, Hospitals, health posts, rivers, roads, altitude, water point,
 - Softwares ARCGIS and Statistical packages softwares
 - GPS to collect some missing GIS files including point data, road data, river data and some water points,
 - Political administrative levels data flow from HFs, district, Zone, regions (Province),
 - Location of Chemical industry Sites
 - CORIN analysis whether the land is industry zone, cultivation zone, urban zone or residential , buildings and etc ...
 - Location of all facilities including HFs, schools, hotels, gas station the chemical zones to identify risk,

II. Main activities

- Using the VRAM techniques producing of risk mapping for that specific community using
 - Surveillance data could be for example
 - Malaria data
 - AWD/Cholera data and etc ...
 - Focusing on the major disease burden in the community or high risk place like chemical industry place including for example
 - Refugee camp main disease burden this is a specific place
 - General disease burden like : Ebola, Meningitis, Malaria, flood prone and etc...
 - Based on the above Surveillance data, we can look in depth the map correlation and association of that specific geographic place with the reported diseases or the location of the chemical industry or the vs. the altitude, rain precipitation, the location of water body and etc ... (Generally what is the association/ correlation specific event happening in one geographic place vs. with a number of IDSR data (the (association),
 - Using different layers calculating **risk level by different kilometer radius**
 - **One layer** could be population density by different political administrative level
 - **Another layer** the location of HFs (Shows equity)
 - **3rd layer** could be the road vs. the above two (shows accessibility),
 - And etc ...
 - Water coverage and latrine coverage by different political administrative level
Water coverage by regions, zones, districts villages. It has a risk of water borne diseases, water related diseases, water based diseases and water scarce diseases (respectively like AWD, typhoid, malaria, yellow fever, river blindness, sleeping sickness, guinea worm, bilharzia, trachoma and scabies),

III. Potential Challenges

- Sometimes, some countries have poor **data repository** sometimes back to 5-10 years
- Some GIS data missing and cost a lot of time, energy, cost to collect (Most of the GIS data freely available in the HDX humanitarian web),
- Some places missing important equipment and softwares including :
 - Computers
 - Plotter machines
 - ARCGIS Softwares
- Human resource gap
 - Most of the time no trained individual on GIS and Mapping. Even capacity building training requires some background,
- **Ownership issues** at the MoH in a given country, **capacity building can be given** but they have to be ready to accept the VRAM process and **they have to cooperate to make ready some data**, (Data belongs to government in the respective country MoH), based on IHR (Data Sharing policy),

IV. Recommendations

- GIS Shape files needs to be ready from X country including :
 - Point data f HFs,
 - Road vs. HFs location (**accessibility**),
 - HFs vs. Population density (**equity**),
- Surveillance Data
 - **Communicable diseases** reported data using Weekly Epidemiological week
 - **Daily IDSR** (immediately reportable diseases,
 - **Non-communicable** reportable diseases n along the road), and accessibility (Could be weekly ...)
 - Location of Water body,
 - Water coverage and laterite coverage,
 - **water-borne diseases** (e.g. AWD, typhoid)
 - **water –related diseases** (e.g. malaria, yellow fever, river blindness, sleeping sickness)
 - **water-based diseases** (e.g. guinea worm and bilharzia) and water-scarce diseases (trachoma and scabies),
 - The final risk mapping shows **resource availability (gaps),** to **appropriate decision making** based on informed decision making, to **put appropriate resource allocation,** **to put appropriate control measures,**

Sample Works

- **From Four (#4) WHO Regions:**

- **AFRO**

- HIV Prevalence from 1992-2012 (20 yrs.)
- Ethiopia (WHO-AFRO)
- Sierra Leone (WHO – AFRO)

- **EMRO**

- Jordan (WHO-EMRO)

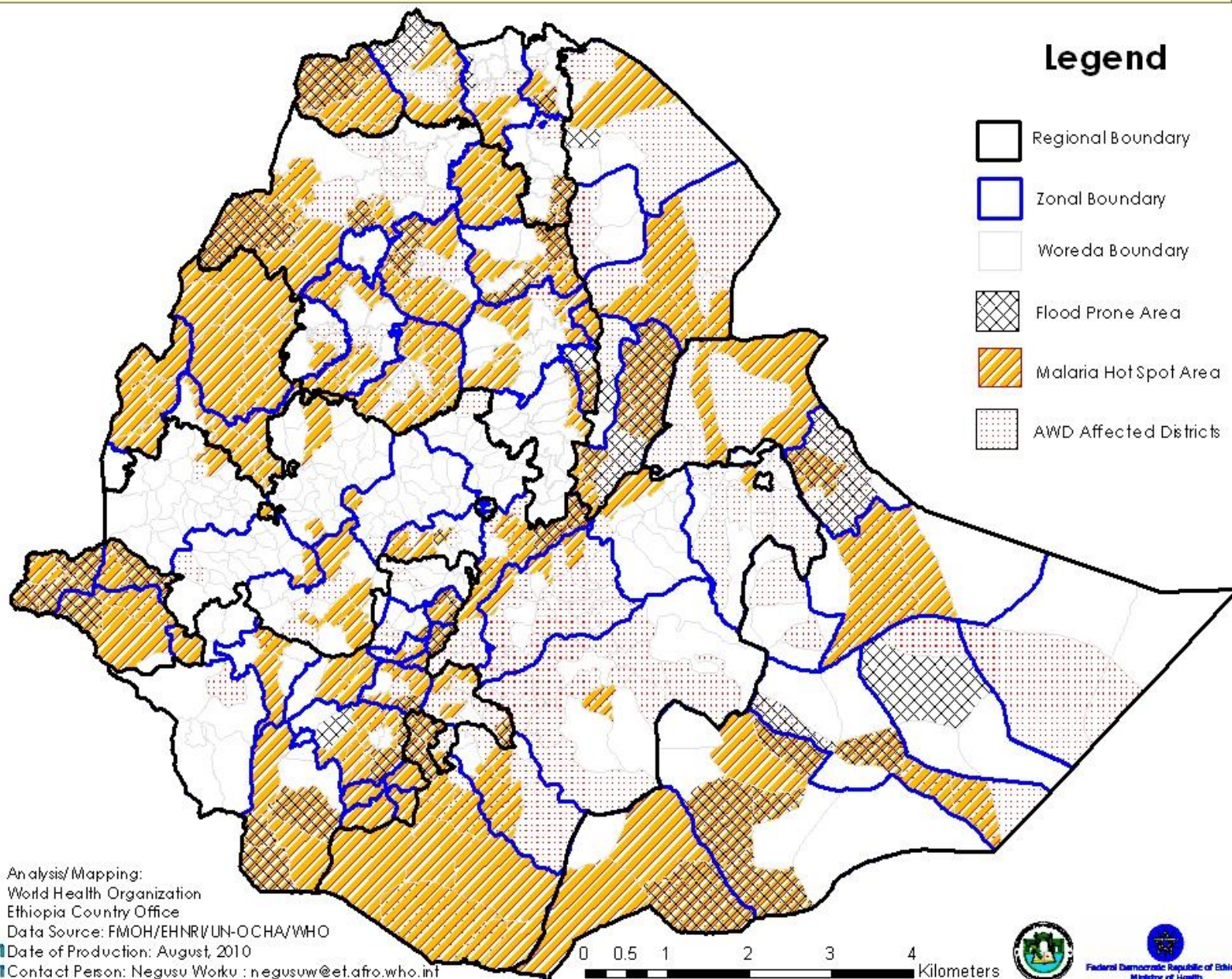
- **SEARO**

- Bangladesh (WHO-SEARO)
- India
- Afghanistan

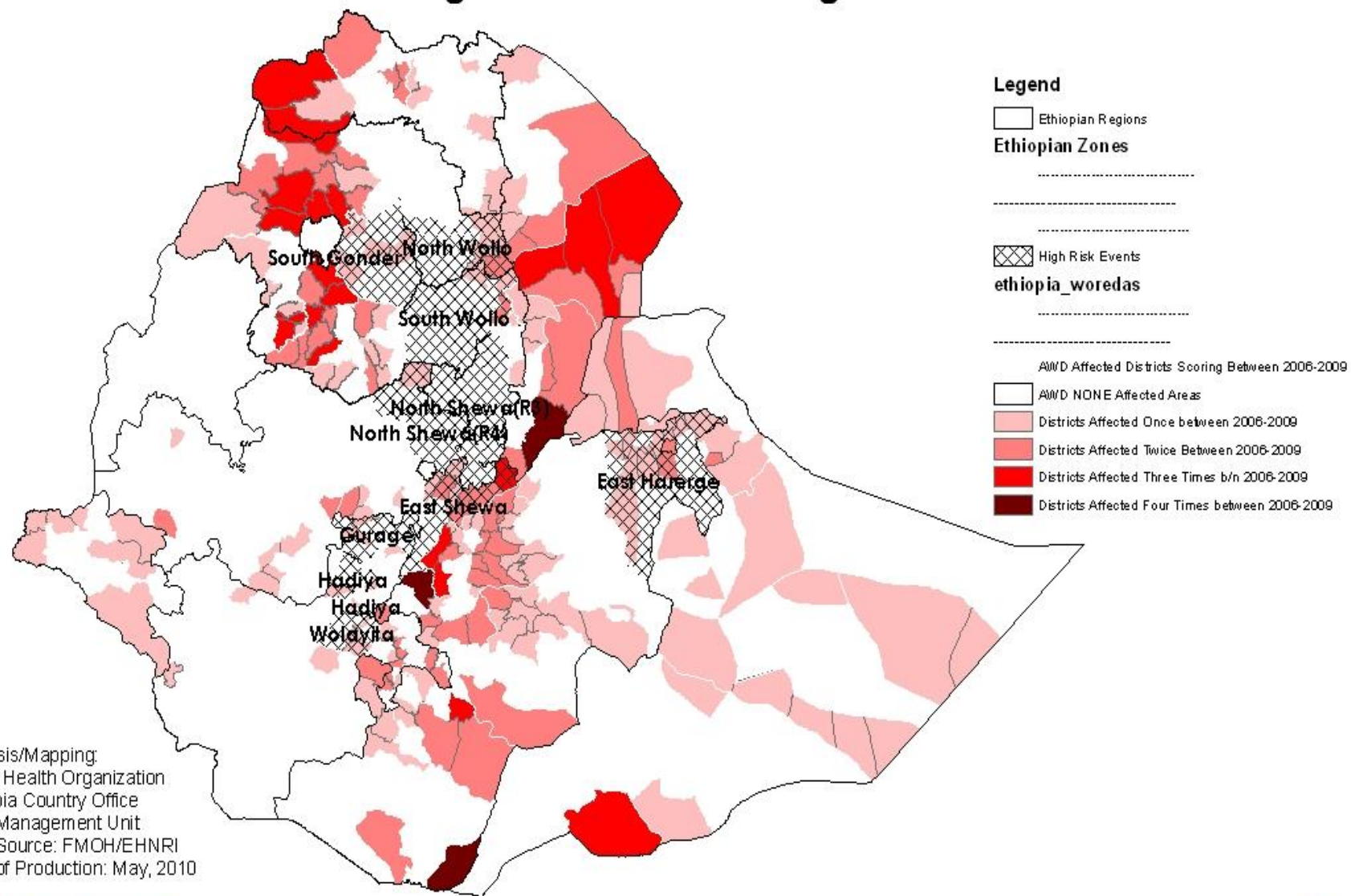
- **EURO**

- Turkey (WHO-EURO)

Map of Ethiopia Showing Correlation of AWD, Malaria and Flood Prone area as of August, 2010



Map of Ethiopia Showing AWD Affected Districts from 2006-2009 along With the National High Risk Events



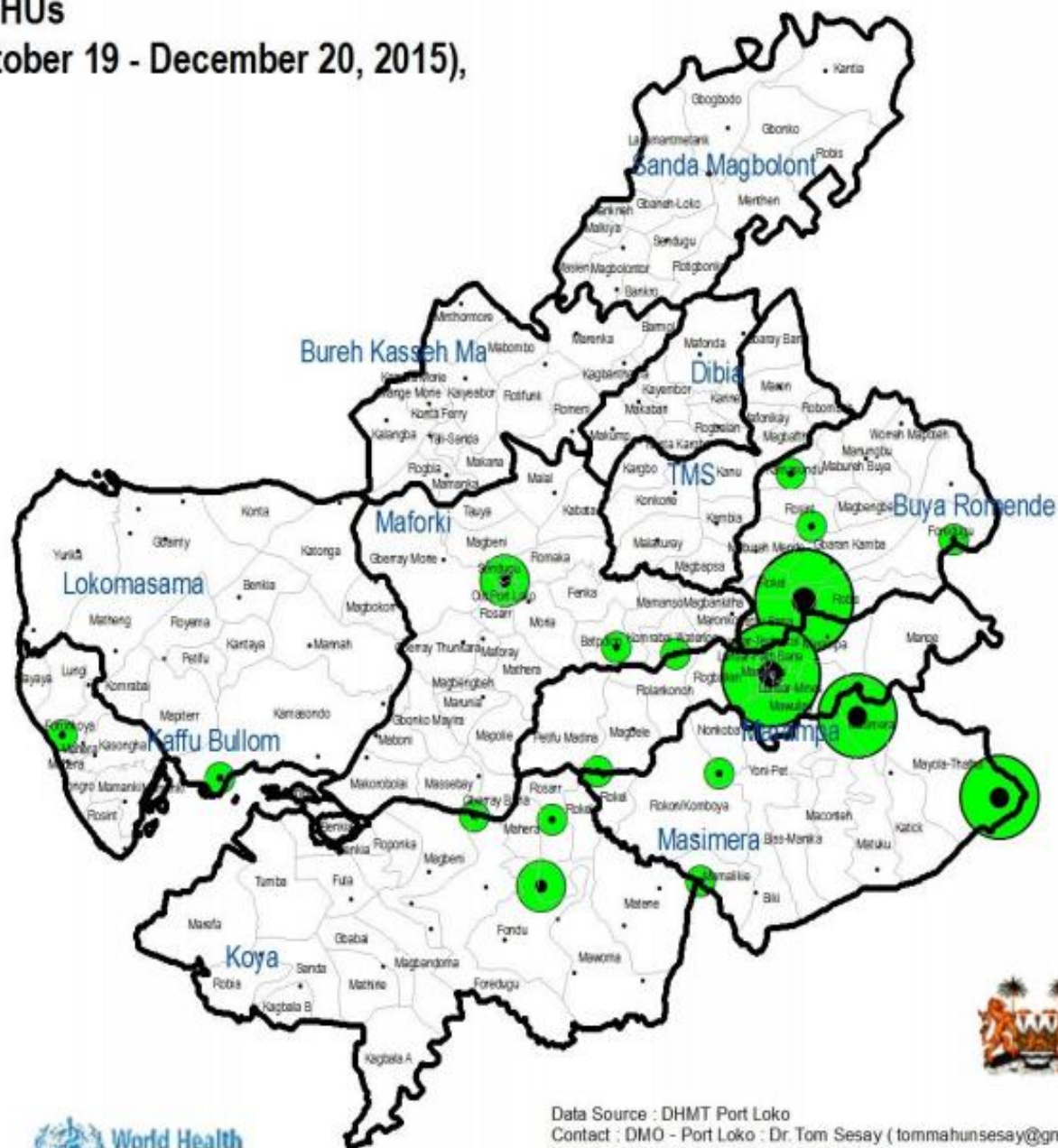
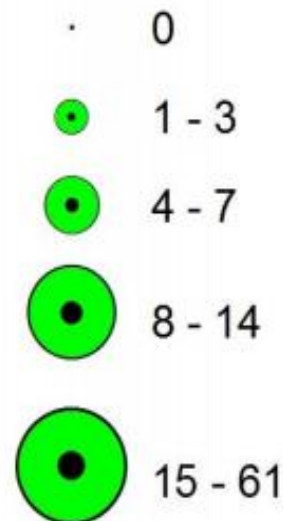
Analysis/Mapping:
World Health Organization
Ethiopia Country Office
Data Management Unit
Data Source: FMOH/EHNRI
Date of Production: May, 2010

Reported Measles Caseload by PHUs from Epidemic Week 43 - 51 (October 19 - December 20, 2015), Port Loko, Sierra Leone

Legend

-  Port Loko Chiefdoms
-  PLK_Sections

Port Loko Measles



Reported Diarrhoea Severe Dehydration Caseload by PHUs from Epidemic Week 43 - 51 (October 19 - December 20, 2015), Port Loko, Sierra Leone

Legend

 Port Loko Chiefdoms

 PLK_Sections

-

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1 - 10



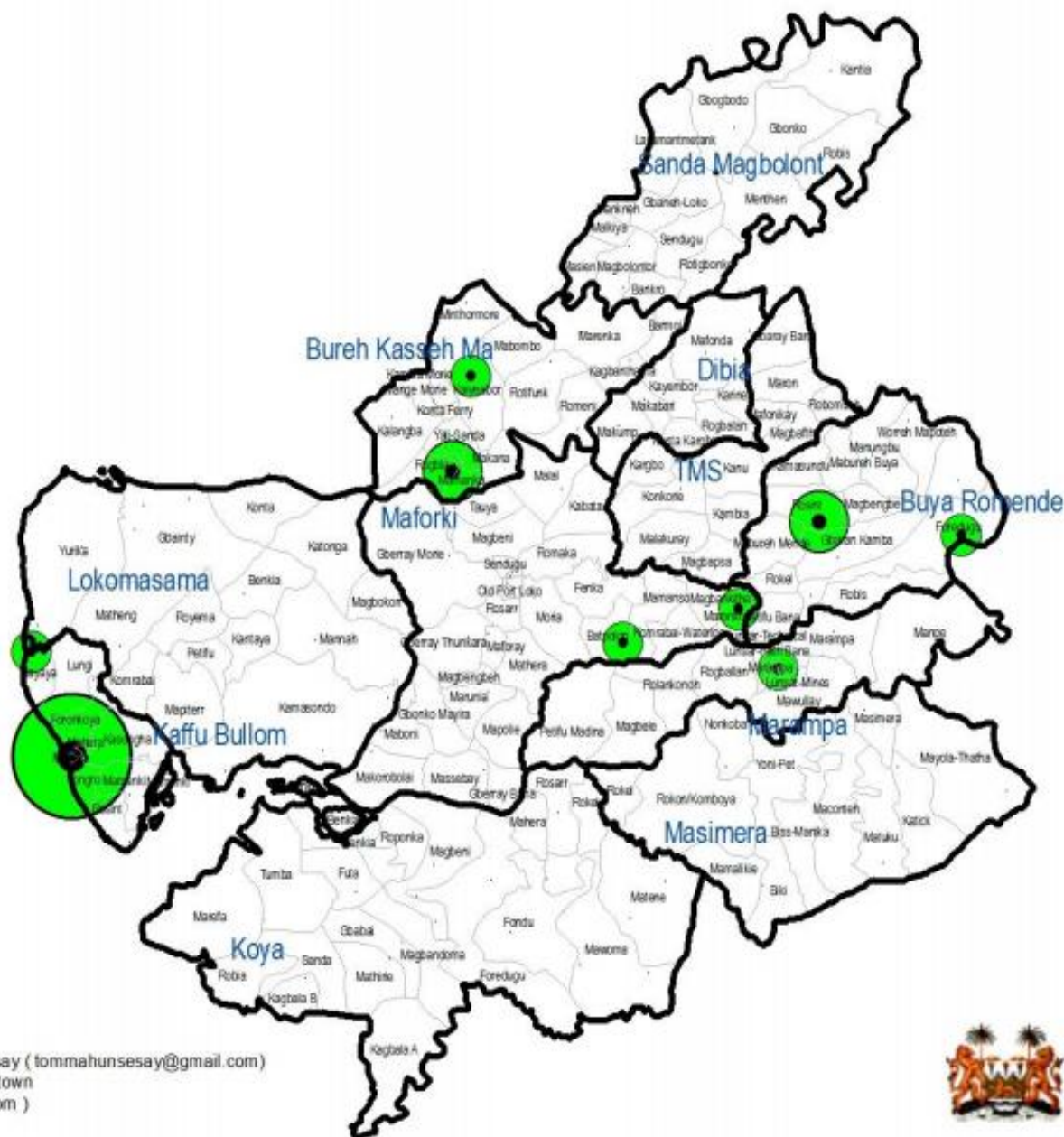
11 - 20



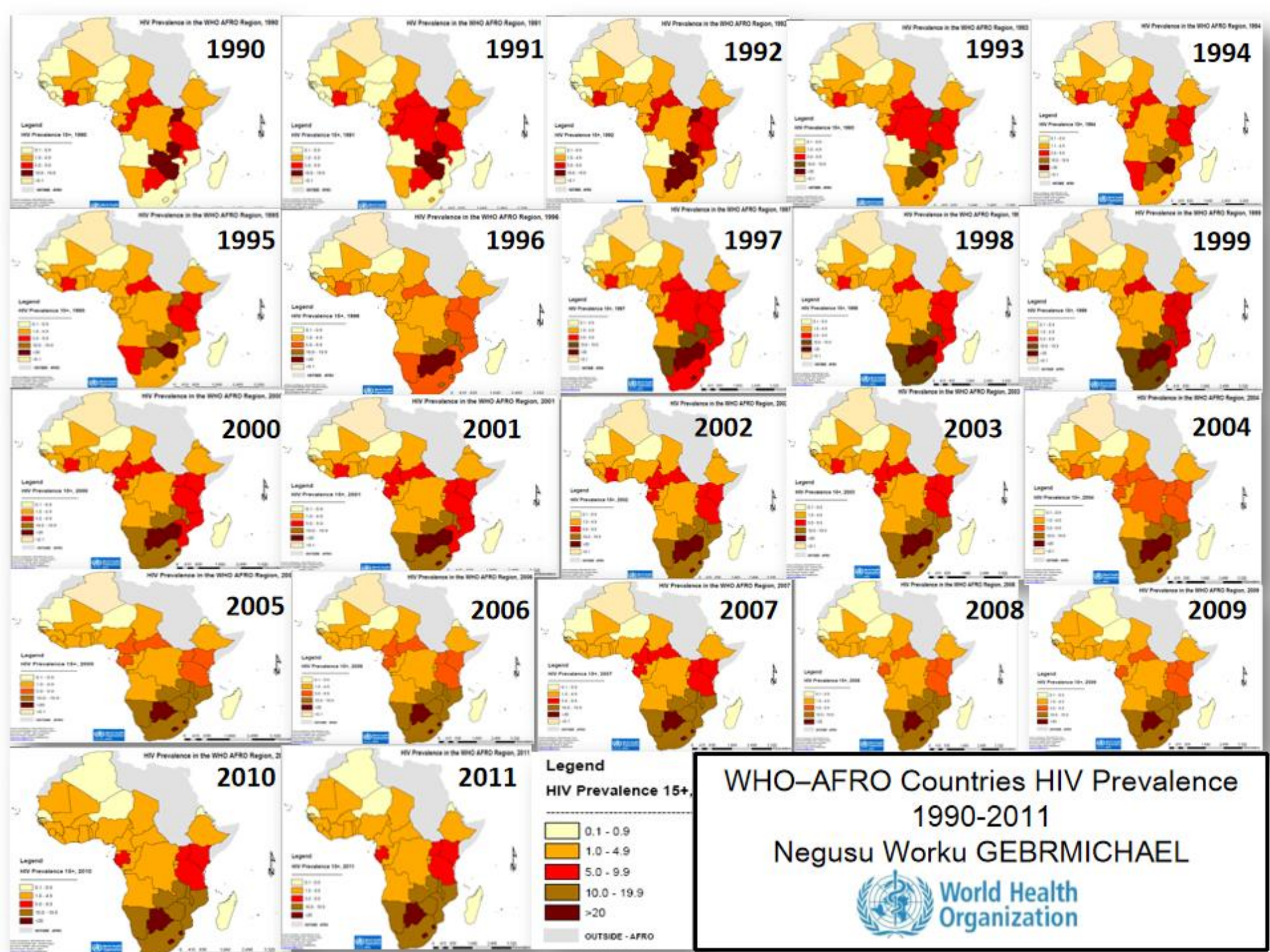
21 - 30



31 - 36



Data Source : DHMT Port Loko
Contact : DMO - Port Loko : Dr. Tom Sesay (tomahunesay@gmail.com)
Data Analysis and Mapping : WHO-Freetown
Negusu Worku (negusuworku@gmail.com)
Date of Production : December, 2015



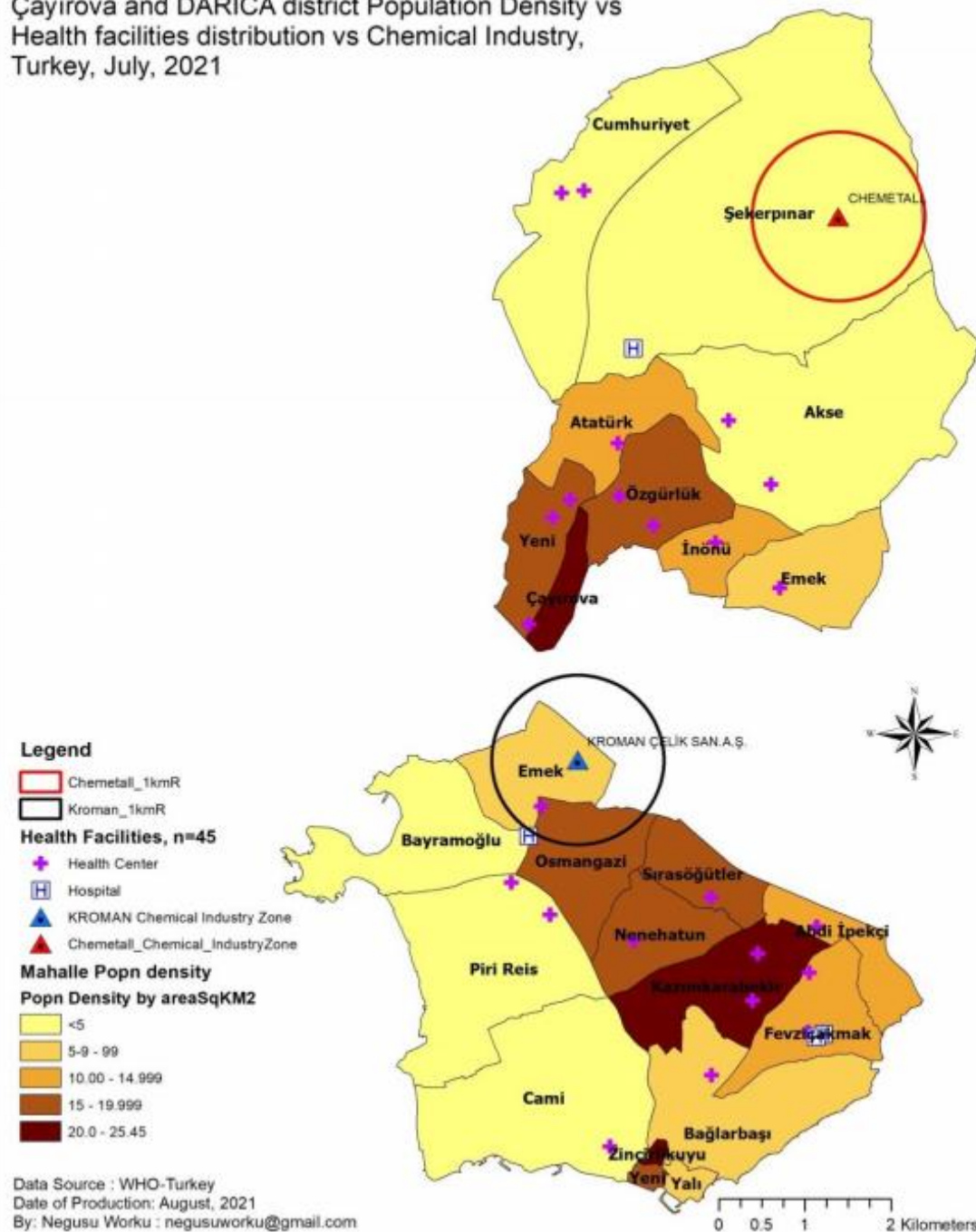
WHO-AFRO Countries HIV Prevalence 1990-2011

Negusu Worku GEBRMICHAEL



World Health Organization

Çayırova and DARICA district Population Density vs Health facilities distribution vs Chemical Industry, Turkey, July, 2021



Health Facilities (Health Center, Hospital and Pharmacy) shows n=44 Observations :

1. Health Center, n=25
2. Hospitals, n=4
3. Pharmacy, n=15

1. Çayırova district : health facilities availability 39% (n=1 hospital, n=12 Health Center and n=3 Hospitals),

2. Darica district : health facilities availability 61% (n=3 Hospitals, n=13 health centers, and n=11 Pharmacy),

3. Çayırova district, within 1kmR no facilities located,

4. Darica, within 1 kmR three (3) health centers available,

5. Çayırova district , within 2kmR radius no Health facilities are available,

6. Darica, within 2kmR, n=10 health facilities are available, of which n=3 Health Centers, n=1 hospitals and n=6 Pharmacies are available.

Çayırova and DARICA district Population Density vs Health facilities distribution vs Chemical Industry, by different risk km radius Turkey, July, 2021

Legend

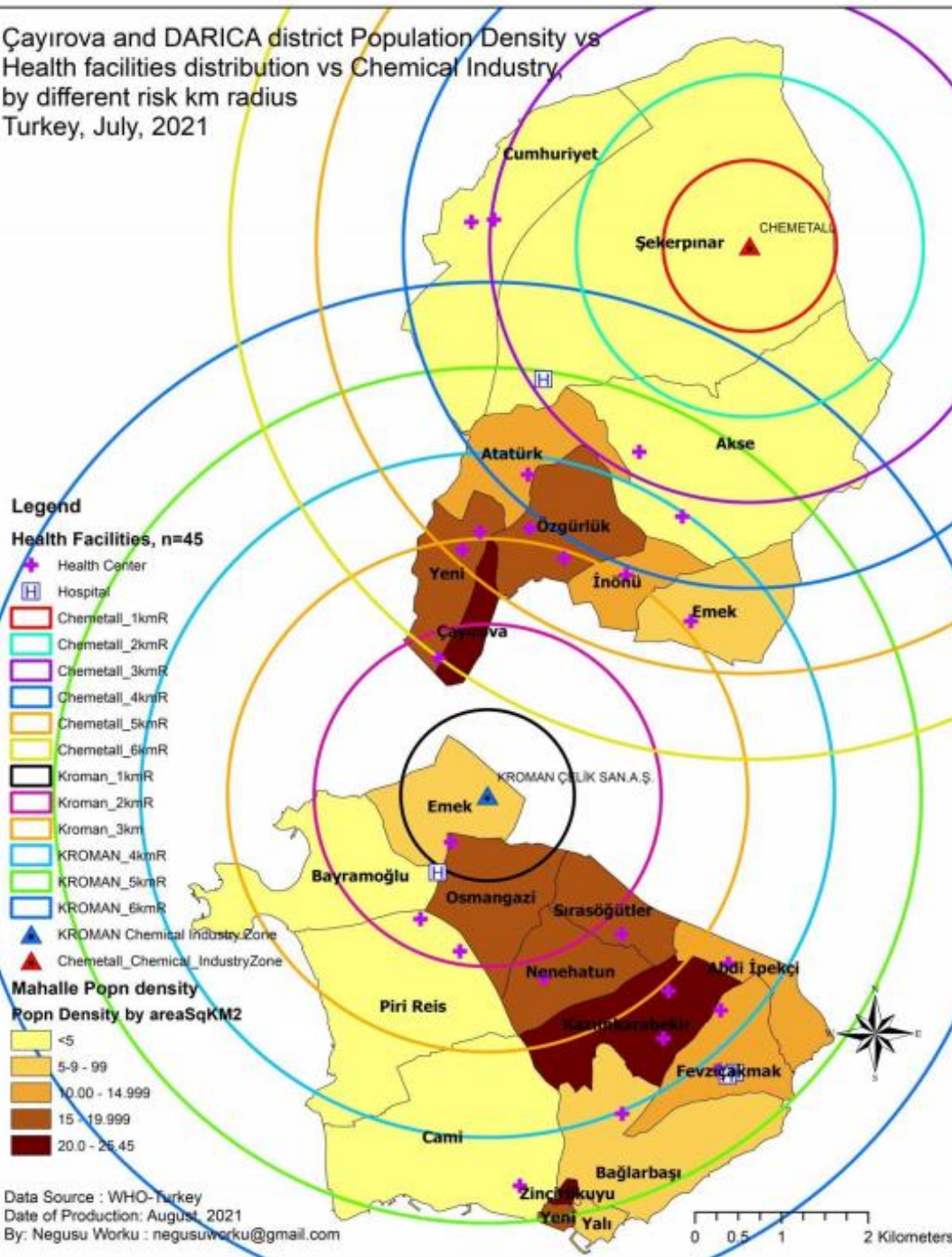
Health Facilities, n=45

- Health Center
- Hospital
- Chemetall_1kmR
- Chemetall_2kmR
- Chemetall_3kmR
- Chemetall_4kmR
- Chemetall_5kmR
- Chemetall_6kmR
- Kroman_1kmR
- Kroman_2kmR
- Kroman_3km
- KROMAN_4kmR
- KROMAN_5kmR
- KROMAN_6kmR
- KROMAN Chemical Industry Zone
- Chemetall_Chemical IndustryZone

Mahalle Popn density Popn Density by areaSqKM2

- <5
- 5-9.99
- 10.00 - 14.999
- 15 - 19.999
- 20.0 - 25.45

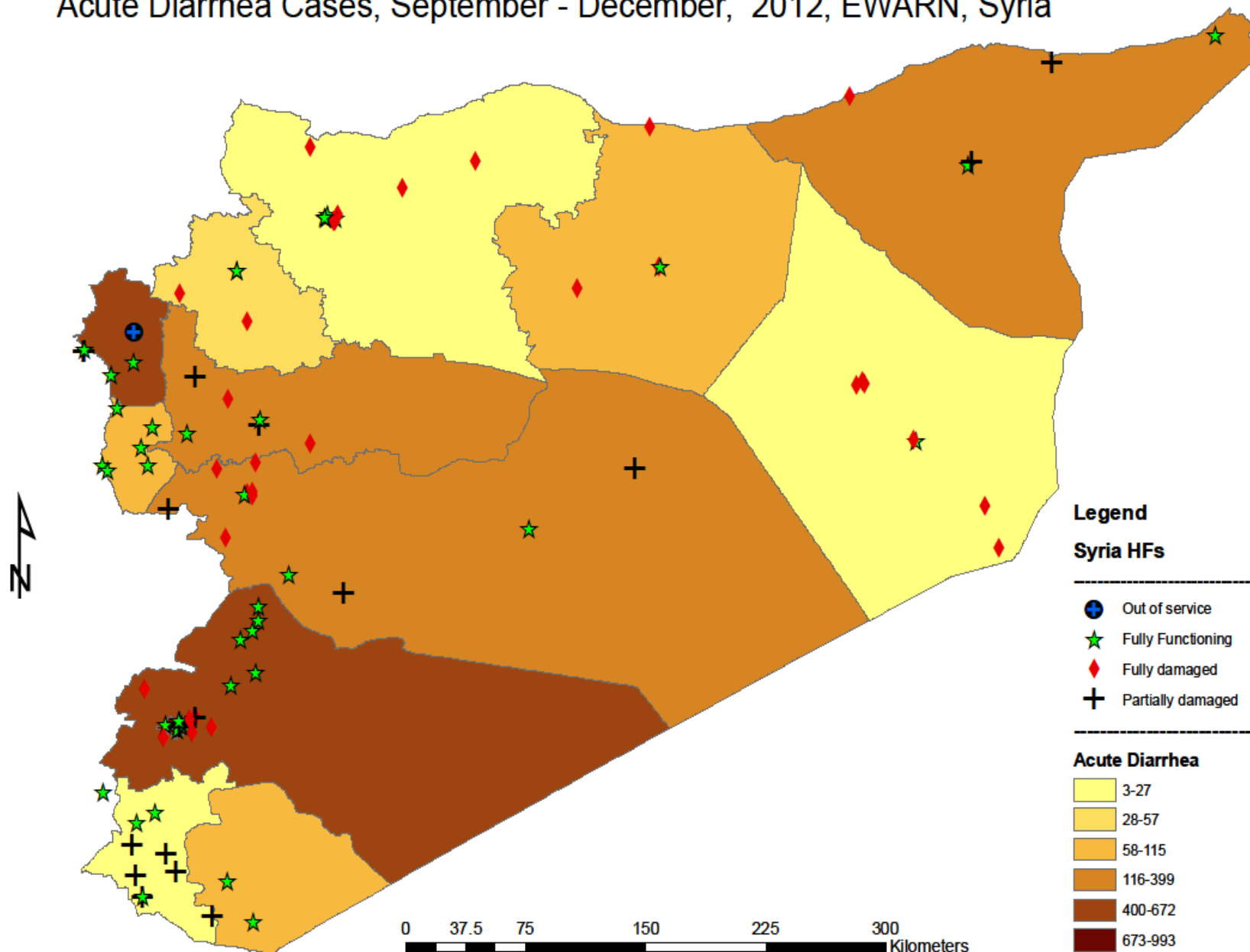
Data Source : WHO-Turkey
Date of Production: August, 2021
By: Negusu Worku : negusuworku@gmail.com



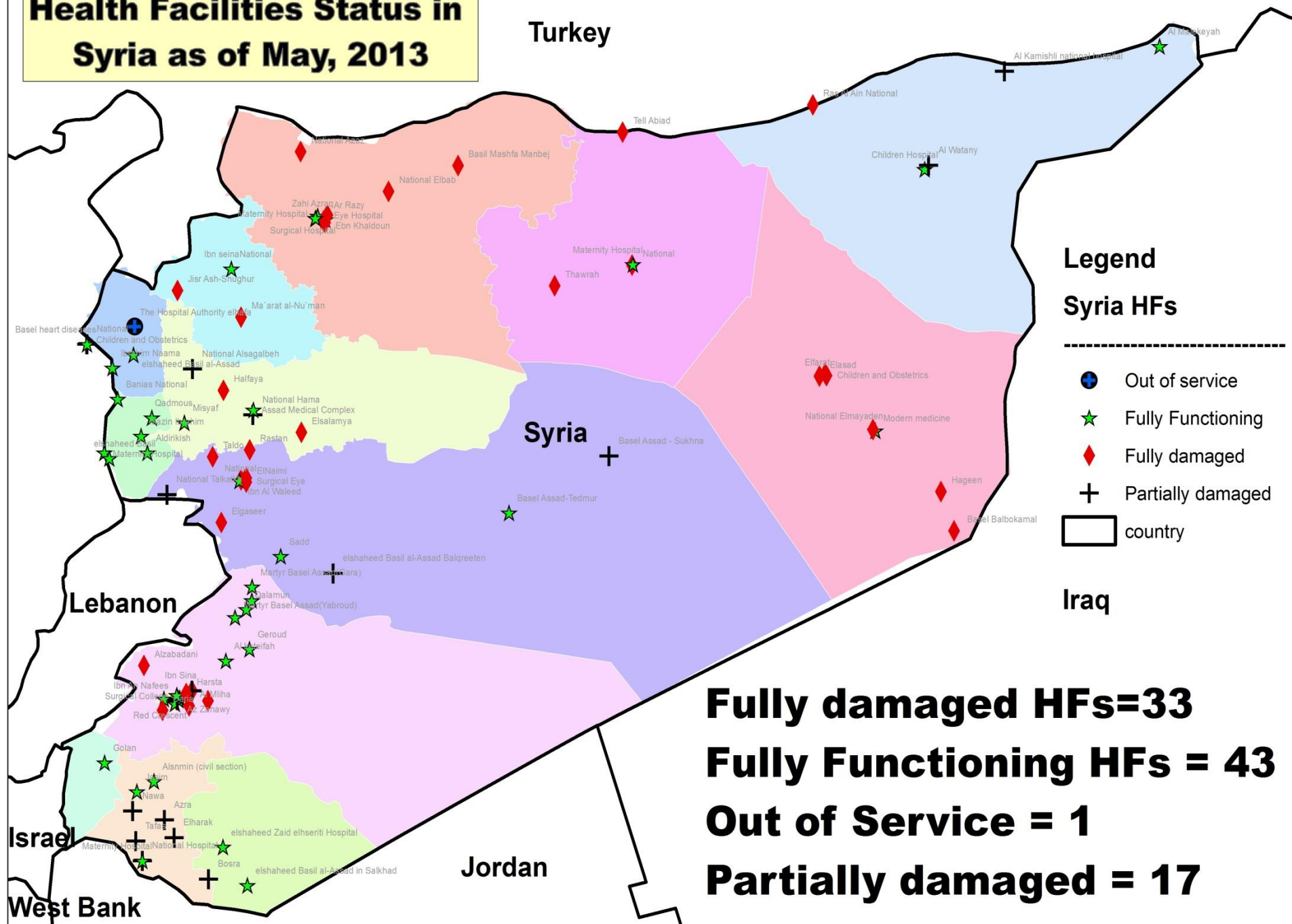
*From Kroman Chemical Industry **n=14** health facilities are available within 2kmR, of which **n=4 facilities** (1 health center and 3 pharmacies are available at CAYIROVA DISTRICT,

* From **3km radius from**, Kroman Chemical industry, **n=19** Health facilities are observed, of which **n=6 of them were from Chemetall Chemical industry** (n=3 Pharmacy and n=3 Health Center,

Acute Diarrhea Cases, September - December, 2012, EWARN, Syria



Health Facilities Status in Syria as of May, 2013



Bangladesh: Health Facilities by Type in Cox's Bazar's Camp Sites (As of 10 November 2018)



HEALTH SECTOR
COX'S BAZAR

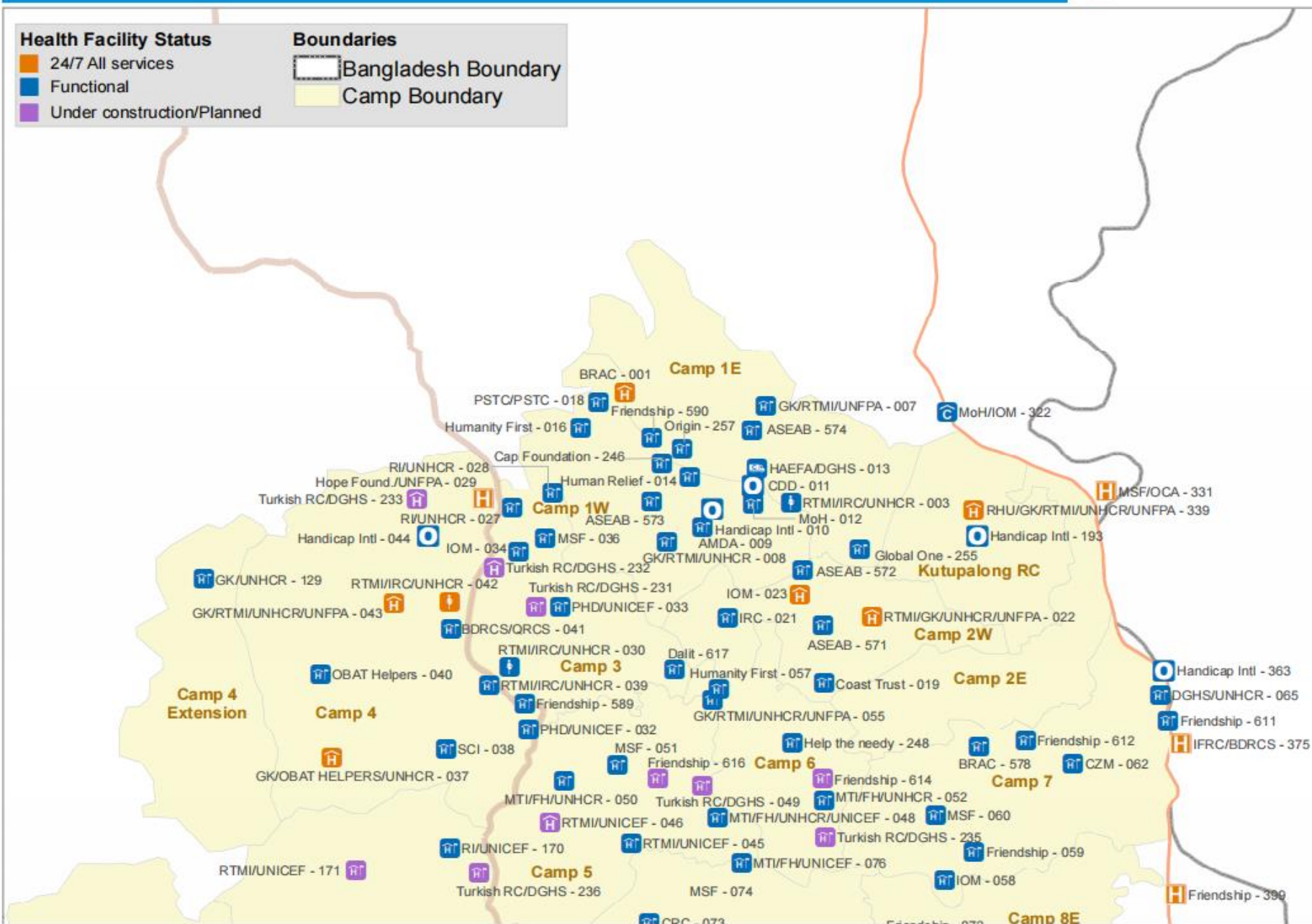


Health Facility Status

- 24/7 All services
- Functional
- Under construction/Planned

Boundaries

- Bangladesh Boundary
- Camp Boundary



References

1. Dr. Boris Pavlin, WHO-HQ, Switzerland, Geneva, Epidemiologist, pavlinb@who.int
2. Dr. Nsenga Ngoy nsengan@who.int Emergency Focal Person, WHO-AFRO, Congo, Brazzaville,
3. Dr. Sharmistha Mishra, MD, MSc, PhD, sharmistha.mishra@utoronto.ca
Associate Professor, Department of Medicine, University of Toronto
4. Dr. Mohamed Diaaeldin Omar, UNICEF-Kenya, Nutrition and Health, mdomer@unicef.org
5. Dr. Anthony Omoko, Epidemiologist, tonyomoko@gmail.com
6. Dr. Mohamed ALI, ALIM@who.int , WHOHQ

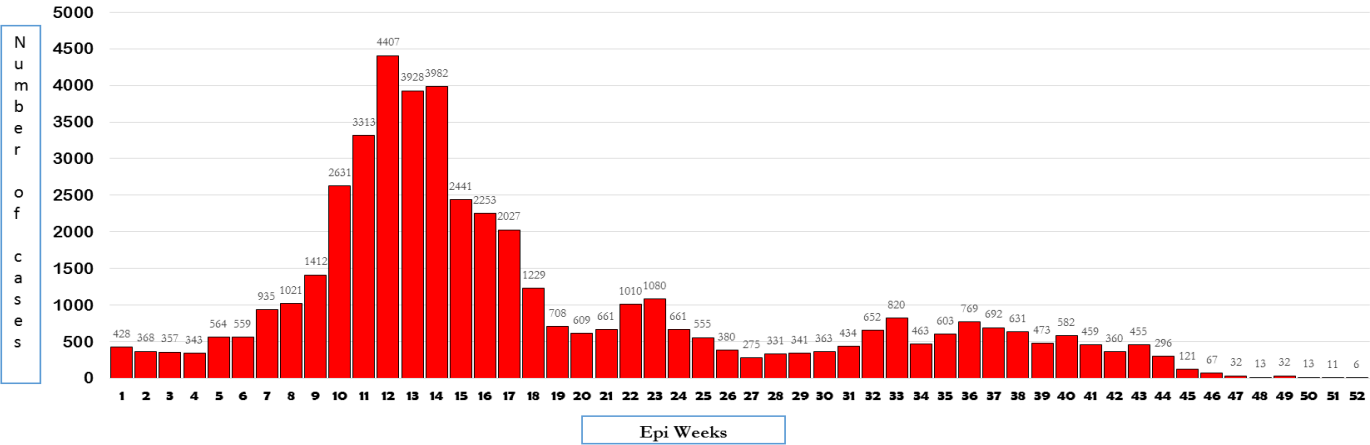
Negusu Worku GEBRMICHAEL

Background

- IT Specialist
- International applied Epidemiology (Emory, USA)
- GIS
- VRAM
- M&E
- Data management
- Information management

AWD Epi Curve (January – December, 2017)

National AWD trend as of Week 52 (January - December), 2017, Ethiopia



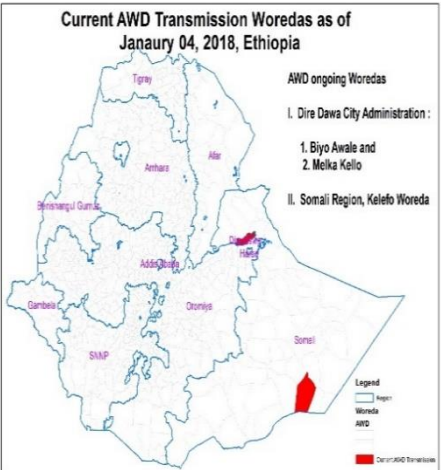
Source : RHBs and EPHI/PHEM

Current AWD transmission as of Week 52, 2017

In 2017 as of Week 52, from January – December 31, 2017, a cumulative of 48,638 AWD Cases and 859 AWD Deaths with CFR of 1.76% were reported from Seven regions in Ethiopia with one city Administrative Council. On week 51, a total of 11 AWD cases were reported from two clusters of Dire Dawa City Administrative Council.

The incidence rate of AWD continued to decrease significantly at national level as well as at regional level especially in Amhara, Oromia and Somali regions. The Only AWD active transmission Woredas is at Kelafo in Somali region and Biyo Awale and Melka Kello cluster from Dire Dawa as of Week 52.

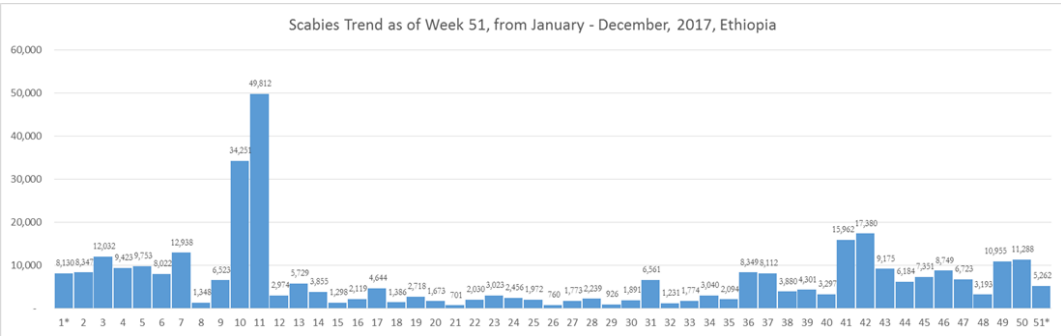
Current AWD Transmission Woredas as of January 04, 2018, Ethiopia



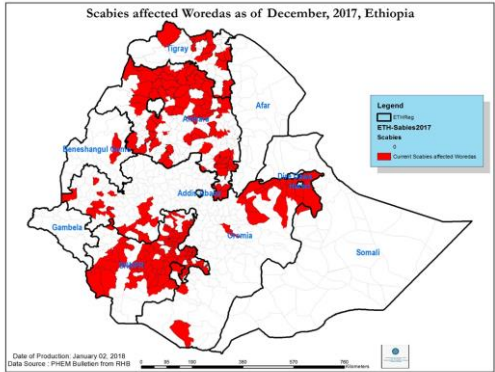
Scabies (From January – December, 2017

Source : Regional Health Bureaus – PHEM (RHBs)

Scabies Trend as of Week 51, from January - December, 2017, Ethiopia



Updated Scabies Case load, trend and Mapping , 2017,
(Data Source RHBs/PHEM Bulletin,) (Jan. – December 31, 2017)



Scabies outbreak is ongoing in five regions from 237 districts from Amhara (62), Oromia (45), SNNPR (118), Beneshangul Gumuz (one woredas (13 Kebeles) and Tigray (11) districts have been affected with a total case of Scabies (n=349,607).

In December, 2017 from Week 47-52, a total case of 37,421 Scabies cases were reported from Oromia, Amhara, SNNP, Tigray and Beneshangul Gumuz. A new outbreak of Scabies cases emerged in Beneshangul Gumuz Since the first week of December, 2017. In 2017, 516,768 Permethrin drugs have been distributed including (SNNP=412,208), Amhara=57,793), Tigray=23,983) and Oromia=22,784 Permethrin lotions have been distributed in the affected districts in 237 districts.

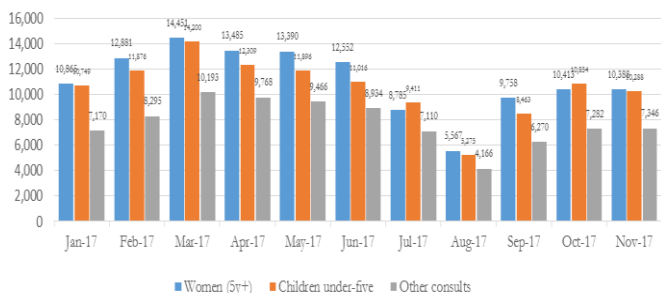
Generally, the Scabies outbreak trend has been increased in the last 16 weeks in all regions which can be attributed to the schools opening last September, 2017.

Distribution of Scabies Cases with Premetrin Lotion as of December, 2017, Ethiopia

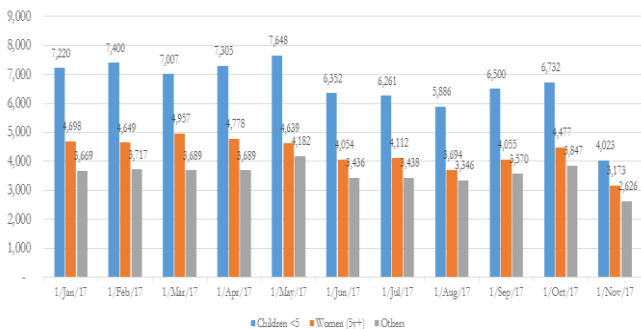
Region	December, 2017 (Week 47-52)	Total Scabies Cases by 2017	Number of Districts Currently affected by Scabies Cases	Permetrin Distribuion, 2017		
				March, 2017	September, 2017	October, 2017
Oromia	5984	28,221	45	10184	12600	
Amhara	21,653	82,201	62	30600	27193	
SNNP	5,540	223,034	118	156768	125840	129600
Tigray	911	12,817	11	10000	6783	7200
B_Gumuz	3334	3334	1			
Total	37,421	349,607	237	166,768	173,407	176,593

Mobile Health and Nutrition (MHNT), Total Consultations,
Afar and Somali Regions, Ethiopia as of December 31, 2017
(613,281 Consultations)

MHNT Consultation as of November, 2017, Ethio-Somali Region, Ethiopia (n=324,872)

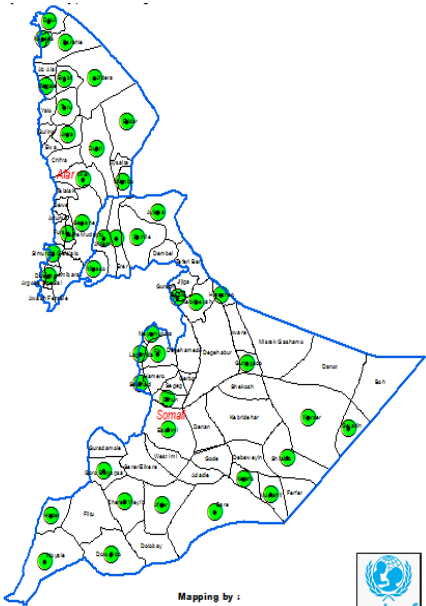
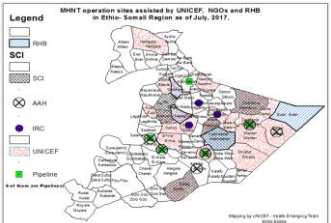


MHNT Consultation as of November, 2017, Afar Region, Ethiopia (n=158,829)



Data Source :RHBs

Location of MHNT Team, Afar and Somali region, Ethiopia



As of December 31, 2017, a cumulative total number of n=483,701 Consultations were made through n=49 Mobile Health and Nutrition Teams (MHNT program) in both Somali and Afar regions; of these, 158,829 from Afar region and 324,872 from Somali region. Out of the total consultation from both regions, 188,671 (39%) for children 5yrs+, 169,821 (35%) for Women and 125,209 (26%) for other groups.

Currently for Somali region, additional n=15 MHNT programs have been implemented through RHB (14) and 19 through NGOs

In Afar the reporting period of November, 2017 the completeness and timeliness of the reporting period coverage was 100%.

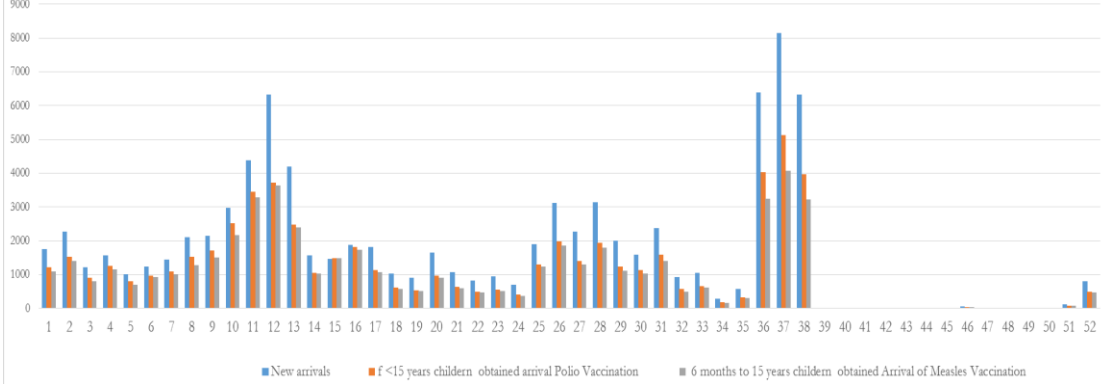


Influx of South Sudan Refugees in Gambella, Ethiopia , as of December 31, 2017

In 2017, a total of 87,678 new South Sudan refugees arrived in Lare and Akobo Woredas in Gambella Region. As of date week 52 (December 31, 2017), a total of 59,121 and 53,231 children less than 15 years received a dose of Polio and Measles Vaccines respectively at the entry points.

Source : Gambella UNICEF and UNHCR Field Offices)

Total new arrival of South Sudan Influx and their vaccination Status as of December, 2017, Gambella, Ethiopia

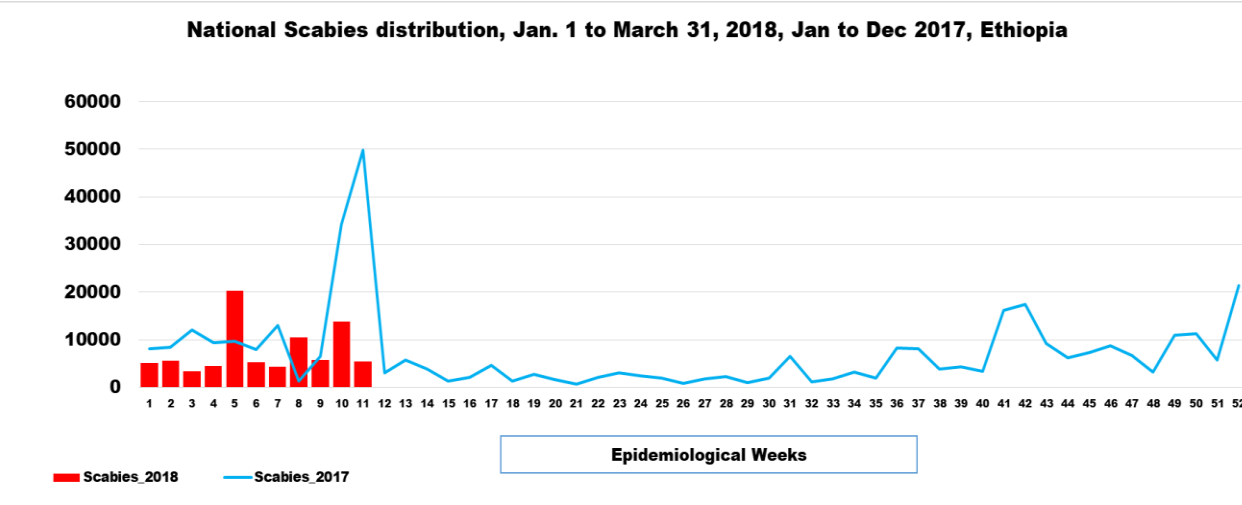
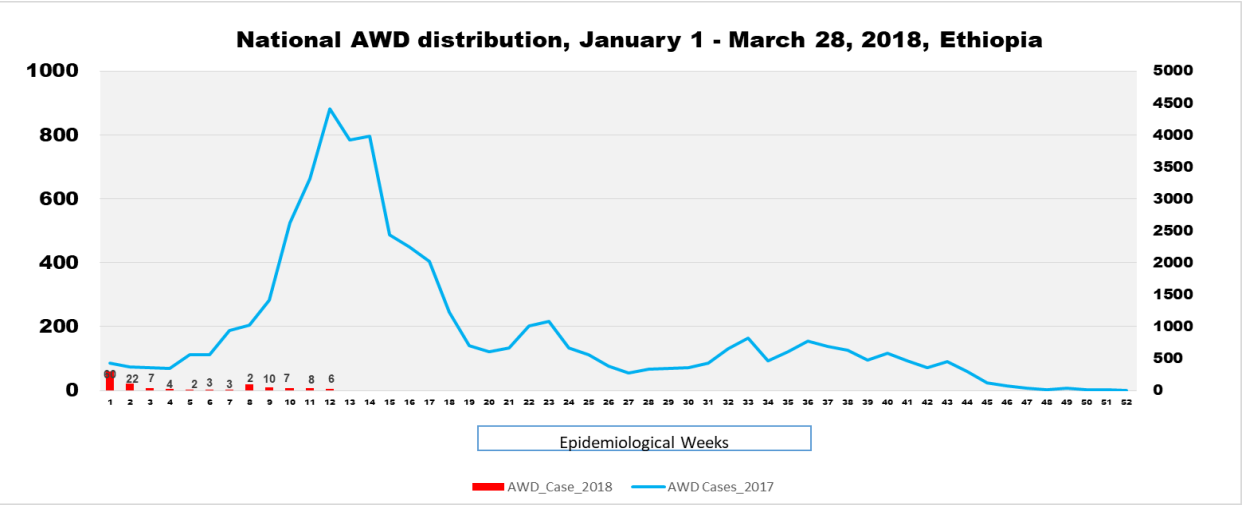


South Sudanese refugees new arrival in Gambella Sites at Gambella , December 31, 2017



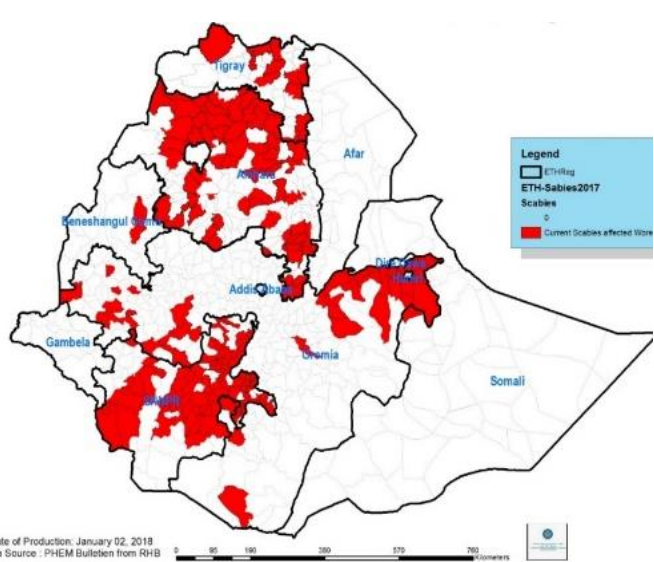
Indicator	Number
# of New South Sudanese arrival in 2017	87,678
# of <15 Children	55,265
# of 6mth. - <15Yrs Children	50,152
# of <15 Children Obtained arrival Polio Vaccination	59,121
# of 6mth to 15Yrs children Obtained Measles Vaccination	53,231

Data Source : UNICEF and UNHCR Gambella Field Offices



Data Source : Regional Health Bureau

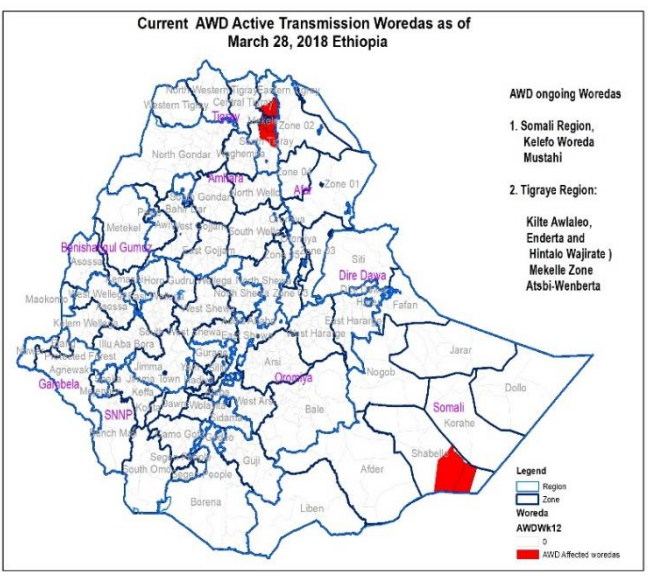
Scabies affected districts as of February, 2018



From March 1 – 31, 2018, a total number of 25,215 new Scabies Cases were reported from five regions. Oromia region continued to be the most affected region with highest number of cases reported. Please find case distribution in the table below. Comparing to the same period in 2017, Scabies cases were reduced by 73%.

Regions	No. of Scabies Cases from March 1 - 31, 2018	Cumulative Scabies Cases from Jan. 1 - March 31, 2018	Scabies Cases 2017
Oromia	14,335	20,055	29,876
Amhara	6,154	50,719	100,301
SNNP	4,210	10,377	225,150
Tigraye	500	2,078	12,962
B_Gumuz	16	1,516	3445
Total	25,215	84,745	371,734

Source : Regional Health Bureaus



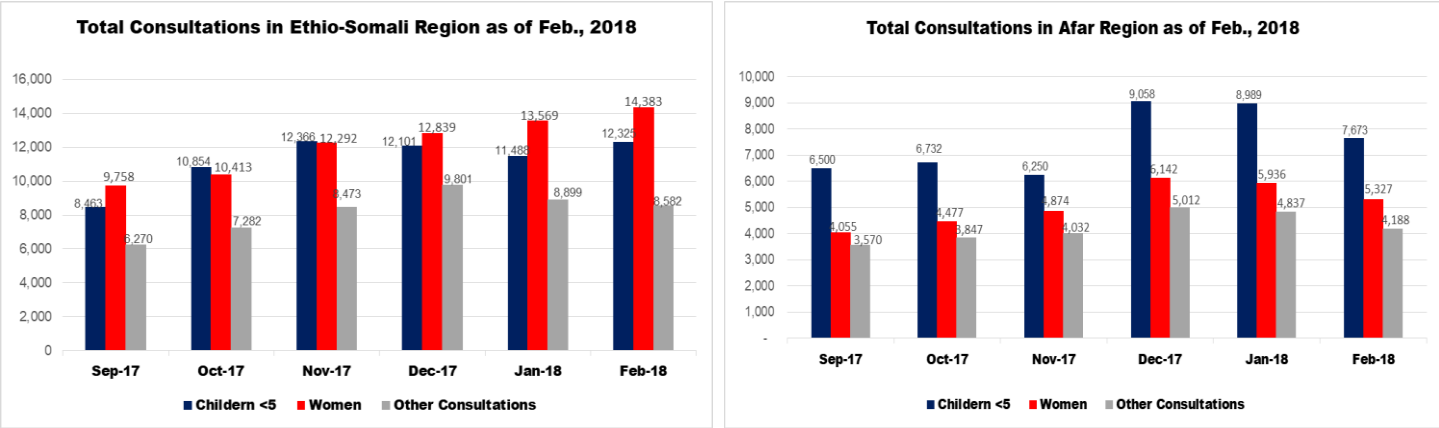
Source : EPHI/PHEM – EOC Center.

On epidemiological Week 12, 2018 four (4) cases were reported from Ethio-Somali and to (2) from Tigray Regions.

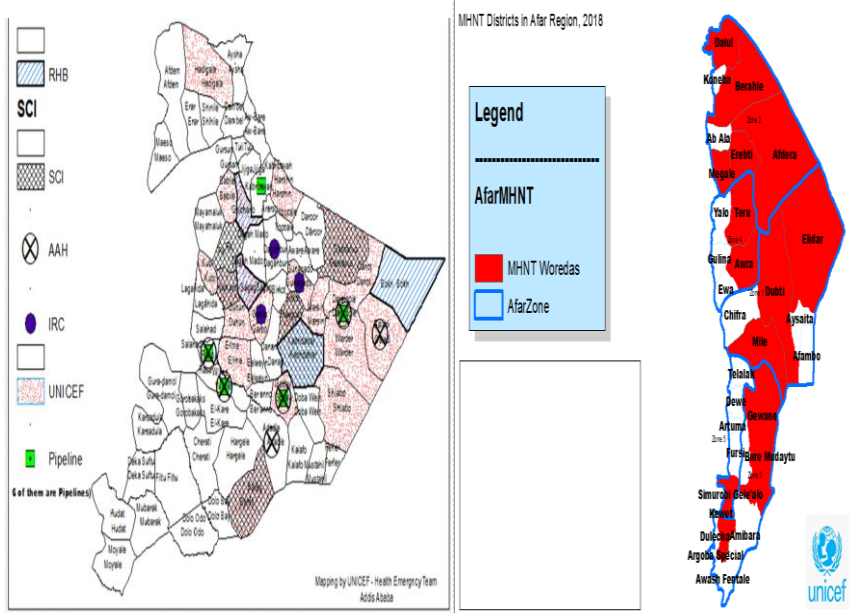
From March 1 – 31, 2018, a total number of 31 AWD Cases with zero death were reported from Somali (8) and Tigraye (23) regions. This is significantly lower compared to the same period in 2017. Total Cumulative AWD cases from January 1 to March 31, 2018 is 151.

In Dire Dawa , zero AWD cases have been reported since January 10, 2018.

Mobile Health and Nutrition Teams, Total Consultations, Afar and Somali Regions, Ethiopia as of March, 31, 2018



Locations of MHNT Team, Afar and Somali region, Ethiopia



Data Source :Regional Health Bureau

From February 1 – 28, 2018, a cumulative 52, 478 medical consultation conducted in hard to reach ears of Afar and Somali regions, of which, 19,998 (38%) children 5yrs+, 19,710 (38%) Women .

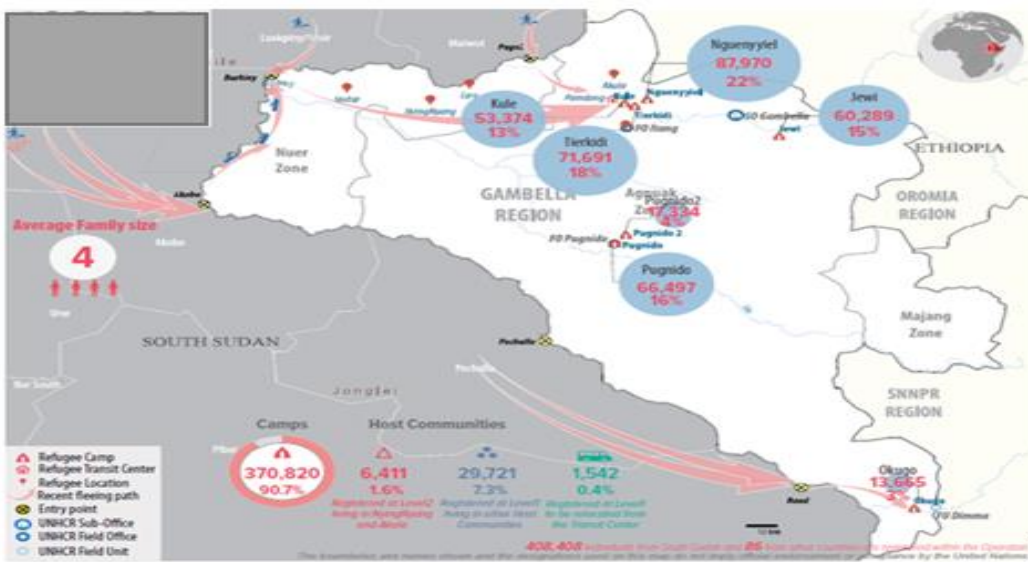
The medical Consultations has been performed by 49 UNICEF supported Mobile Health and Nutrition Teams plus 11 International NGOs and 6 RHBs in Somali in region ; Of these consultations, 35,290 were from Somali (46 teams) and 17,188 from Afar with 20 Teams. In addition, from January to February 2018, a total 10,817 medical consultation has been conducted by 15 Sustainable Outreach Service Team (SOS). The consultation has been increasing in both regions

Due to delays in reporting, no March data is received as of date.

South Sudanese Refugees Response, Jan. 1 – March 28, 2018, Ethiopia

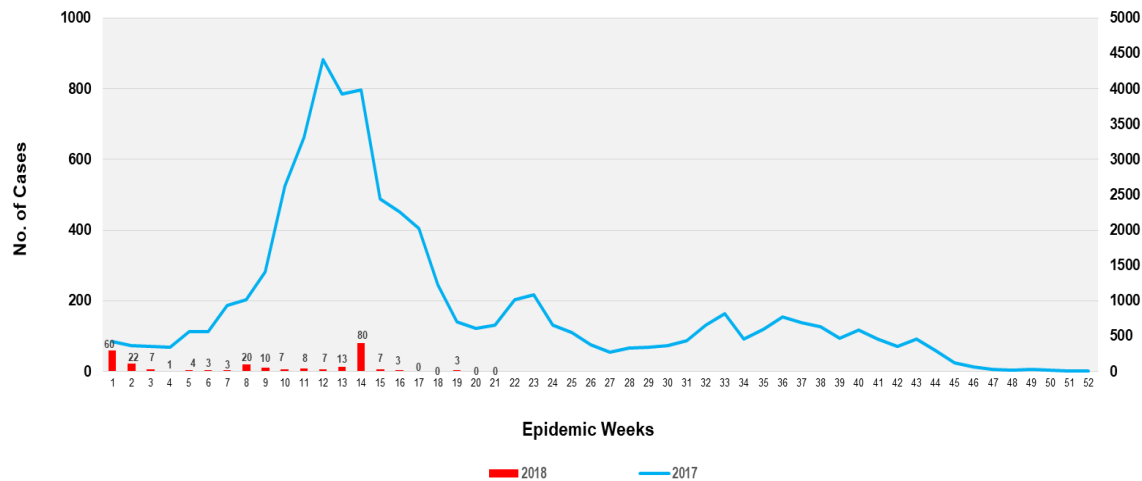
Indicator	2017	March, 2018
# of New South Sudanese arrival in 2017	87,678	15,778
# of <15 Children	55,265	8,186
# of 6mth. - <15Yrs Children	50,152	7,782
# of <15 Children Obtained arrival Polio Vaccination	59,121	7,782
# of 6mth to 15Yrs children Obtained Measles Vaccination	53,231	7,782

South Sudanese refugees, Gambella Sites, March, 2018



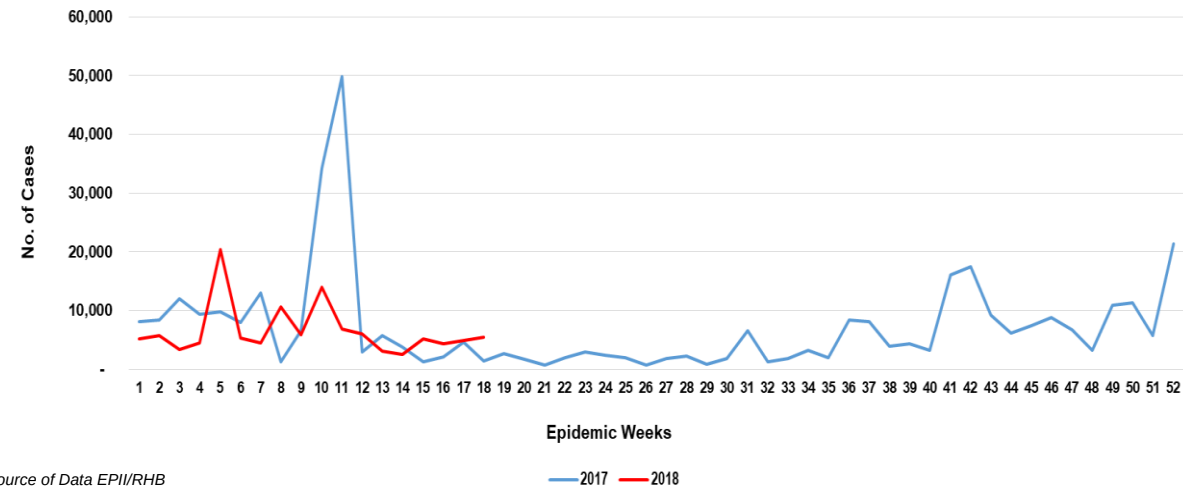
Data Source : Gambella UNHCR Field Office

National trend of reported AWD Cases, 2017/2018, Ethiopia



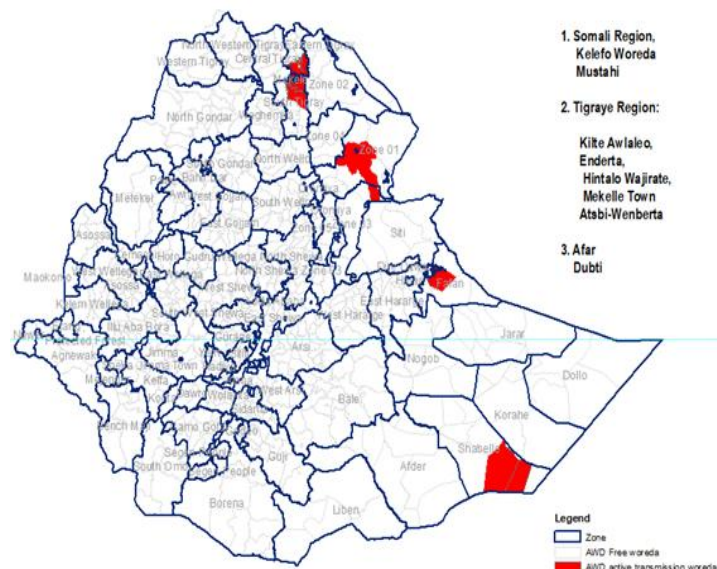
Source of Data EPII/RHB

National trend of reported Scabies cases 2017/2018, Ethiopia



Source of Data EPII/RHB

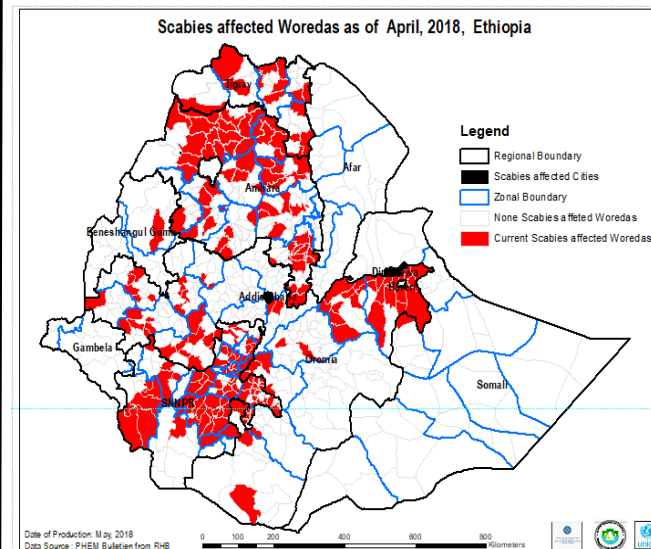
AWD active woredas as of April 30, 2018, Ethiopia



From April 1 – 30, 2018, a total number of 91 AWD Cases with zero death were reported from Somali (26), Tigraye (1) and Afar (64) regions. This is significantly lower compared to the same period in 2017. Total Cumulative AWD cases from January 1 to April 30, 2018 is 255.

On April 04, 2018, a new AWD outbreak occurred in Afar region. On epidemiological Week 17, 2018 Zero AWD cases were reported from the affected regions of Afar, Tigray and Ethio-Somali regions.

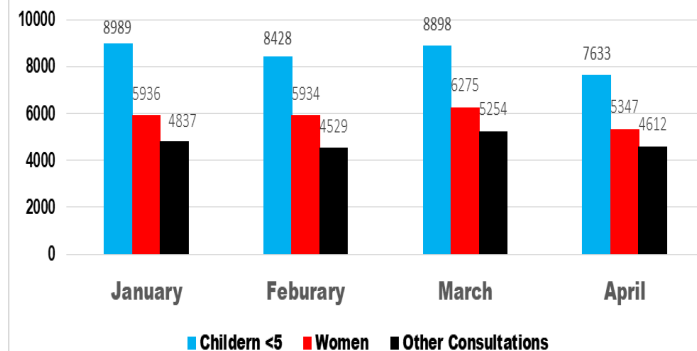
Scabies active woredas 2018, Ethiopia



From April 1 – 30, 2018, a total number of 20,365 new Scabies Cases were reported from Seven regions. Of which Dire Dawa and Addis Ababa regions became a newly Scabies affected city administrative regions. Amhara, SNNP and Oromia regions reported highest number of cases. Please find case distribution in the table below. Comparing to the same period in 2017, Scabies cases were reduced by 65%.

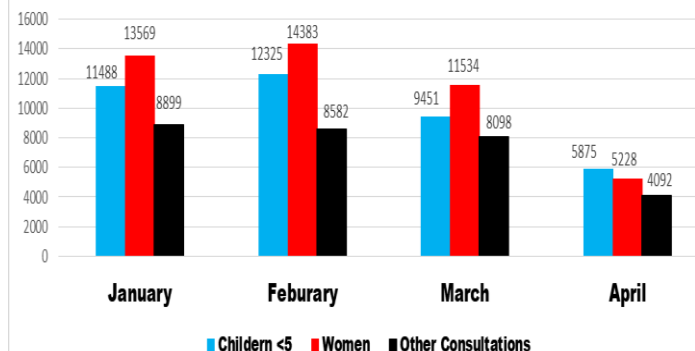
Regions	April, 2018	Cummulative 2018	Cummulative 2017
Oromia	6,070	28,075	29,876
Amhara	6,405	61,295	100,301
SNNP	6,044	17,076	225,150
Tigraye	856	3,384	12,962
B_Gumuz	8	1,528	3445
Addis Ababa	515	1,009	0
Dire Dawa	467	467	0
Total	20,365	112,834	371,734

MHNT : Medical Consultation in Afar Region, April, 2018



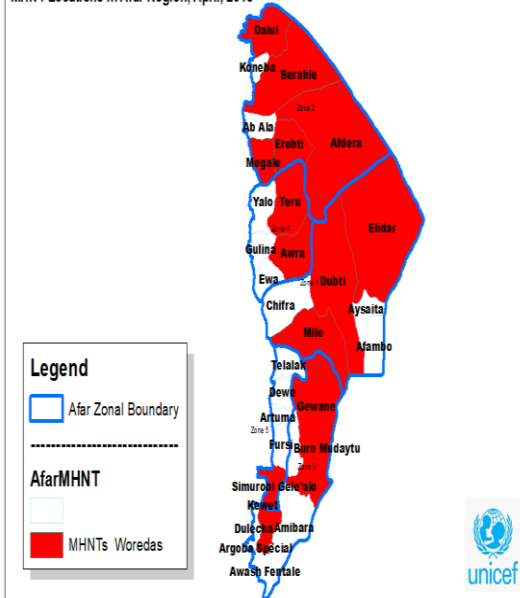
Source: Regional Health Bureaus

MHNT : Medical Consultation in Ethio-Somali Region, 2018.

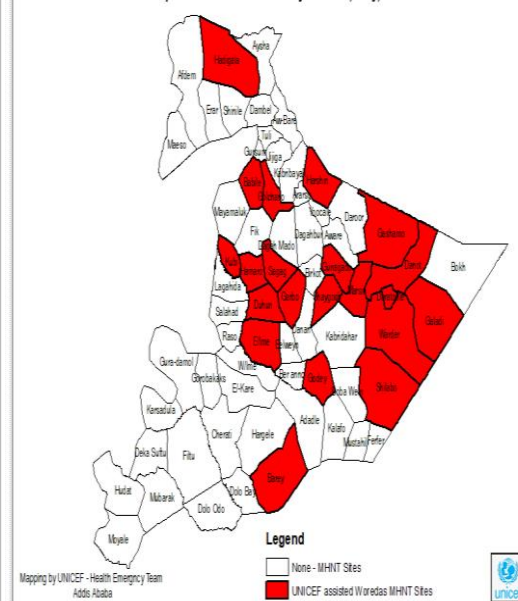


Locations of MHNTs, Afar and Ethio-Somali Regions, April, 2018.

MHNT Locations in Afar Region, April, 2018



MHNT operation sites assisted by UNICEF, May, 2018



From April 1–30, 2018 a cumulative total of 32,787 medical consultations conducted in hard to reach areas of Afar and Somali regions, of which, 13,508 (41%) under five children and 10,575 (32%) were Women.

From January 1–April 30, 2018, a total cumulative 190,196 MHNT consultations were made 76,672 from Afar and 113,524 from Ethio-Somali region. Of which 73,087 (38%) Under five children, 68,206 (36%) Women and 48,903 (26%) were from Other adults.

The medical Consultation has been performed by 49 UNICEF supported Mobile Health and Nutrition Teams plus 11 International NGOs and 6 RHBS in Somali in region.

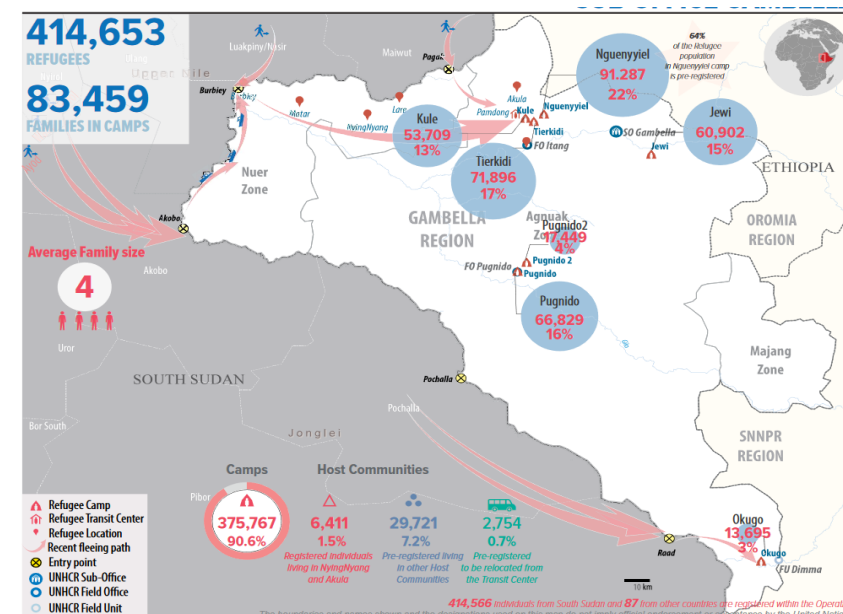
Moreover, total MHNT beneficiaries from both regions became nearly 2.8 million. Of which 1,689,651 from Ethio-Somali and the rest 1,083,044 from Afar region.

South Sudanese Refugees: Selective Screening Vaccination, 2017/2018

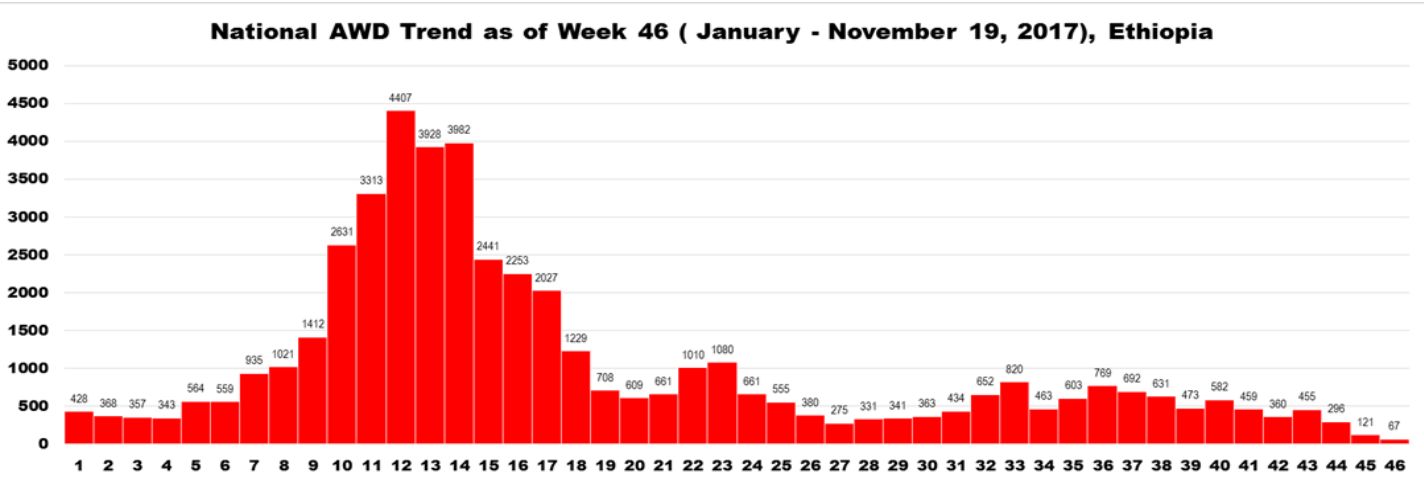
Indicators	April 2018	Cummulative 2018	Cummulative 2017
Total No. of new arrivals	2777	19,178	87,678
No. of Children Under 15 years	1742	10,333	55,265
No children 6 months - 14 years	1498	9,629	50,152
No. of children received polio vaccination	1711	10,344	59,121
No of children received Measles Vaccination	1521	9,684	53,231

Source: UNICEF and UNHCR Gambella Field Offices

South Sudanese Refugees in Gambella Sites, April, 2018

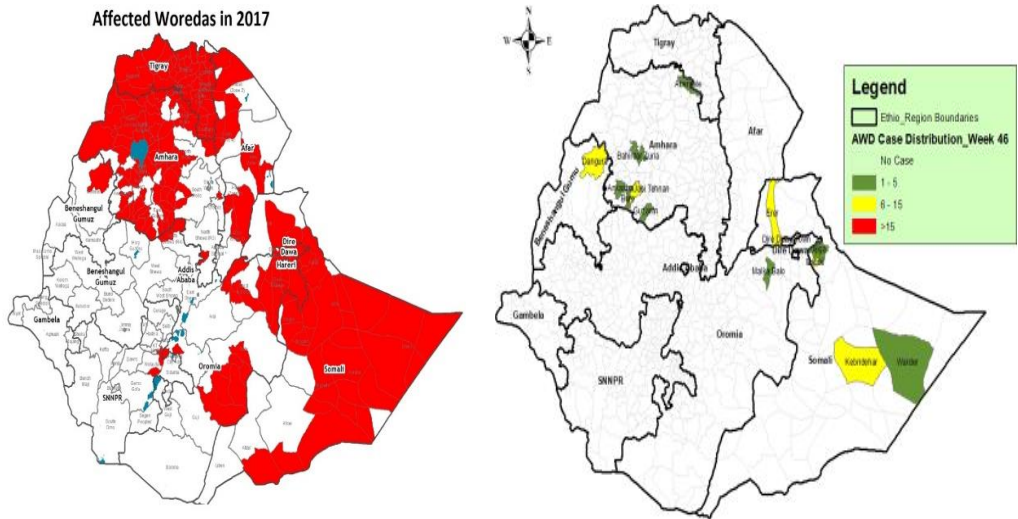


AWD Epi Curve (January – November 30, 2017)



Source : RHBs, PHEM and HCC

Current AWD transmission as of week 46, 2017



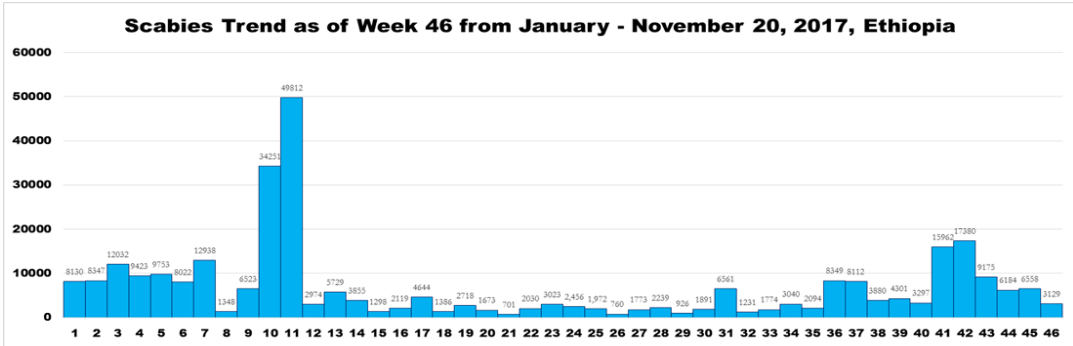
In 2017 as of Week 46, from January – November 19, 2017, a cumulative total of 48,638 AWD Cases and 859 AWD Deaths with CFR of 1.8% were reported from Seven regions in Ethiopia with one city Administrative Council. On week 46, a total of 67 AWD cases from 16 woredas were reported at national level; the distribution of the cases are: Oromia (4), Amhara (22), Somali (26), Dire Dawa (8), BGumuz (6), and Tigraye (1).

The incidence rate of AWD continued to decrease significantly at national level as well as at regional level especially in Amhara, Oromia and Somali regions.

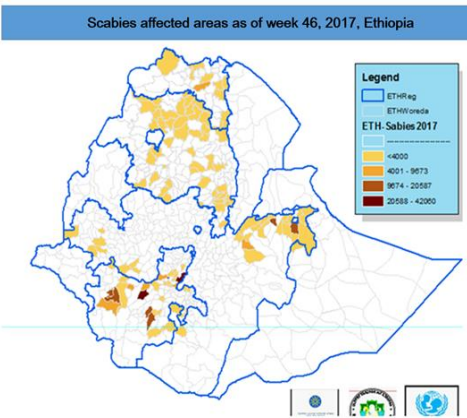
Current AWD hotspot with Caseload as of November 30, 2017 Source : EPHI/PHEM

Scabies (From January – November 30, 2017

Source : Regional Health Bureaus – PHEM (RHBs)



Updated Scabies Case load, trend and Mapping , 2017, (Data Source RHBs/PHEM Bulletin,) (Jan. – November 30, 2017)

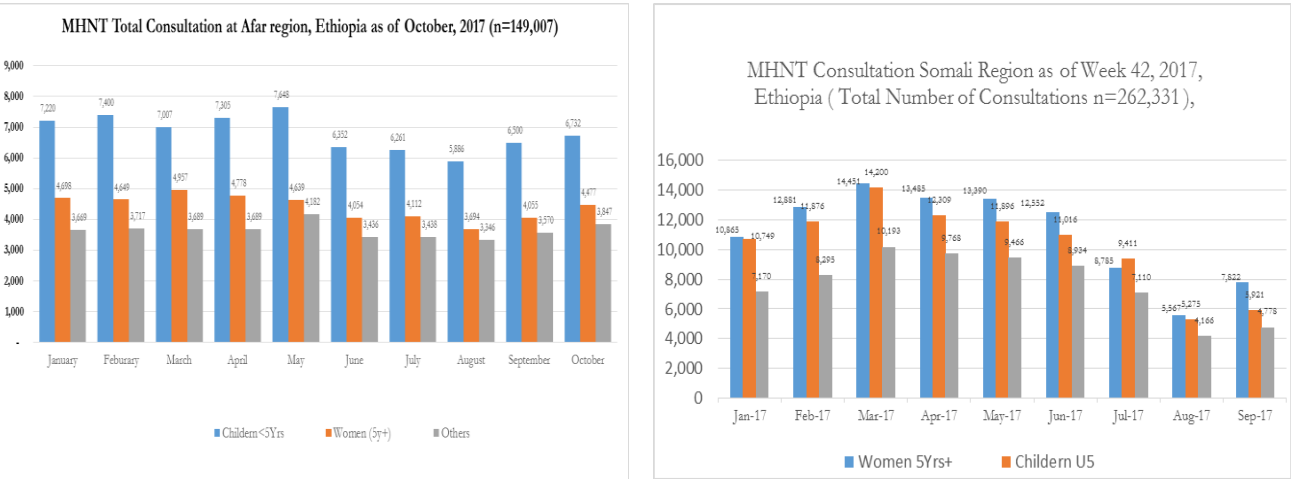


Scabies outbreak is ongoing in four regions from 74 districts from Amhara (33), Oromia (12), SNNPR (20), and Tigray (9) districts. A total of 42,426 new scabies cases were reported for the month November, 2017 (Epidemic Weeks 42-46). The total cumulative reported scabies cases in 2017 became 305,773 from the four regions: Amhara (56,358), Oromia (20,015), SNNP (217,494) and Tigray (11,906). Generally, the Scabies outbreak trend has increased in the last Six weeks in all regions which can be attributed to the schools opening last September.

Region	November, 2017 (Week 42 - 46)	2017
Amhara	11,832	56,358
Oromia	1509	20,015
SNNP	28,014	217,494
Tigray	1,071	11,906
Total	42,426	305,773

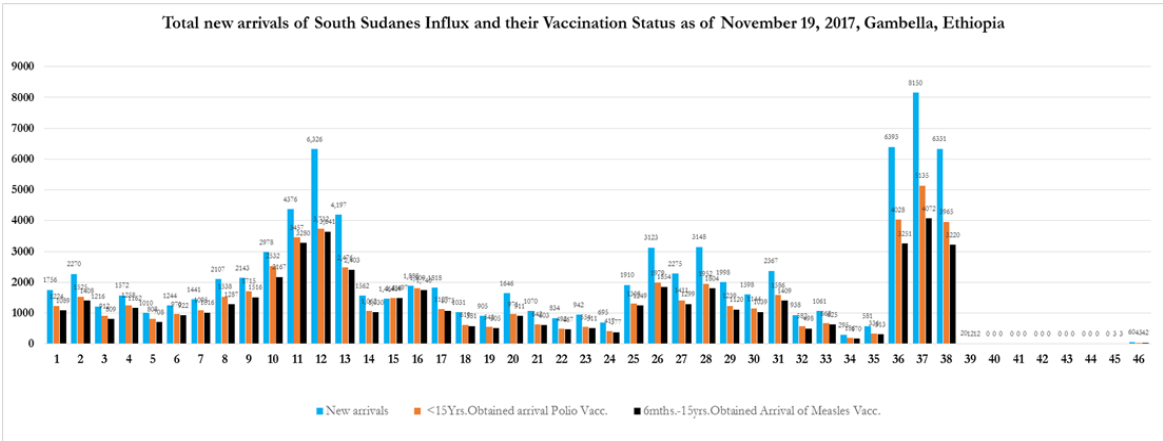
Response interventions are ongoing in the ground which including: active surveillance case finding, case management, social mobilization activities in the affected woredas and distribution of 350,000 Permethrin drugs (SNNP=255,440), Amhara=57,793), Tigraye=13,983) and Oromia=22,784 Permethrin lotions have been distributed in the affected districts.

Mobile Health and Nutrition (MHNT), Total Consultations, Afar and Somali Regions, Ethiopia as of November 30, 2017

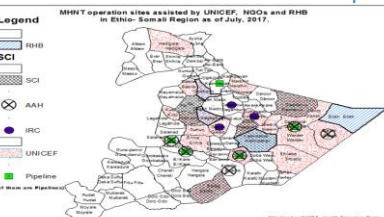


Influx of South Sudan Refugees in Gambella, Ethiopia , as of November 30, 2017

In 2017, a total of 86,739 new South Sudan refugees arrived in Lare and Akobo Woredas in Gambella Region. As of date week 46 (November 19, 2017), a total of 58,495 and 52,683 children less than 15 years received a dose of Polio and Measles Vaccines respectively at the entry points.
Source : Gambella UNICEF and UNHCR Field Offices)

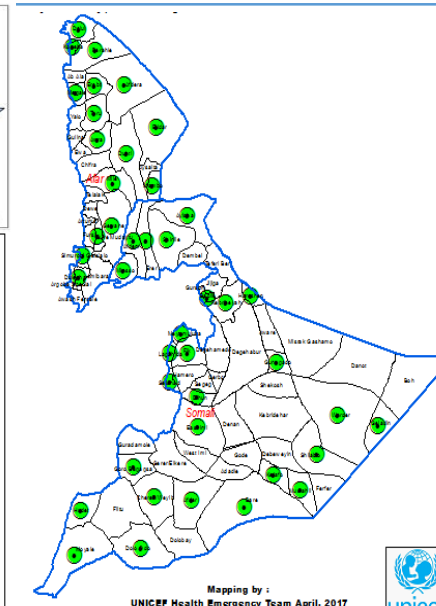


Data Source :RHBs



Mapping by :
UNICEF Health Emergency Team April, 2017

Location of MHNT Team, Afar and Somali region, Ethiopia



Mapping by :
UNICEF Health Emergency Team April, 2017

In 2017, a cumulative total of n=411,338 Consultations were made through n=49 Mobile Health and Nutrition Teams (MHNT program) in both Somali and Afar regions; of these, 149,007 from Afar and 262,331 from Somali. Out of the total consultation from both regions, 143,911 (35%) were for Women, 160,964 5yrs+ Children under 5 years and 106,463 (26%) for other groups.

Currently for Somali region, additional n=15 MHNT programs have been implemented through RHB (5), NGOs (10) while 9 additional teams from the NGOs are in the pipeline.

Please take note that the October report on the consultations is still incomplete and will be updated during the next reporting period including the timeliness and completeness.

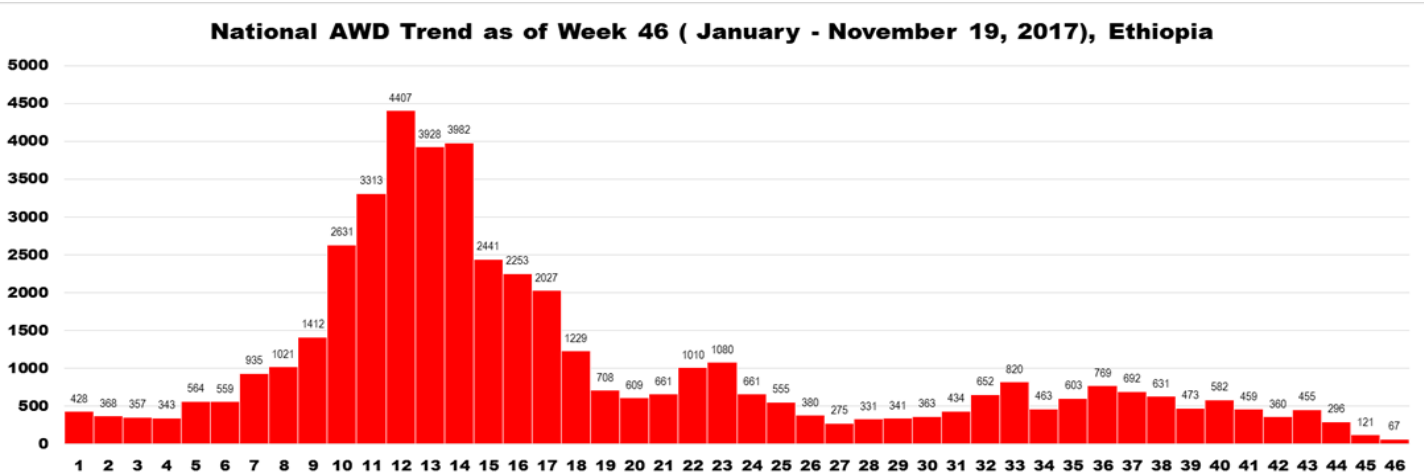
South Sudanese refugees new arrival in Gambella Sites at Gambella , November 30, 2017



Indicator	Number
# of New South Sudanese arrival in 2017	86,739
# of <15 Children	54,687
# of 6mth. - <15Yrs Children	46,391
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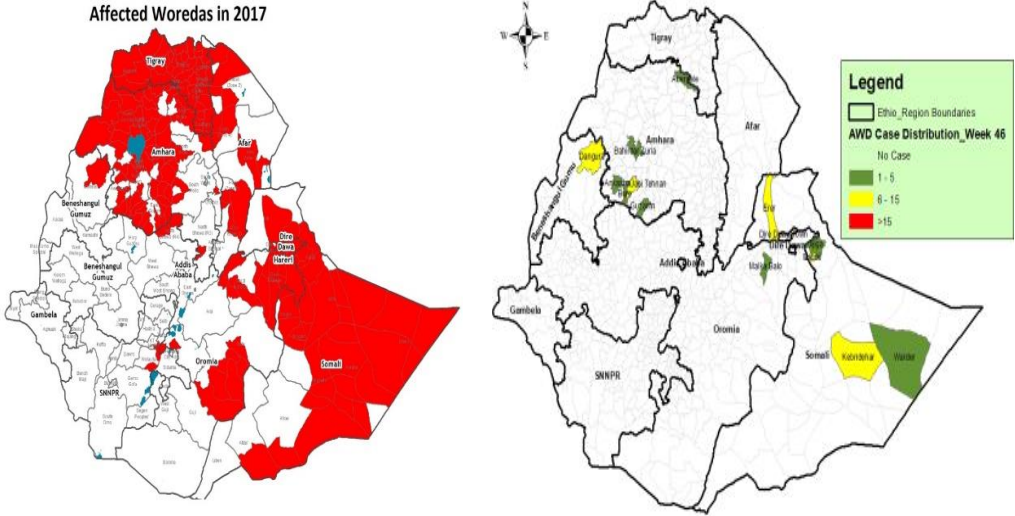
Data Source : UNICEF and UNHCR Gambella Field Offices

AWD Epi Curve (January – November 30, 2017)



Source : RHBs, PHEM and HCC

Current AWD transmission as of week 46, 2017



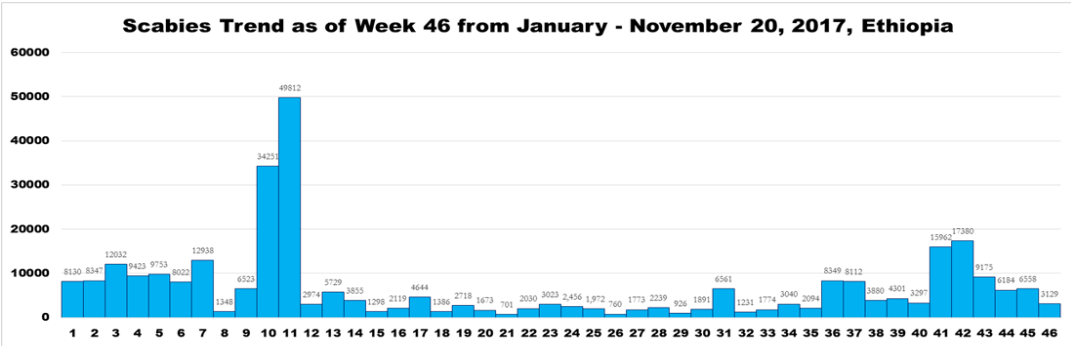
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The incidence rate of AWD continued to decrease significantly at national level as well as at regional level especially in Amhara, Oromia and Somali regions.

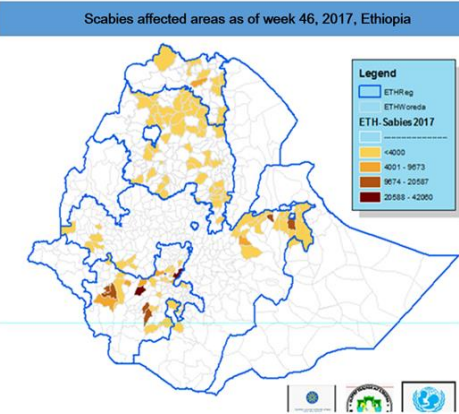
Current AWD hotspot with Caseload as of November 30, 2017 Source : EPHI/PHEM

Scabies (From January – November 30, 2017

Source : Regional Health Bureaus – PHEM (RHBs)



Updated Scabies Case load, trend and Mapping , 2017,
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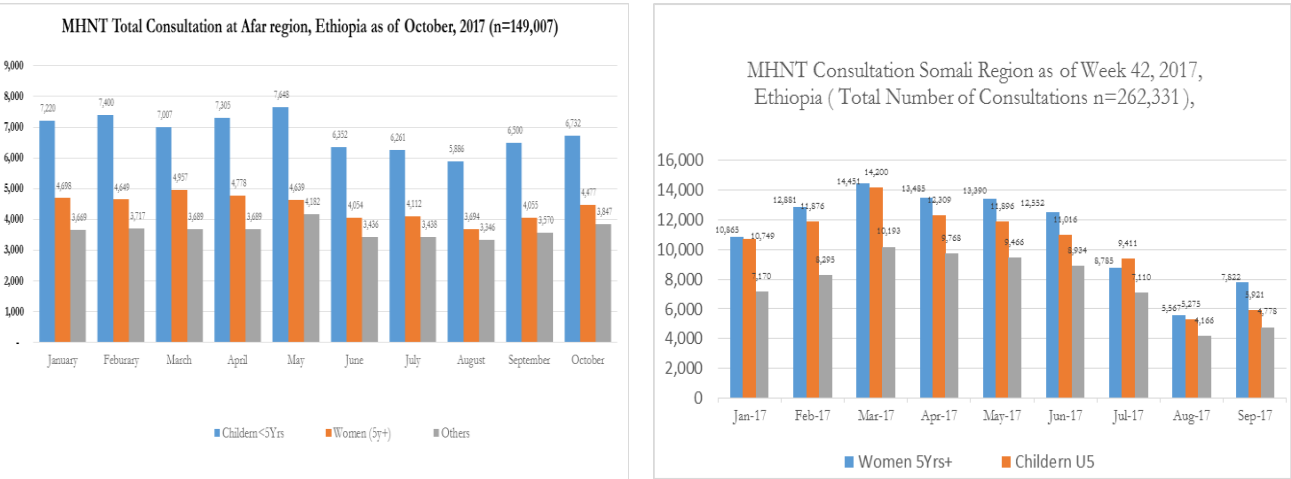


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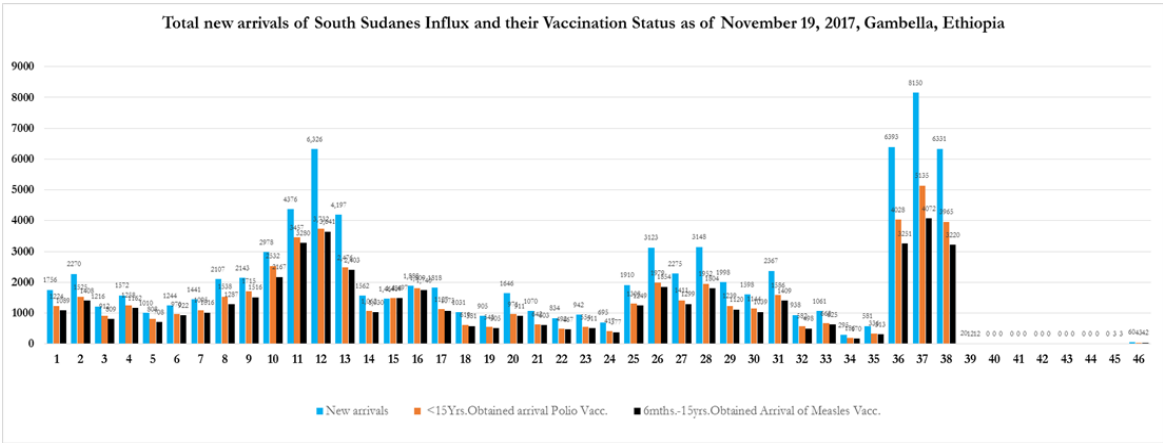
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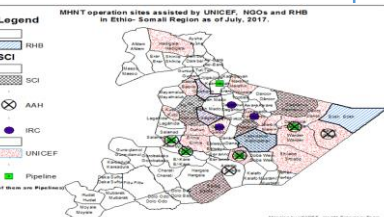


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Source : Gambella UNICEF and UNHCR Field Offices)

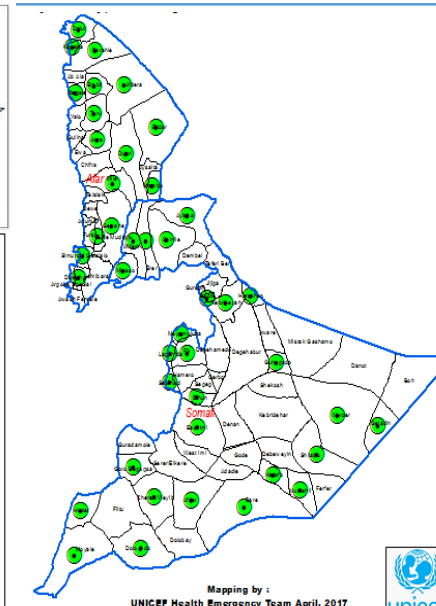


Data Source :RHBs



Mapping by :
UNICEF Health Emergency Team April, 2017

Location of MHNT Team, Afar and Somali region, Ethiopia



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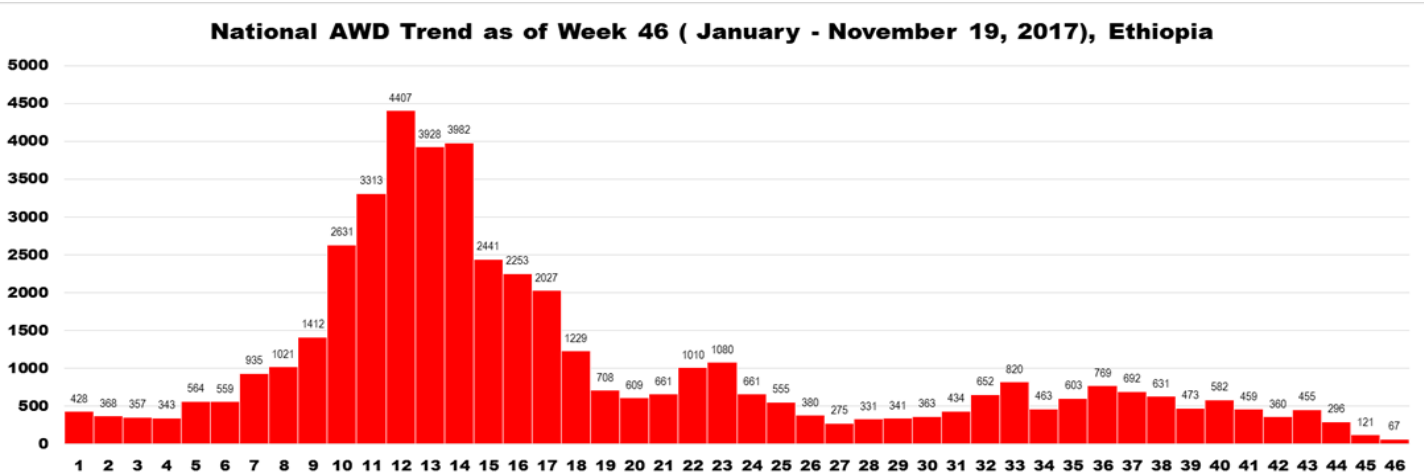
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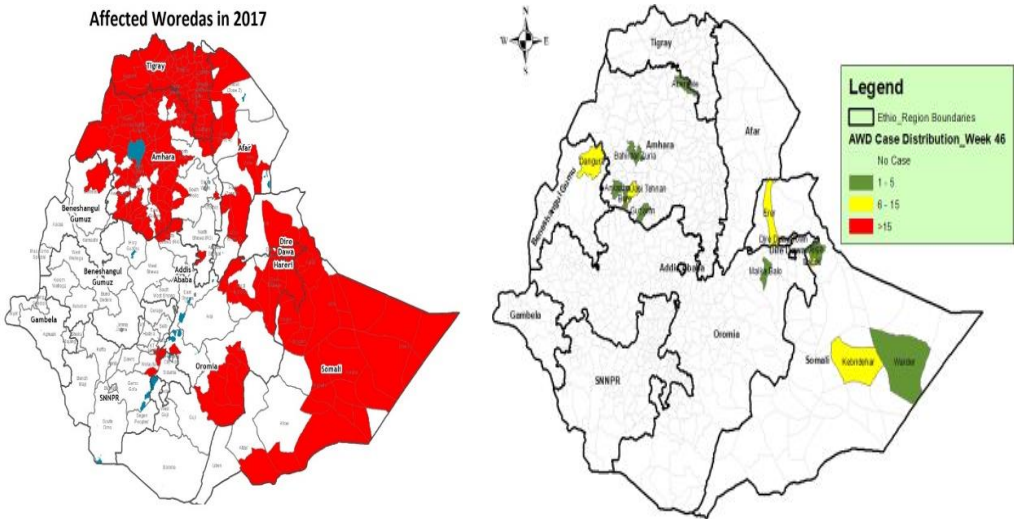
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AWD Epi Curve (January – November 30, 2017)



Source : RHBs, PHEM and HCC

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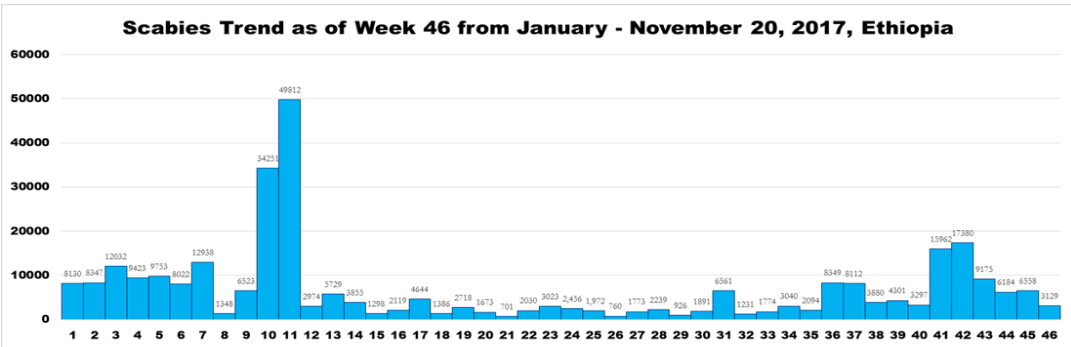
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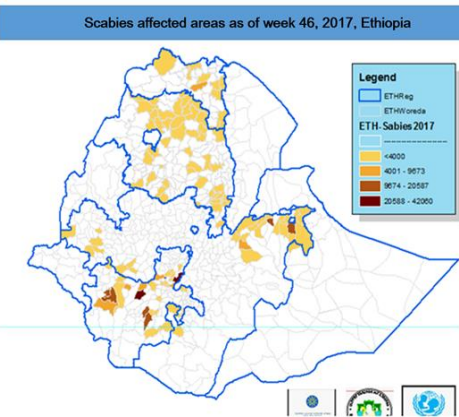
Current AWD hotspot with Caseload as of November 30, 2017 Source : EPHI/PHEM

Scabies (From January – November 30, 2017

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Updated Scabies Case load, trend and Mapping , 2017,
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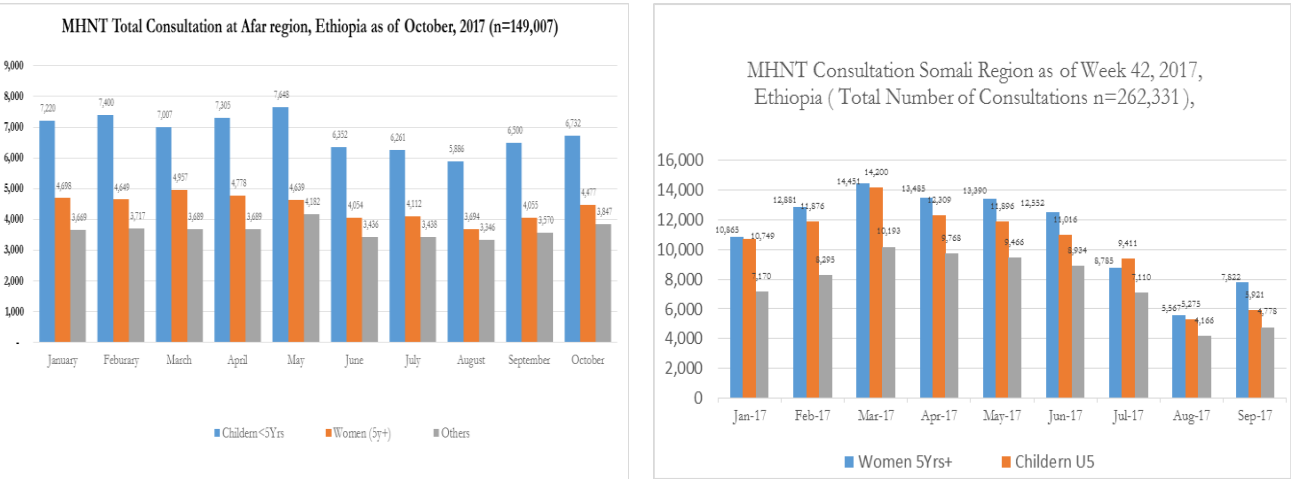


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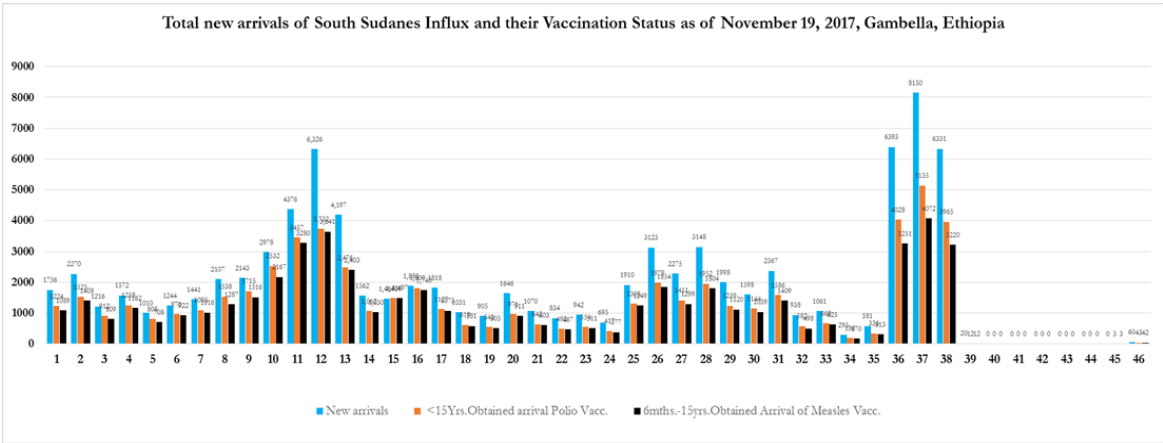
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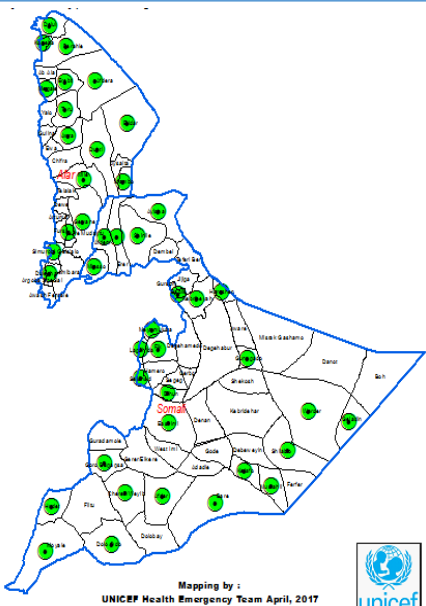
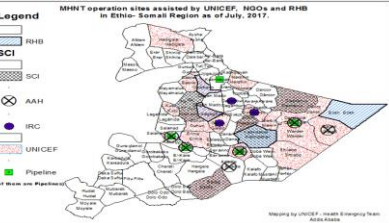
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Source : Gambella UNICEF and UNHCR Field Offices)



Data Source :RHBs

Location of MHNT Team, Afar and Somali region, Ethiopia



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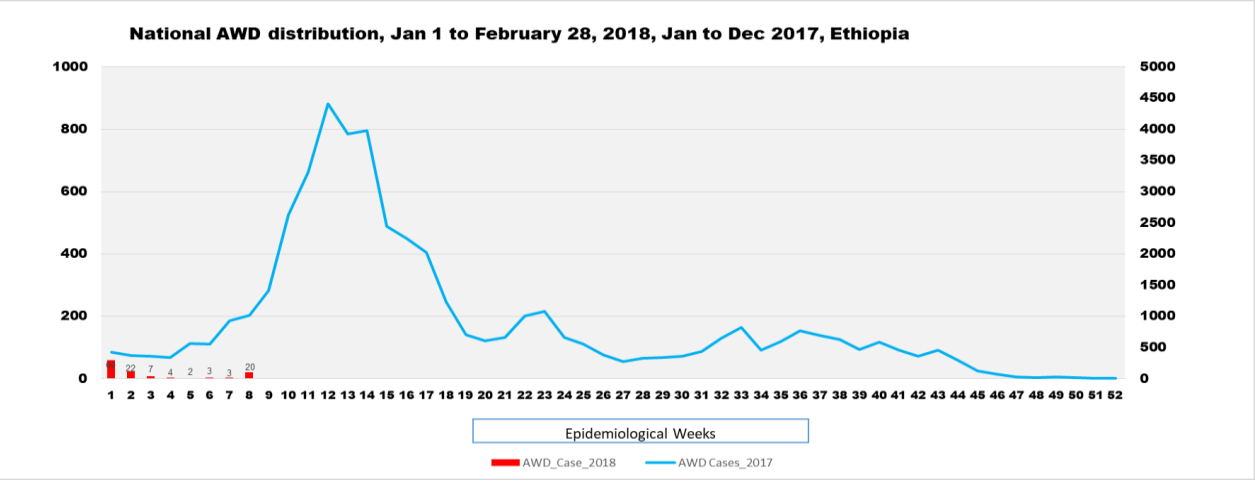
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South Sudanese refugees new arrival in Gambella Sites at Gambella , November 30, 2017

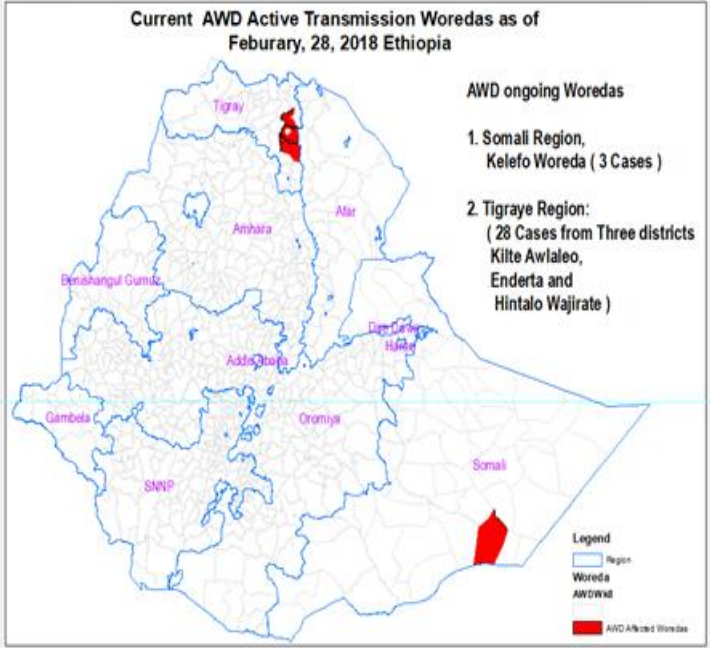


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Source : Regional Health Bureaus

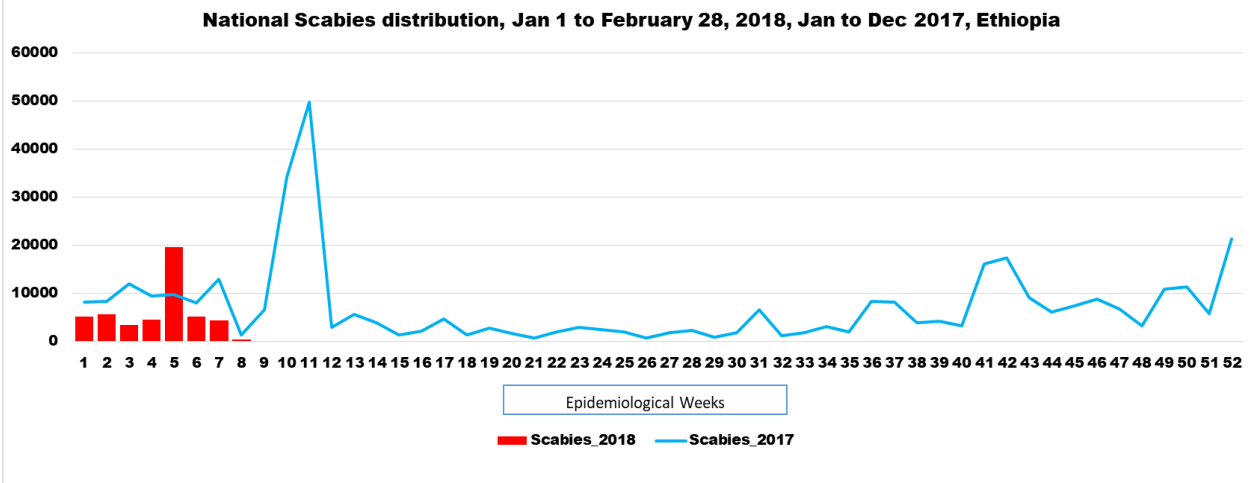


Source : EPHI/PHEM – EOC Center.

From February 1 – 28, 2018, a total number of 28 AWD Cases with zero death were reported from Somali (8) and Tigraye (20) regions. This is significantly lower compared to the same period in 2017. Total Cumulative AWD cases from January 1 to February 28, 2018 is 121.

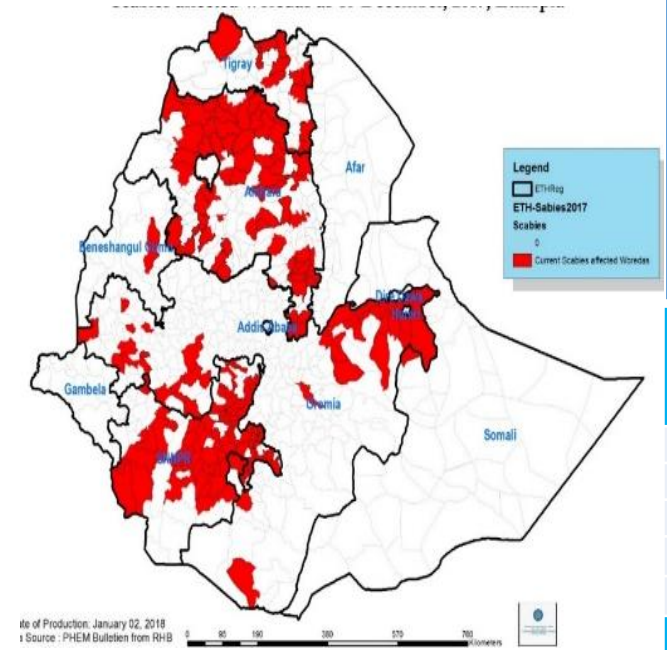
On epidemiological week 8, 2018 only one case was reported in Kelafo Woreda, Shabelle Zone in Ethio-Somali region.

In Dire Dawa , zero AWD cases have been reported since January 10, 2018.



Data Source : Regional Health Bureau

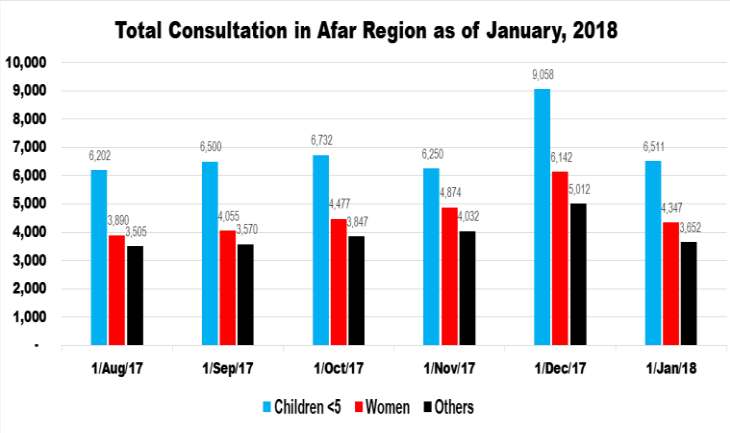
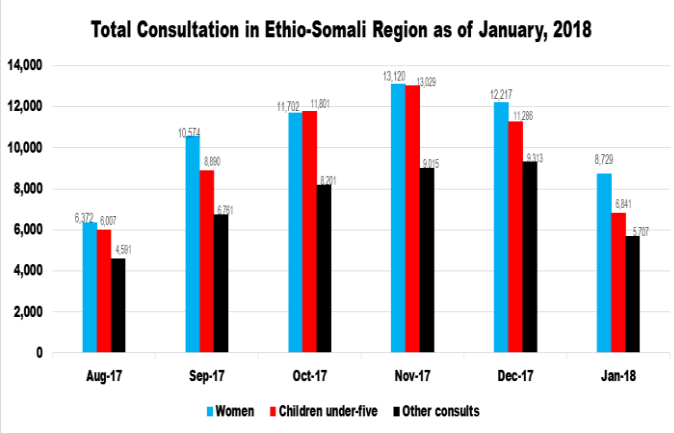
Scabies affected districts as of February, 2018



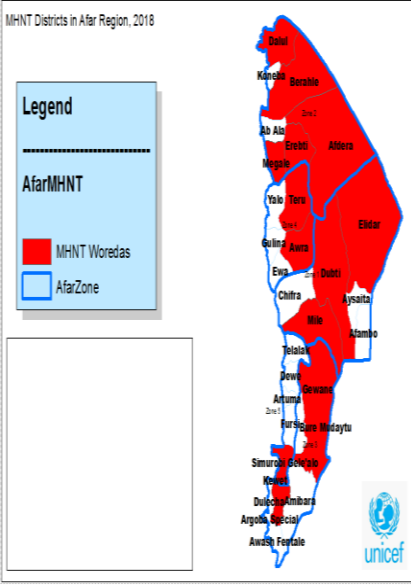
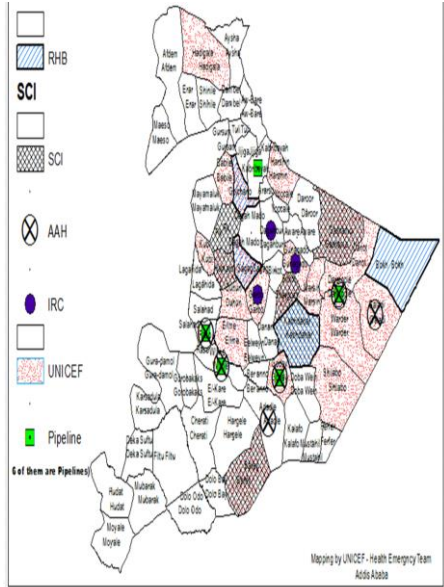
From February 1-28, 2018, a total number of 29,918 new Scabies Cases were reported in regions of the country. Amhara continued to be the most affected region with highest number of cases reported. Please find case distribution in the table below. Comparing to the same period in 2017, Scabies cases were reduced by 43%.

Region	No. of Scabies Cases from Feb 1-28,2017	Cumulative Scabies Cases from Jan. 1 - Feb. 28, 2018	Cumulative Scabies Cases 2017
Amhara	25,294	36,068	100,301
SNNP	3,104	6,167	225,150
Oromia	1,025	3,849	29,876
Tigraye	485	1,299	12,962
B_Gumuz	10	1,498	3445
Total	29,918	48,881	371,734

Mobile Health and Nutrition Teams, Total Consultations, Afar and Somali Regions, Ethiopia as of February 28, 2018



Locations of MHNT Team, Afar and Somali region, Ethiopia



Data Source :Regional Health Bureau

From January 1 – 31,2018, a cumulative total number of 13,352 (37%) children 5yrs+, 13,076 (37%) Women and 9,359 (26%) for other age groups were reached with essential life saving health services in Afar and Somali region.

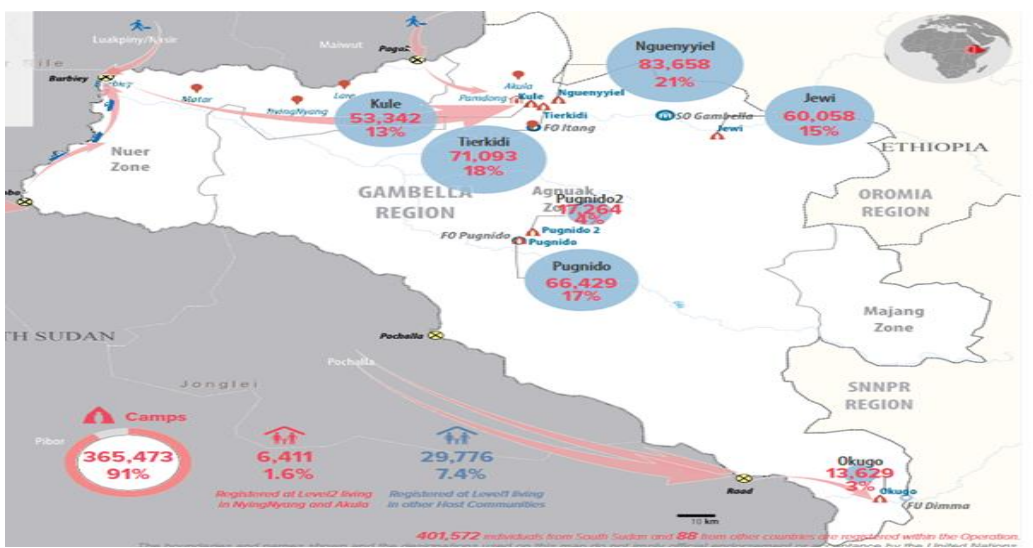
The services reported a total of 35,787 Consultations performed by the Mobile Health and Nutrition Teams: 49 UNICEF supported, 11 International NGOs and 6 RHBs in Somali and Afar regions; Of these consultations, 21,277 were from Somali and 14,510 from Afar.

Due to delays in reporting, no February data is received as of date.

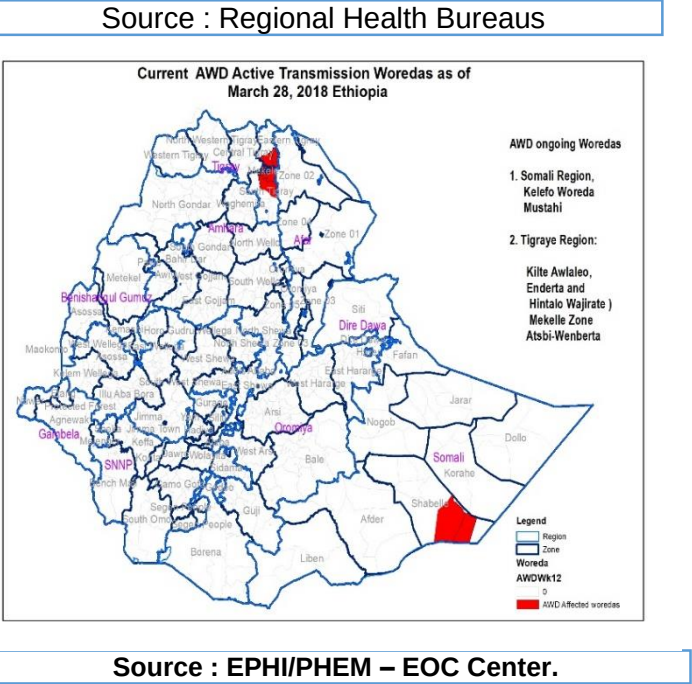
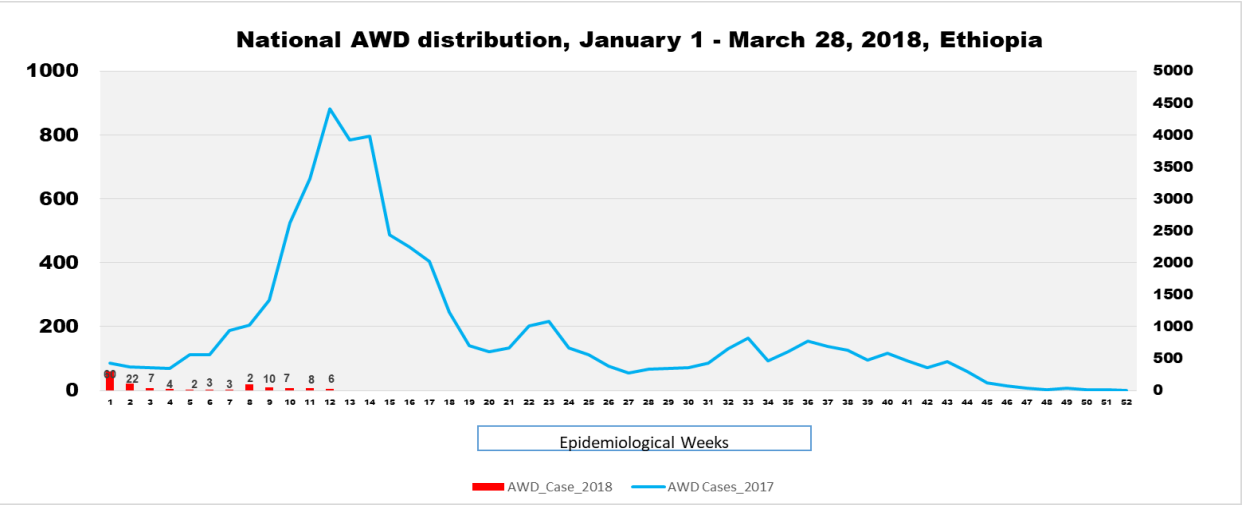
South Sudanese Refugees Response, Jan. 1 – Feb. 28, 2018, Ethiopia

Indicator	2017	Feb. 28, 2018
# of New South Sudanese arrival in 2017	87,678	12,945
# of <15 Children	55,265	6,643
# of 6mth. - <15Yrs Children	50,152	6,337
# of <15 Children Obtained arrival Polio Vaccination	59,121	6,679
# of 6mth to 15Yrs children Obtained Measles Vaccination	53,231	6,366

South Sudanese refugees, Gambella Sites, February, 2018

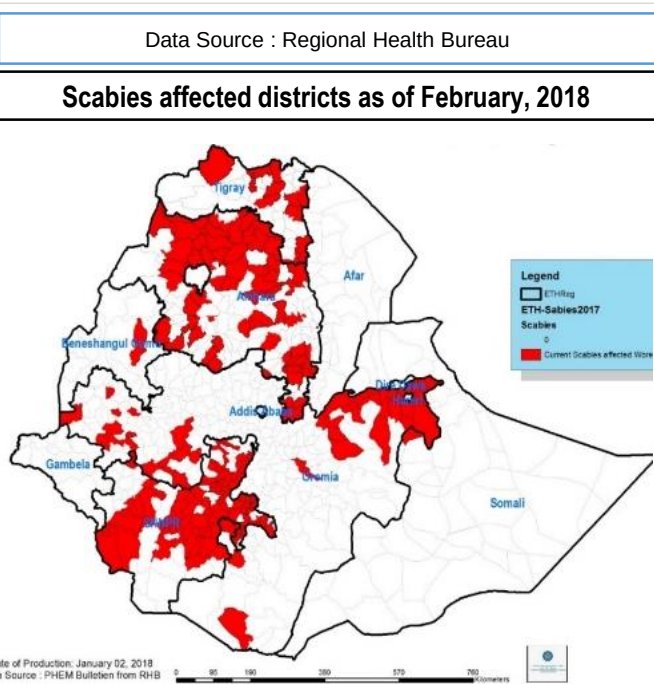
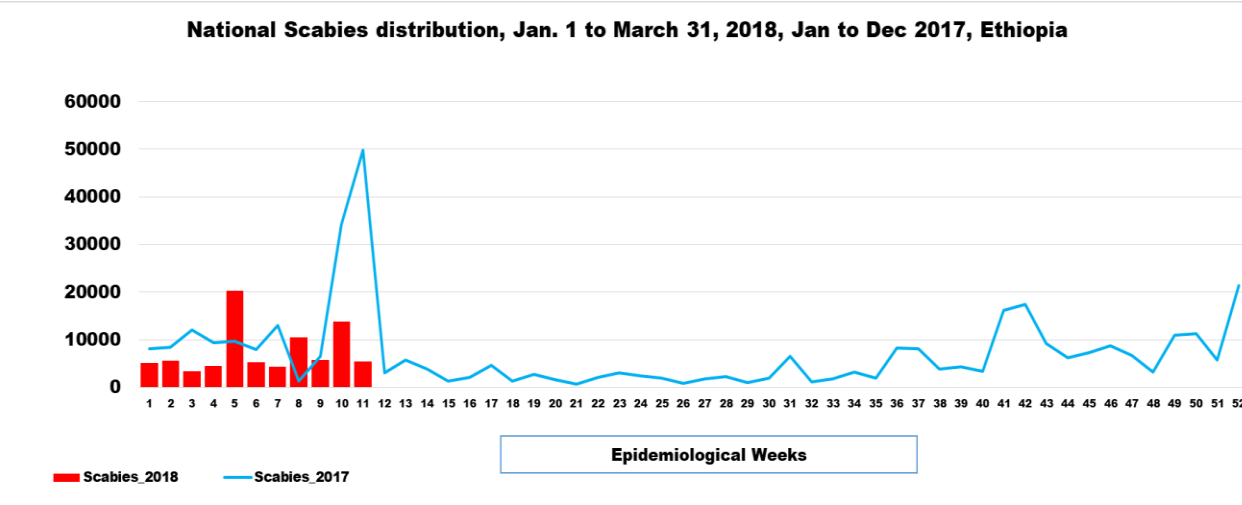


Data Source : Gambella UNICEF Field Office



From March 1 – 31, 2018, a total number of 31 AWD Cases with zero death were reported from Somali (8) and Tigraye (23) regions. This is significantly lower compared to the same period in 2017. Total Cumulative AWD cases from January 1 to March 31, 2018 is 152.

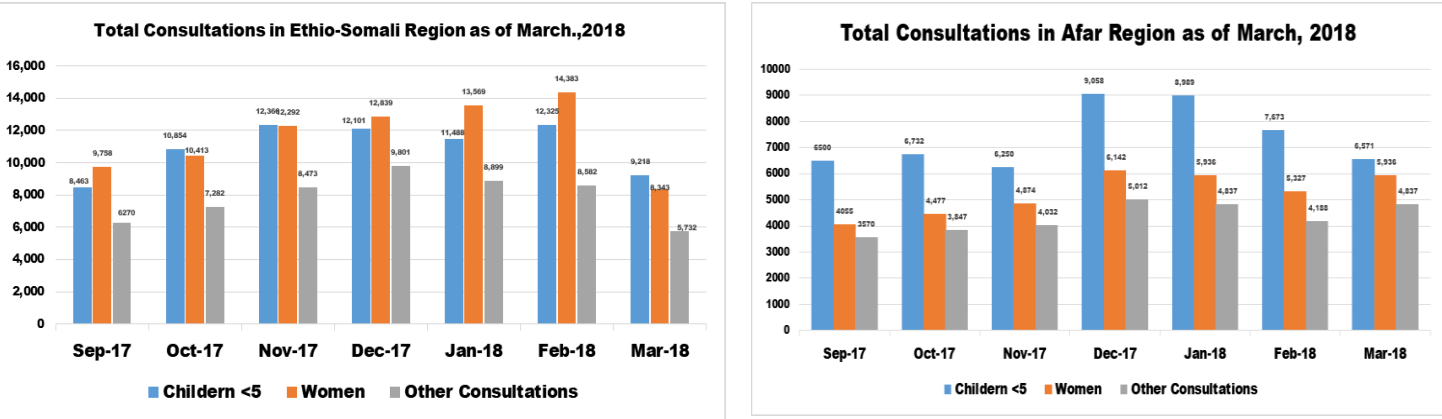
On epidemiological Week 12, 2018 four (4) cases were reported from Ethio-Somali and to (3) from Tigray Regions.



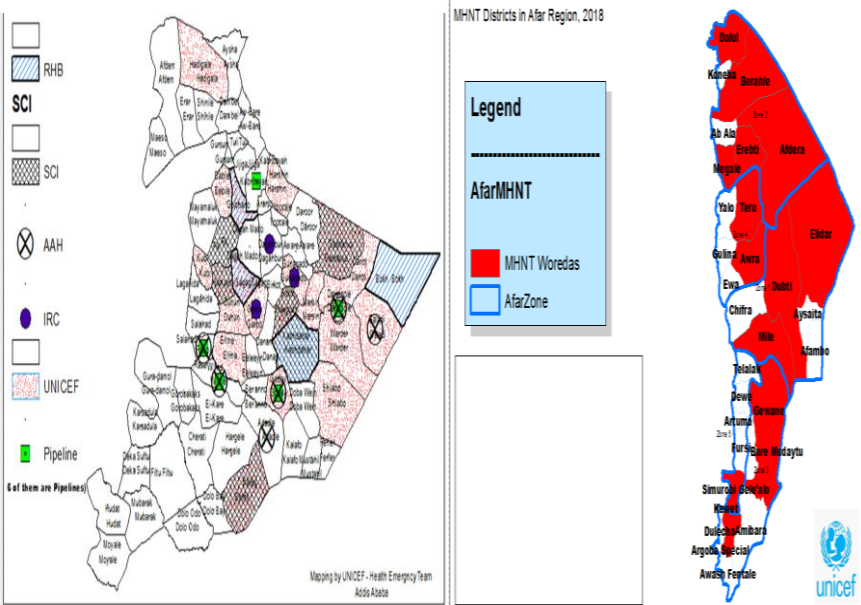
From March 1 – 31, 2018, a total number of 25,215 new Scabies Cases were reported from five regions. Oromia region reported highest number of cases. Please find case distribution in the table below. Comparing to the same period in 2017, Scabies cases were reduced by 73%.

Regions	No. of Scabies Cases from March 1 - 31, 2018	Cumulative Scabies Cases from Jan. 1 - March 31, 2018	Scabies Cases 2017
Oromia	14,335	20,055	29,876
Amhara	6,154	50,719	100,301
SNNP	4,210	10,377	225,150
Tigraye	500	2,078	12,962
B_Gumuz	16	1,516	3445
Total	25,215	84,745	371,734

Mobile Health and Nutrition Teams, Total Consultations, Afar and Somali Regions, Ethiopia as of March, 31, 2018



Locations of MHNT Team, Afar and Somali region, Ethiopia



Data Source :Regional Health Bureau

From March 1–31, 2018 a cumulative total of 40,637 medical consultations conducted in hard to reach areas of Afar and Somali regions, of which, 15,789 (39%) under five children and 14,279 (35%) were Women.

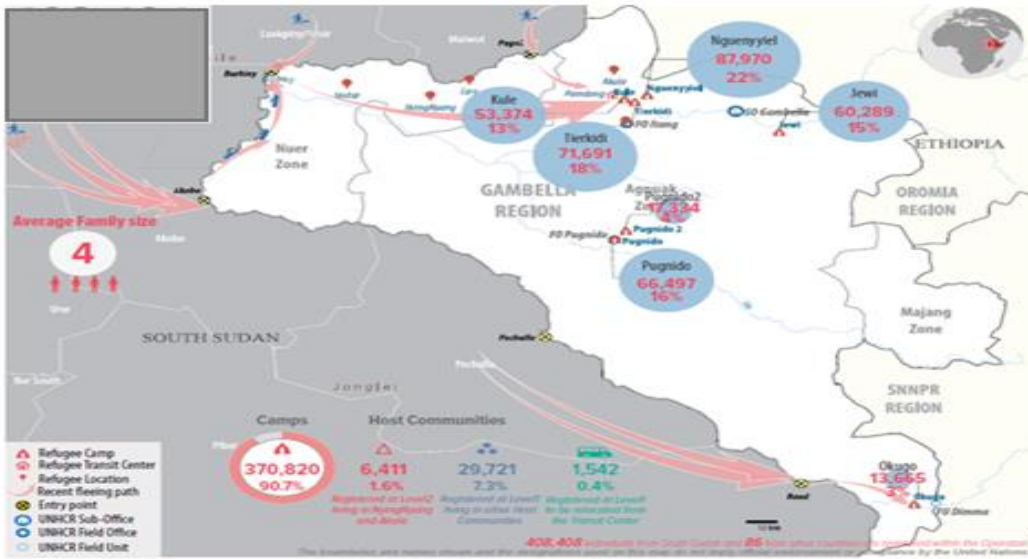
From January 1–March 31, 2018, a total cumulative 146,833 MHNT consultations were made (54,294 from Afar and 92,539 from Ethio-Somali region). Of which 56,264 (38%) Under five children, 53,494 (36%) Women and 37,075 (25%) were from Other adults.

The medical Consultation has been performed by 49 UNICEF supported Mobile Health and Nutrition Teams plus 11 International NGOs and 6 RHBs in Somali in region ; Of these consultations, 17,561 were from Somali (46 teams) and 12,507 from Afar with 20 Teams. In addition, from January to February 2018, a total 10,817 medical consultation has been conducted by 15 Sustainable Outreach Service Team (SOS). The consultation has been increasing in both regions.

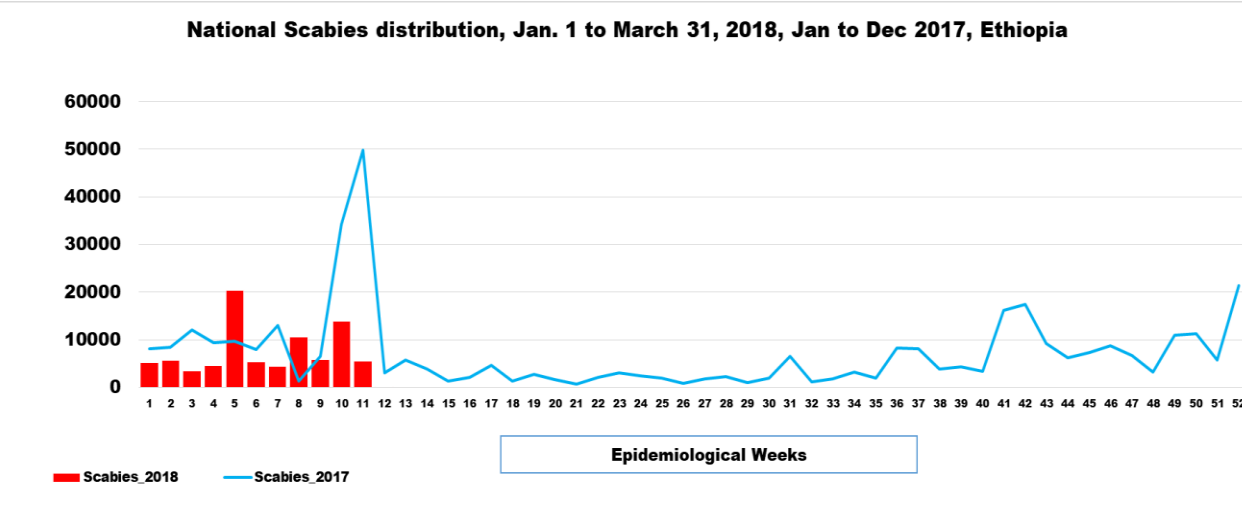
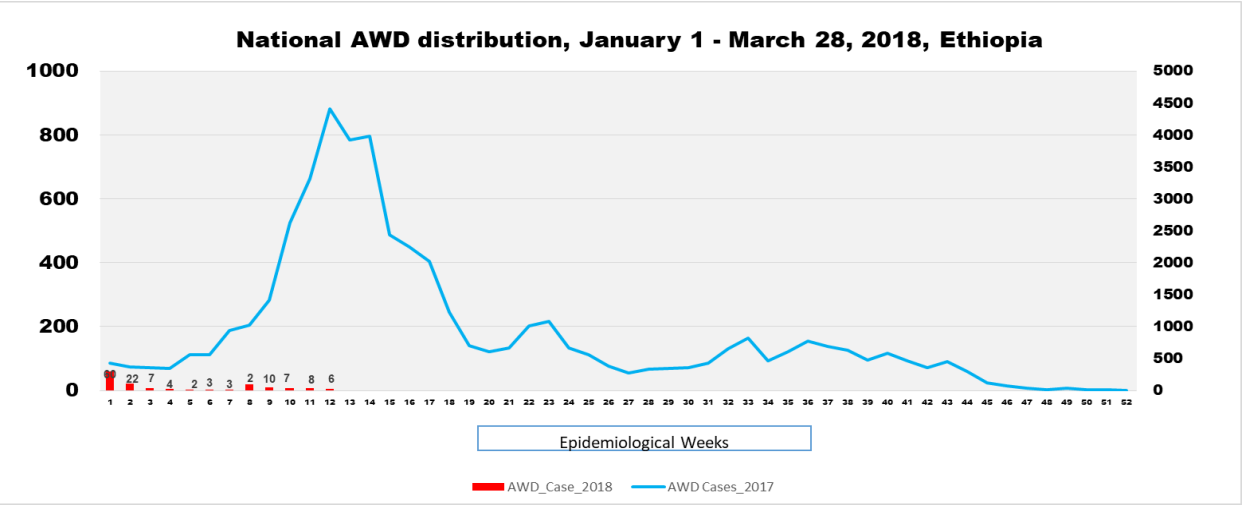
South Sudanese Refugees Response, Jan. 1 – March 28, 2018, Ethiopia

Indicator	2017	March, 2018
# of New South Sudanese arrival in 2017	87,678	15,778
# of <15 Children	55,265	8,186
# of 6mth. - <15Yrs Children	50,152	7,782
# of <15 Children Obtained arrival Polio Vaccination	59,121	7,782
# of 6mth to 15Yrs children Obtained Measles Vaccination	53,231	7,782

South Sudanese refugees, Gambella Sites, March, 2018



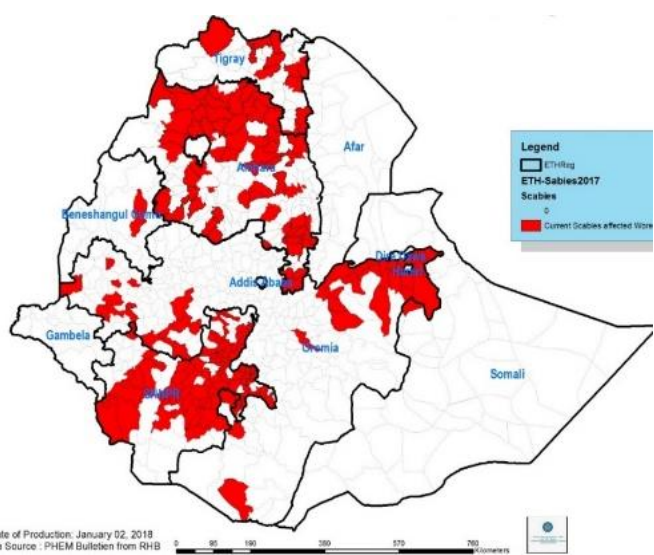
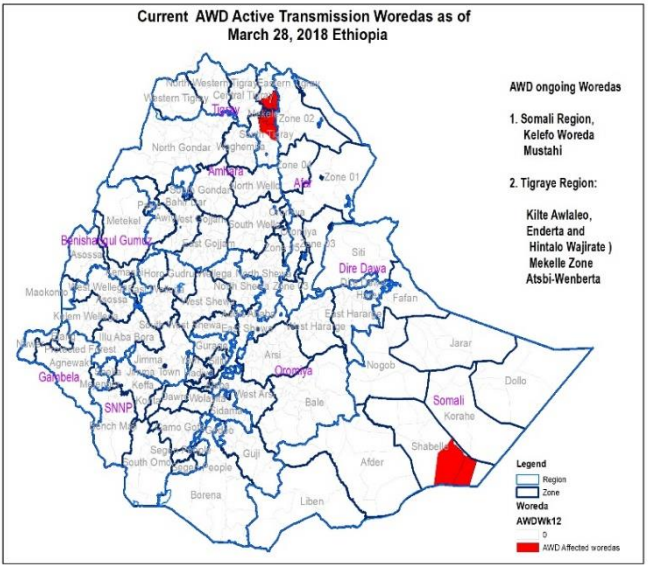
Data Source : Gambella UNHCR Field Office



Source : Regional Health Bureaus

Data Source : Regional Health Bureau

Scabies affected districts as of February, 2018



Source : EPHI/PHEM – EOC Center.

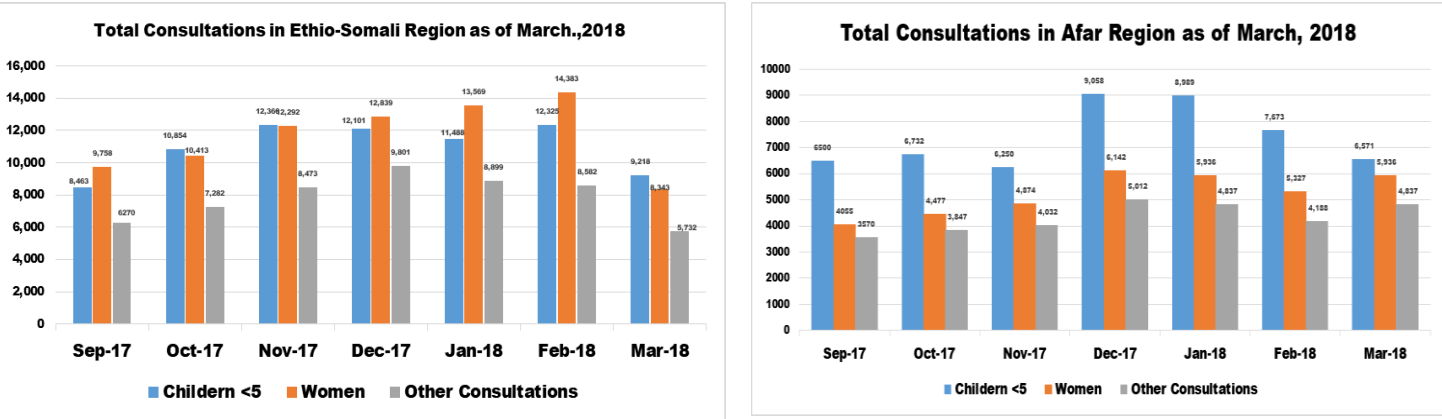
From March 1 – 31, 2018, a total number of 31 AWD Cases with zero death were reported from Somali (8) and Tigraye (23) regions. This is significantly lower compared to the same period in 2017. Total Cumulative AWD cases from January 1 to March 31, 2018 is 152.

On epidemiological Week 12, 2018 four (4) cases were reported from Ethio-Somali and to (3) from Tigray Regions.

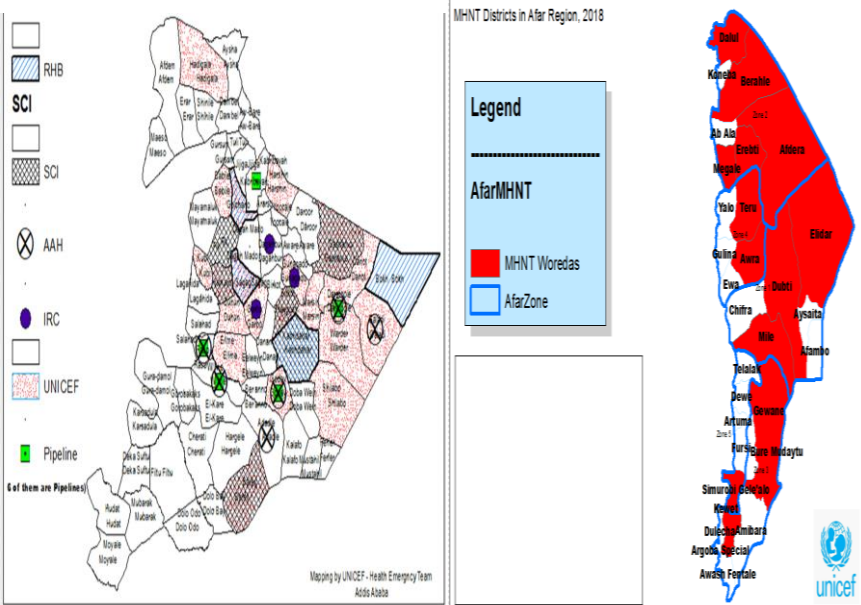
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Oromia	14,335	20,055	29,876
Amhara	6,154	50,719	100,301
SNNP	4,210	10,377	225,150
Tigraye	500	2,078	12,962
B_Gumuz	16	1,516	3445
Total	25,215	84,745	371,734

Mobile Health and Nutrition Teams, Total Consultations, Afar and Somali Regions, Ethiopia as of March, 31, 2018



Locations of MHNT Team, Afar and Somali region, Ethiopia



Data Source :Regional Health Bureau

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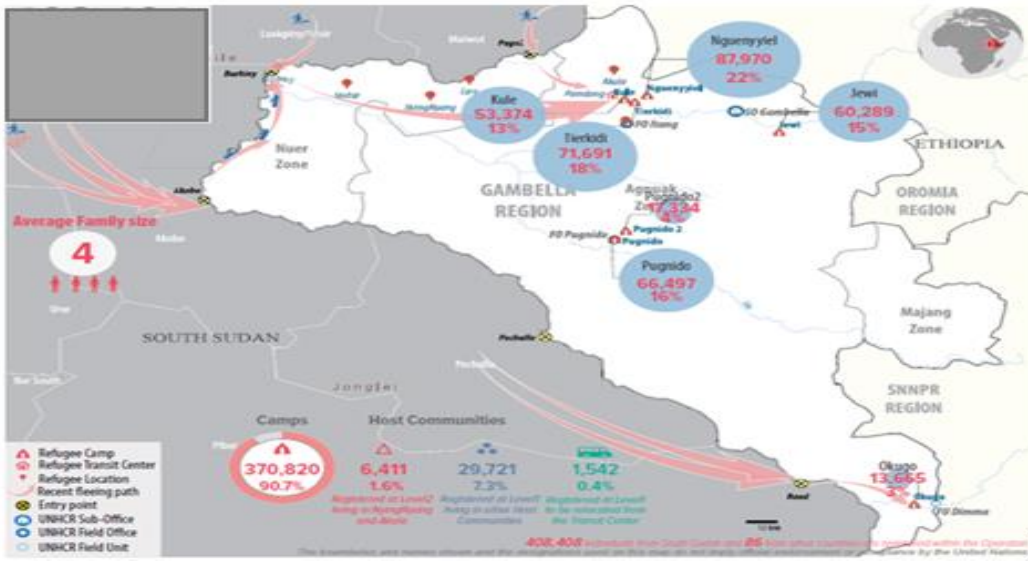
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The medical Consultation has been performed by 49 UNICEF supported Mobile Health and Nutrition Teams plus 11 International NGOs and 6 RHBs in Somali in region ; Of these consultations, 17,561 were from Somali (46 teams) and 12,507 from Afar with 20 Teams. In addition, from January to February 2018, a total 10,817 medical consultation has been conducted by 15 Sustainable Outreach Service Team (SOS). The consultation has been increasing in both regions.

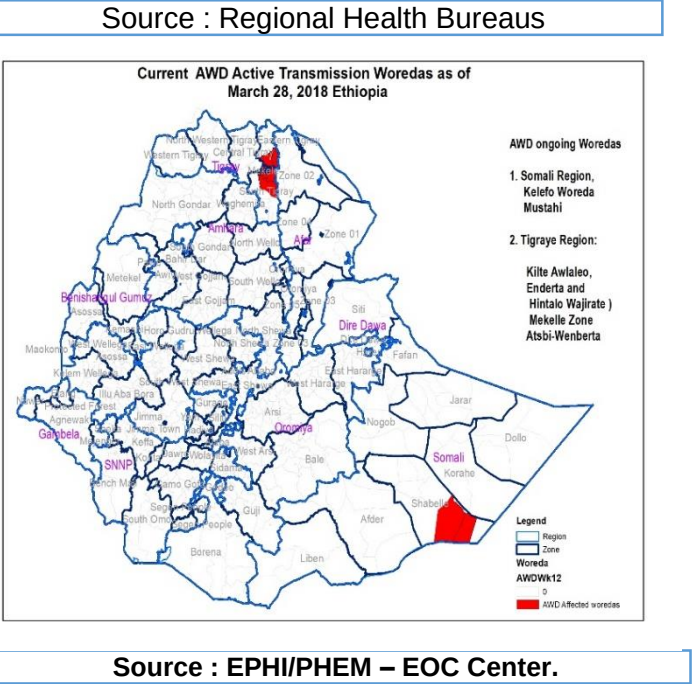
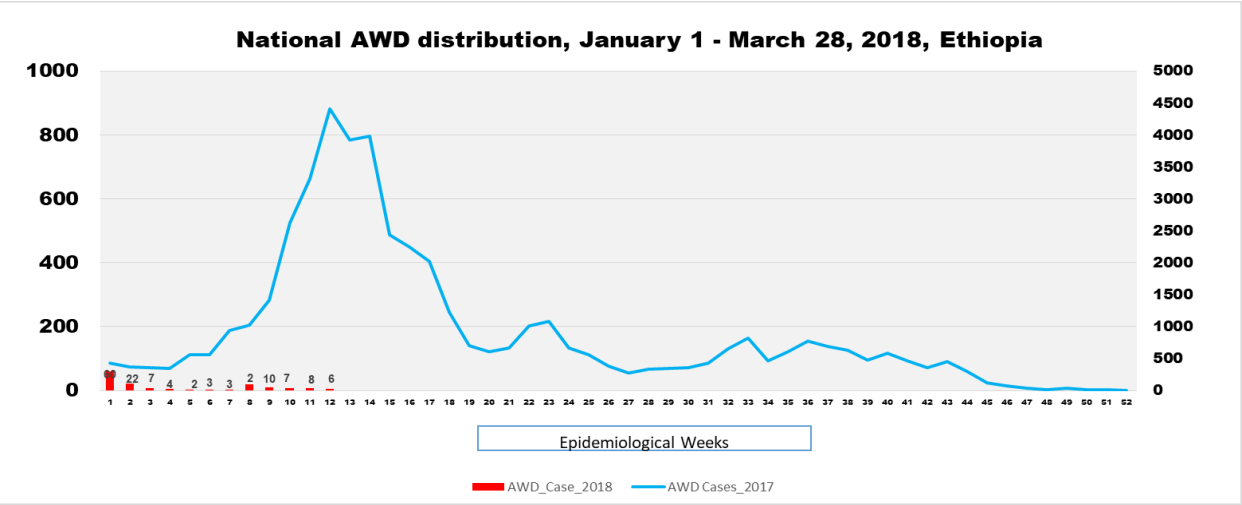
South Sudanese Refugees Response, Jan. 1 – March 28, 2018, Ethiopia

Indicator	2017	March, 2018
# of New South Sudanese arrival in 2017	87,678	15,778
# of <15 Children	55,265	8,186
# of 6mth. - <15Yrs Children	50,152	7,782
# of <15 Children Obtained arrival Polio Vaccination	59,121	7,782
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South Sudanese refugees, Gambella Sites, March, 2018

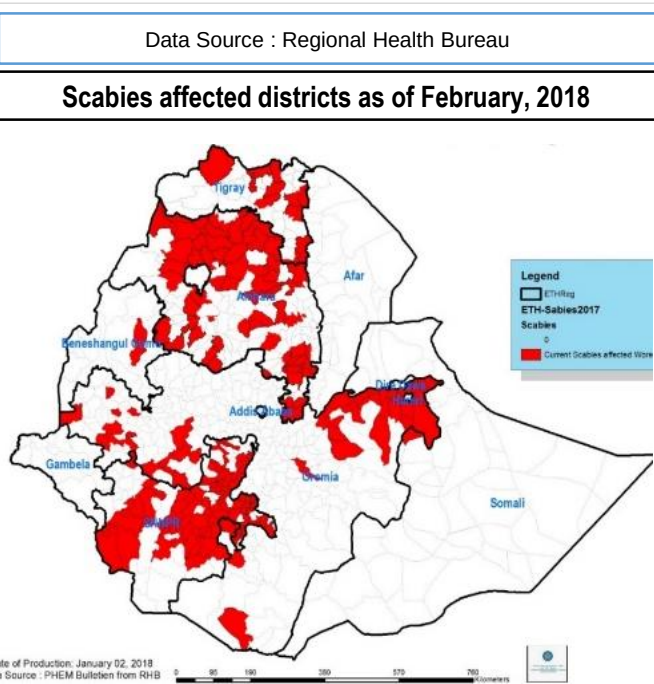
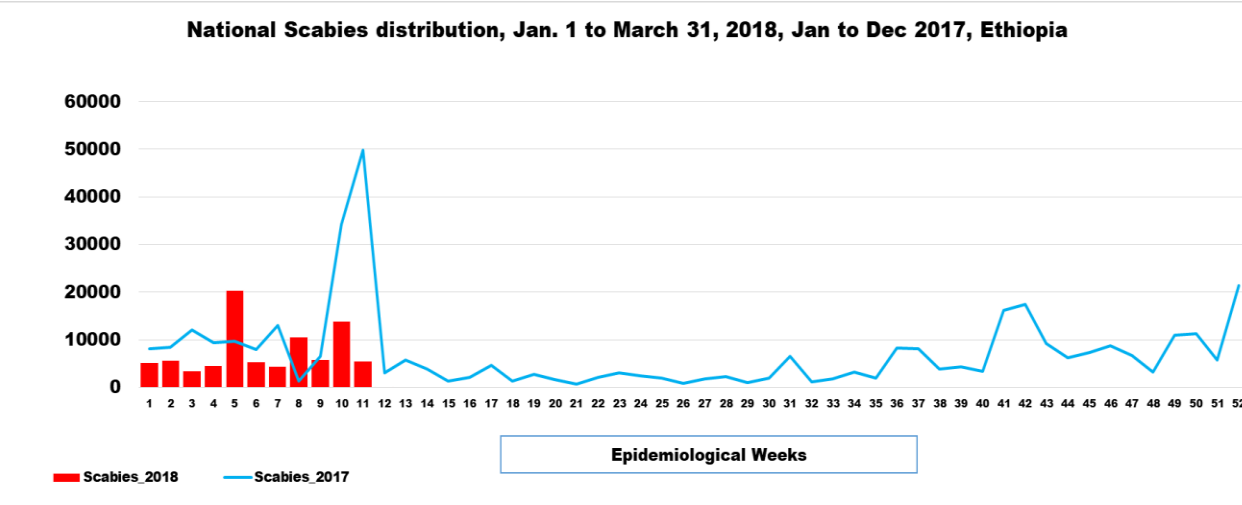


Data Source : Gambella UNHCR Field Office



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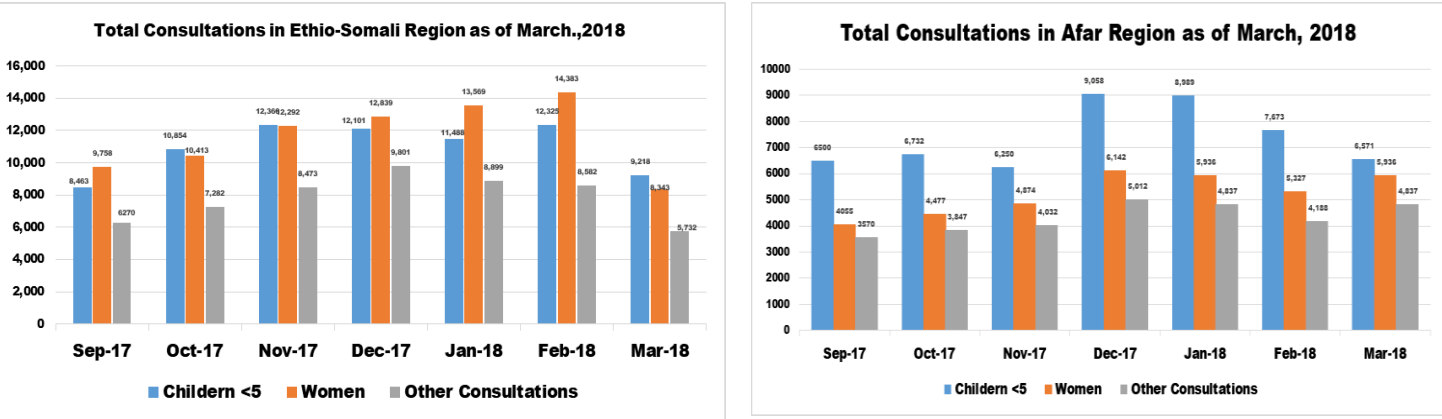
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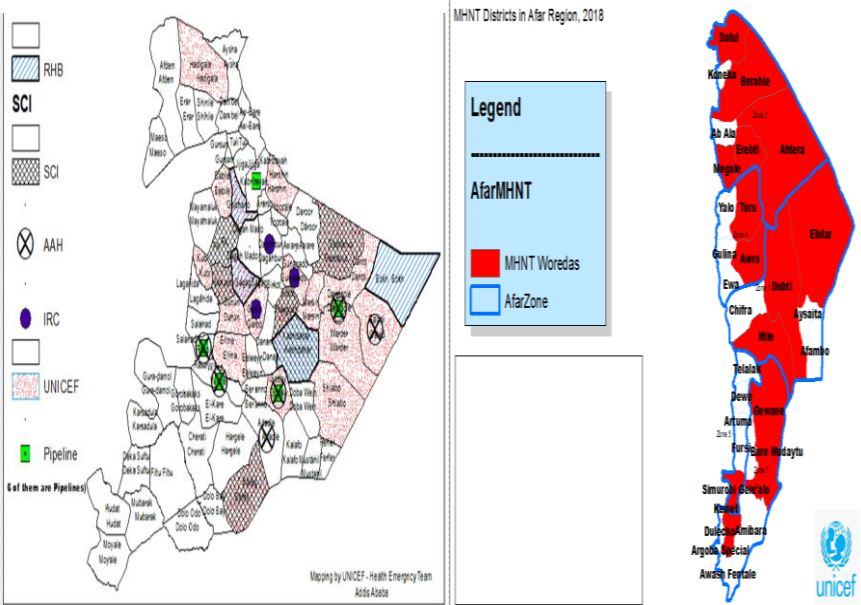
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Data Source :Regional Health Bureau

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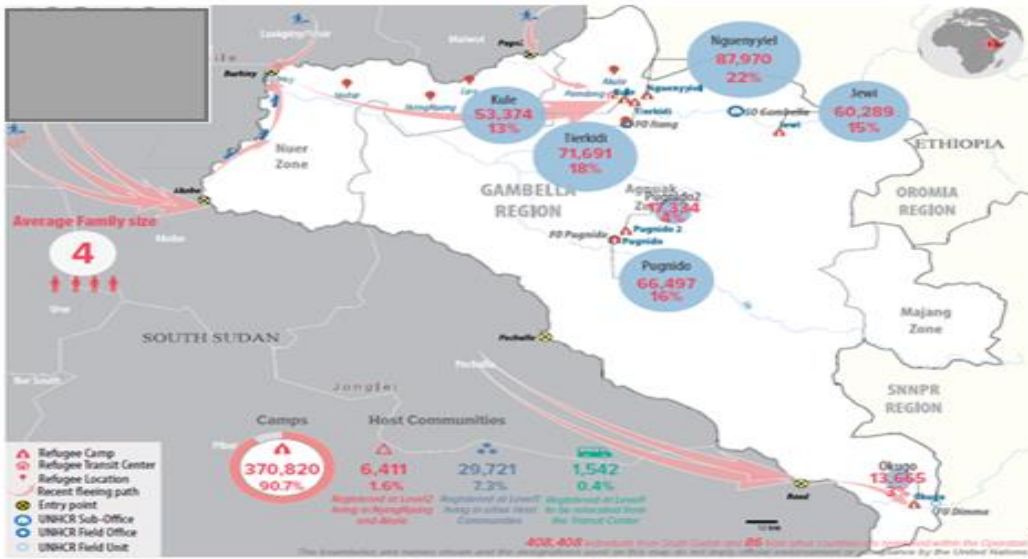
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South Sudanese Refugees Response, Jan. 1 – March 28, 2018, Ethiopia

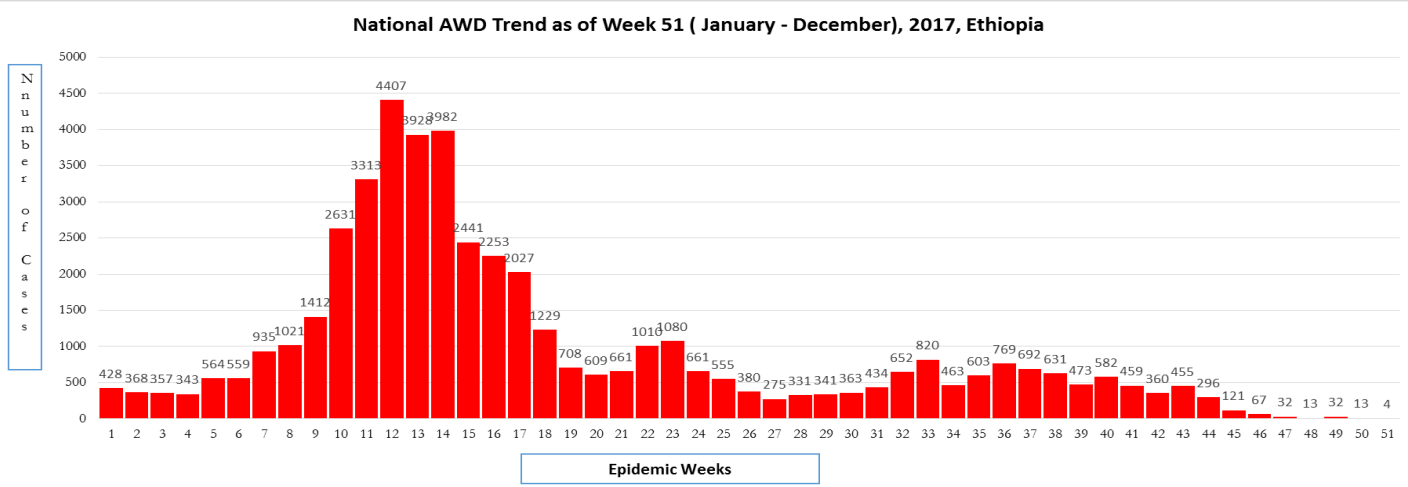
Indicator	2017	March, 2018
# of New South Sudanese arrival in 2017	87,678	15,778
# of <15 Children	55,265	8,186
# of 6mth. - <15Yrs Children	50,152	7,782
# of <15 Children Obtained arrival Polio Vaccination	59,121	7,782
# of 6mth to 15Yrs children Obtained Measles Vaccination	53,231	7,782

South Sudanese refugees, Gambella Sites, March, 2018



Data Source : Gambella UNHCR Field Office

AWD Epi Curve (January – December, 2017)



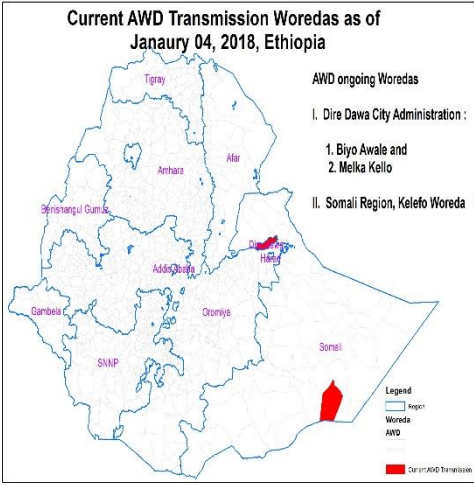
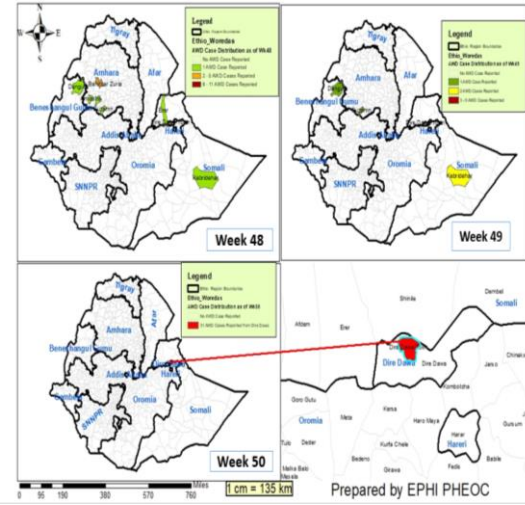
Source : RHBs and EPHI/PHEM

Current AWD transmission as of Week 51, 2017

In 2017 as of Week 52, from January – December 31, 2017, a cumulative of 48,638 AWD Cases and 859 AWD Deaths with CFR of 1.8% were reported from Seven regions in Ethiopia with one city Administrative Council. On week 51, a total of 4 AWD cases were reported 2 clusters of Dire Dawa City Administrative Council.

The incidence rate of AWD continued to decrease significantly at national level as well as at regional level especially in Amhara, Oromia and Somali regions. The Only AWD active transmission Woredas is at Kelafo in Somali region and Biyo Awale and Melka Kello.

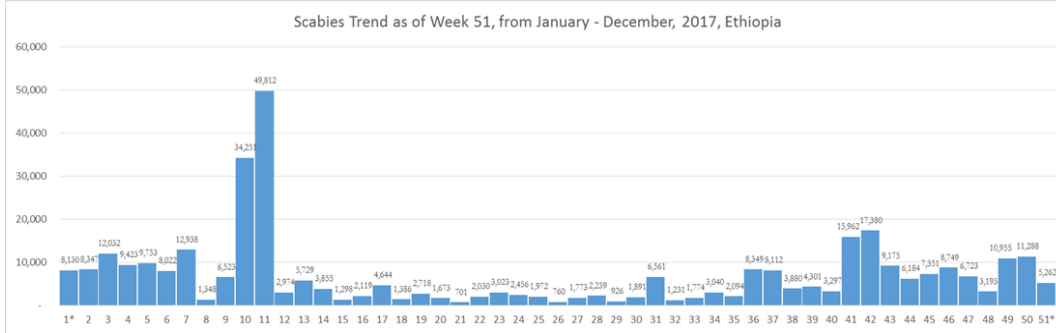
Map of Acute Watery Diarrhea Case Distribution as of Week 48, 49 and 50, Ethiopia, December.



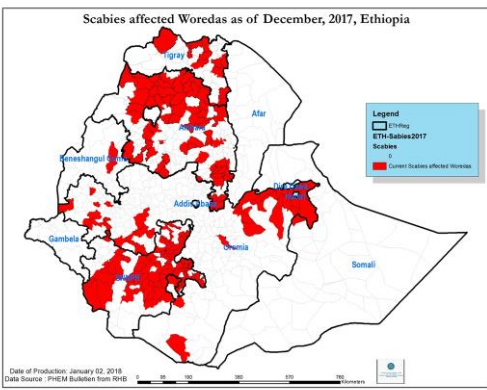
Current AWD transmission Sites as of Week 52 Source : EPHI/PHEM – EOC Center.

Scabies (From January – December, 2017

Source : Regional Health Bureaus – PHEM (RHBs)



Updated Scabies Case load, trend and Mapping , 2017, (Data Source RHBs/PHEM Bulletin,) (Jan. – December 31, 2017)

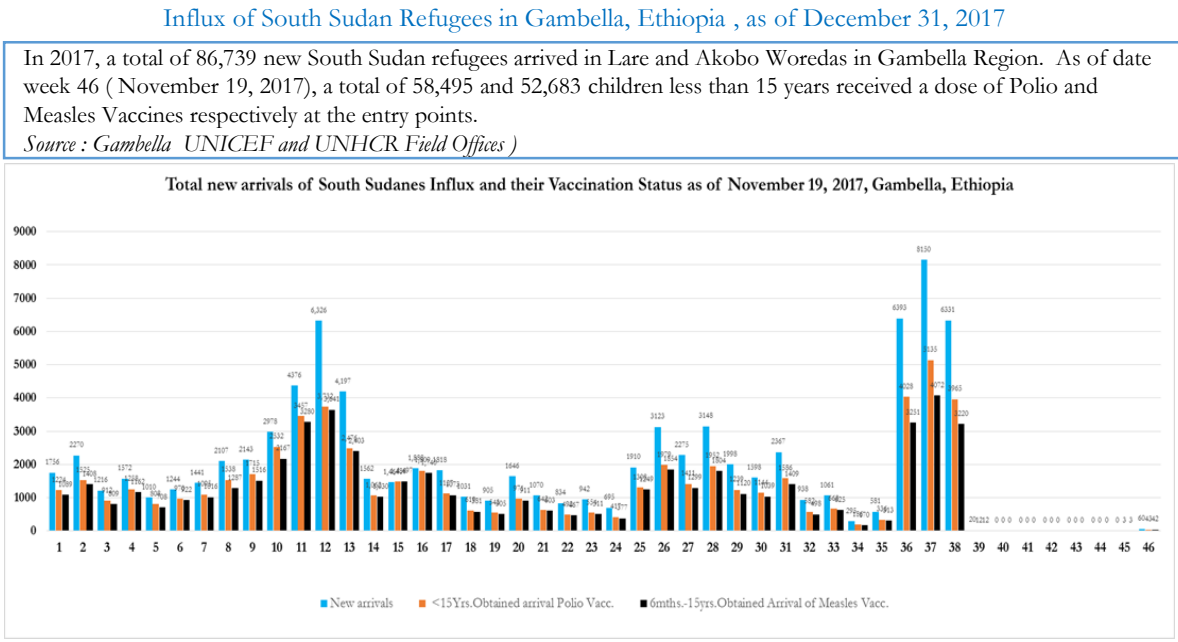
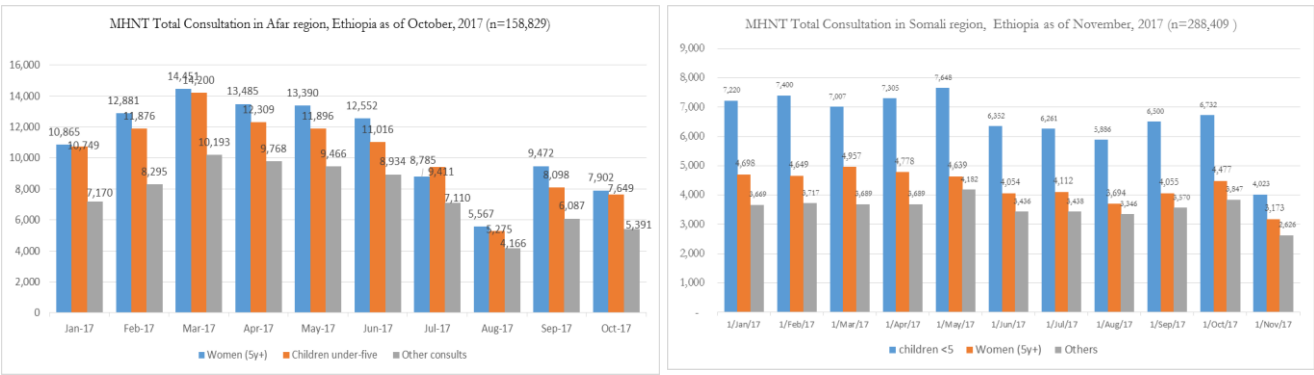


Region	December, 2017 (Week 47-51)	2017
Oromia	5,984	28,221
Amhara	21,653	82,201
SNNP	5540	223,034
Tigray	911	12,817
B_Gumuz	3334	3,334
Total	37,422	349,607

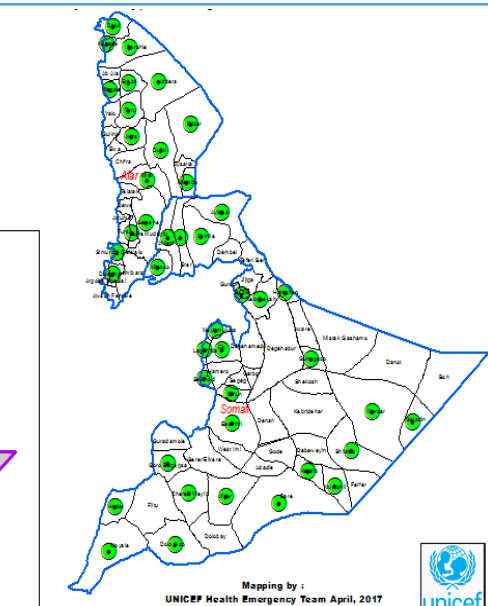
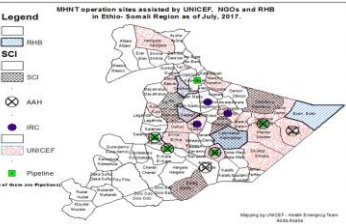
Scabies outbreak is ongoing in five regions from 237 districts from Amhara (62), Oromia (45), SNNPR (118), Beneshangul Gumuz (one woredas (13 Kebeles) and Tigray (11) districts have been affected with a total case of Scabies (n=349,607).

In December, 2017 from Week 47-52, a total case of 37,421 Scabies cases were reported from Oromia, Amhara, SNNP, Tigraye and Beneshangul Gumuz. A new outbreak of Scabies cases emerged in Beneshangul Gumuz Since the first week of December, 2017. The total cumulative reported scabies cases in 2017 became 349,607 from the five regions: Amhara (82,201), Oromia (28,221), SNNP (223,034) Tigray (12,817) and Beneshangul Gumuz (3,334). Generally, the Scabies outbreak trend has been increased in the last 16 weeks in all regions which can be attributed to the schools opening last September, 2017.

Mobile Health and Nutrition (MHNT), Total Consultations, Afar and Somali Regions, Ethiopia as of December 31, 2017



Data Source :RHBs



Mapping by : UNICEF Health Emergency Team April, 2017

Location of MHNT Team, Afar and Somali region, Ethiopia

In 2017, a cumulative total number of n=447,238 Consultations were made through n=49 Mobile Health and Nutrition Teams (MHNT program) in both Somali and Afar regions; of these, 158,829 from Afar region and 288,409 from Somali region. Out of the total consultation from both regions, 174,813 (39%) for children 5yrs+, 156,636 (35%) for Women and 115,789 (26%) for other groups.

Currently for Somali region, additional n=15 MHNT programs have been implemented through RHB (5), NGOs (10) while 9 additional teams from the NGOs are in the pipeline.

In Afar the reporting period of November, 2017 the completeness and timeliness of the reporting period coverage was 70%.

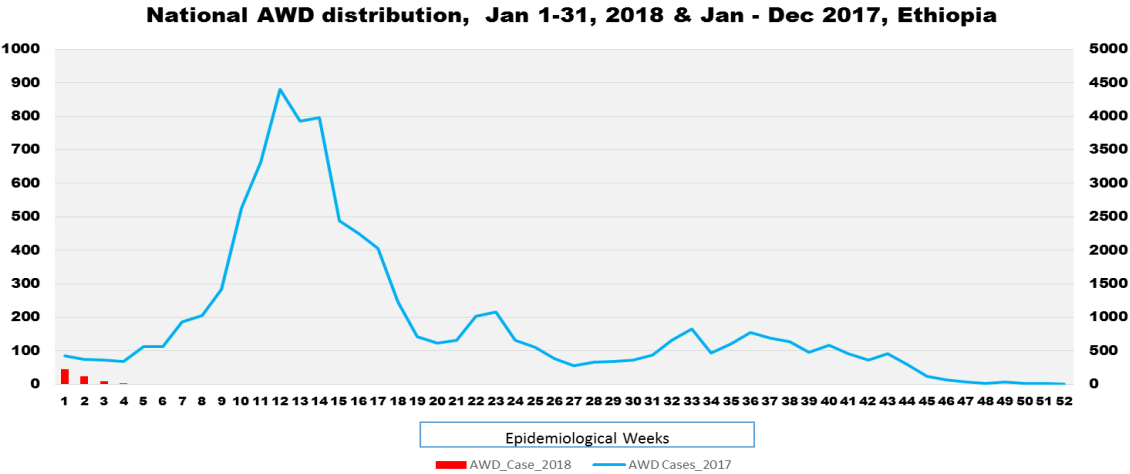
South Sudanese refugees new arrival in Gambella Sites at Gambella , December 31, 2017



Legend: UNHCR Sub office, UNHCR Field Office, Refugee camp, Refugee location, Refugee transit centre, Entry point, Main town, Village.

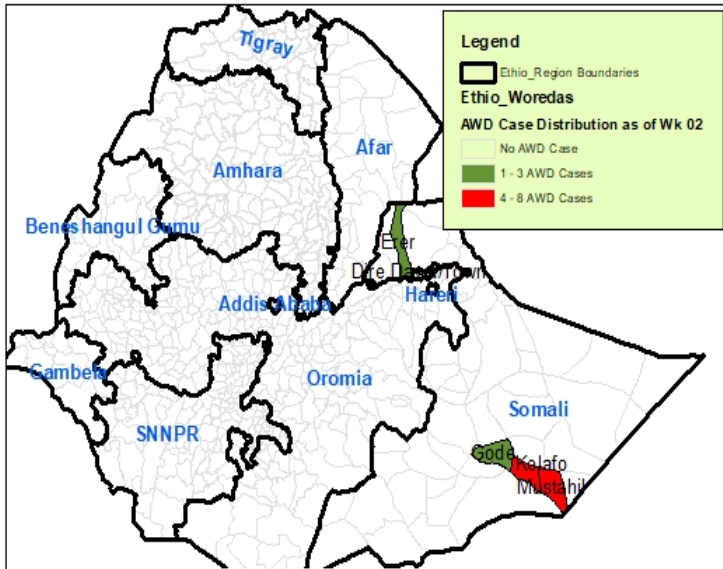
Indicator	Number
# of New South Sudanese arrival in 2017	86,739
# of <15 Children	54,687
# of 6mth. - <15Yrs Children	46,391
# of <15 Children Obtained arrival Polio Vaccination	58,495
# of 6mth to 15Yrs children Obtained Measles Vaccination	52,683

Data Source : UNICEF and UNHCR Gambella Field Offices



Source : Regional Health Bureaus

Current AWD transmission, Week 04, 2018

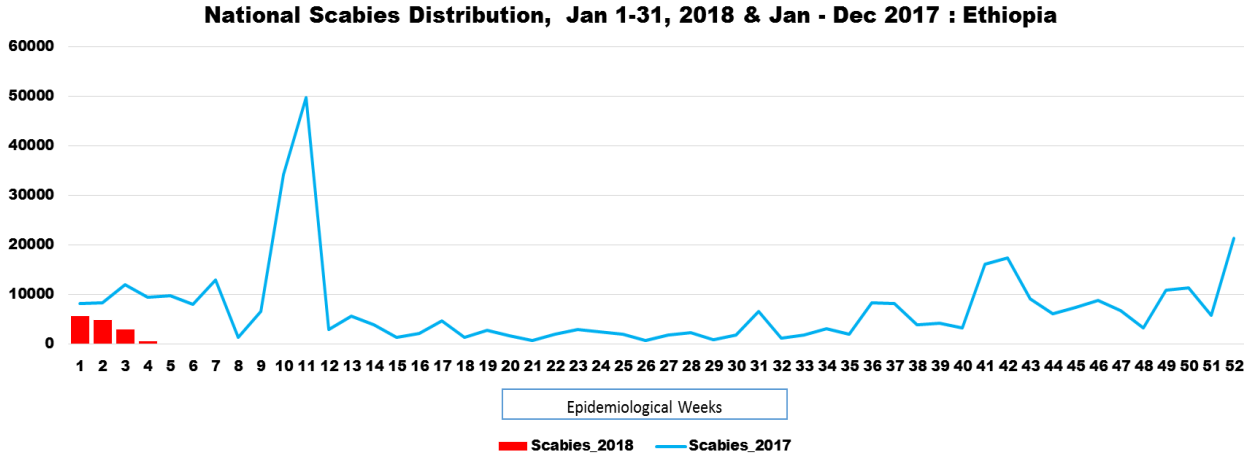


Source : EPHI/PHEM – EOC Center.

From January 1 – 31, 2018, a total number of 90 AWD Cases with zero death were reported from Somali region and Dire Dawa City Administrative City Council. This is significantly lower compared to the same period in 2017.

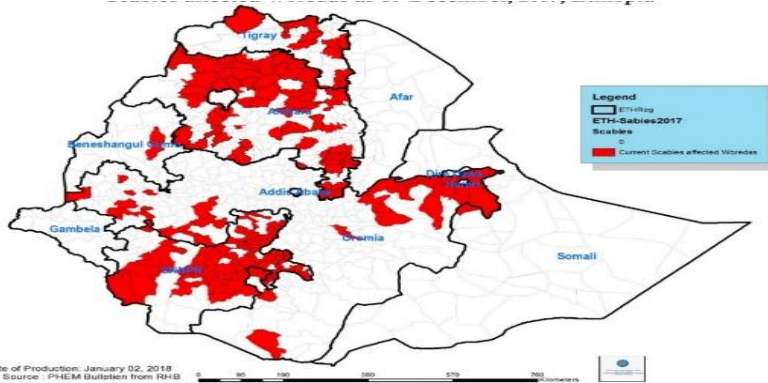
On epidemiological week 4, 2018 a total of three (3) cases were reported in Gode city and Mustahi Woredas from Shabella Zone in Ethio-Somali region.

In Dire Dawa no AWD cases have been reported since January 10, 2018 or in the last 20 days.



Data Source : Regional Health Bureau

Scabies affected districts as of January, 2018

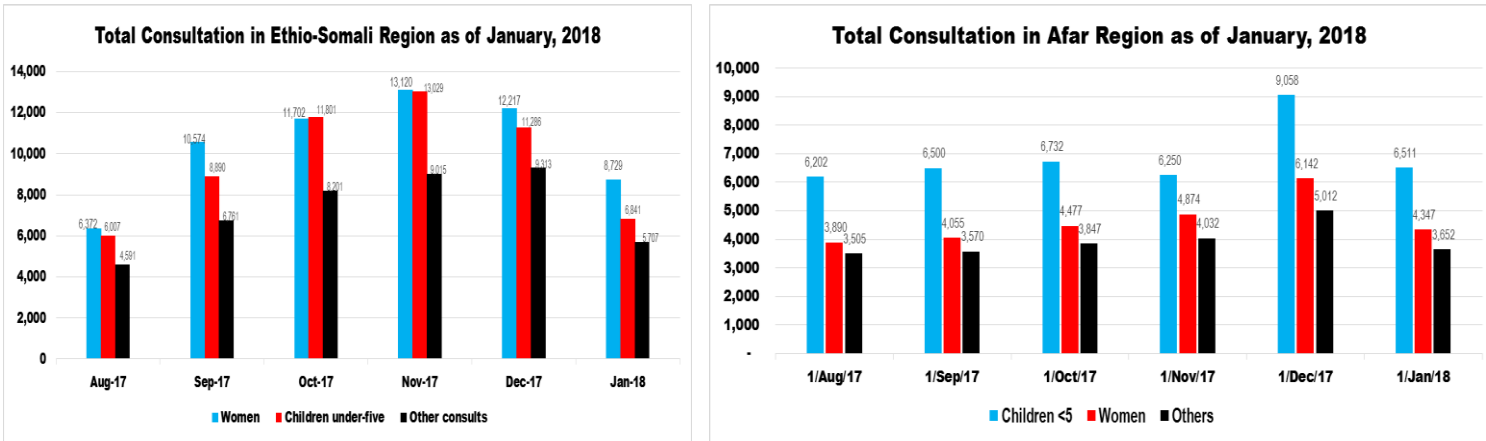


Regions	Scabies Cases Reported from January 1 -31, 2018	Scabies Cases 2017
Oromia	2824	29,876
Amhara	7018	100,301
SNNP	2957	225,150
Tigraye	797	12,962
B_Gumuz	682	3445
Total	14278	371,734

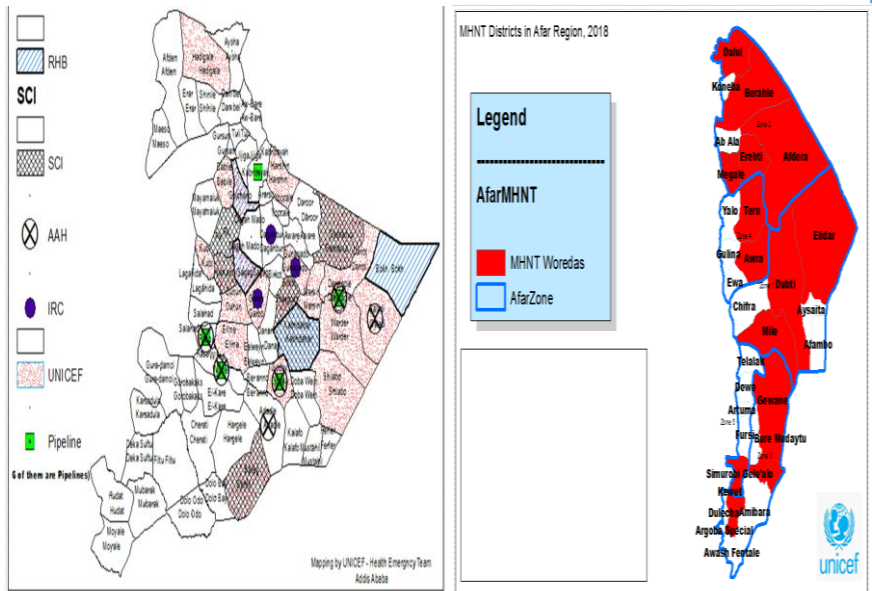
From January 1 -31, 2018, a total number of 14,278 Scabies cases were reported from 237 districts in 5 regions: Amhara (62), Oromia (45), SNNPR (118), Tigray (24), Beneshangul Gumuz (1). Refer to Table below on distribution of cases.

Comparing to same period in 2017, scabies cases is lower in 2018 by 37.6%.

Mobile Health and Nutrition Teams, Total Consultations, Afar and Somali Regions, Ethiopia as of January 31, 2018



Locations of MHNT Team, Afar and Somali region, Ethiopia



Data Source :Regional Health Bureau

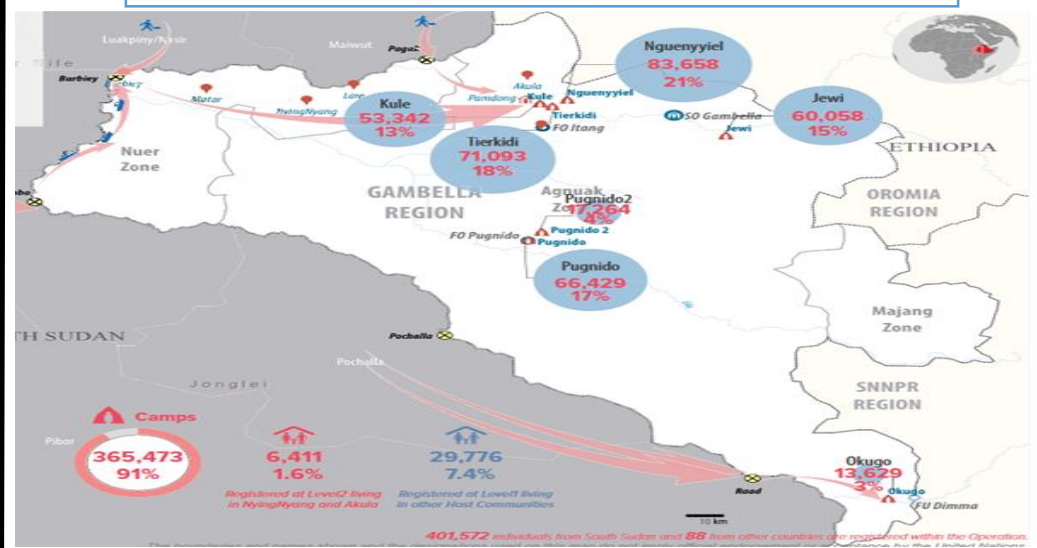
From January 1 – 31, 2018, a total number of 13,352 (37%) children 5yrs+, 13,076 (37%) Women and 9,359 (26%) for other age groups were reached with essential life saving health services.

The services reported a total of 35,787 Consultations performed by the Mobile Health and Nutrition Teams: 49 UNICEF supported, 11 International NGOs and 6 RHBs in Somali and Afar regions; Of these consultations, 21,277 were from Somali and 14,510 from Afar.

South Sudanese Refugees Response as of January, 2018, Ethiopia

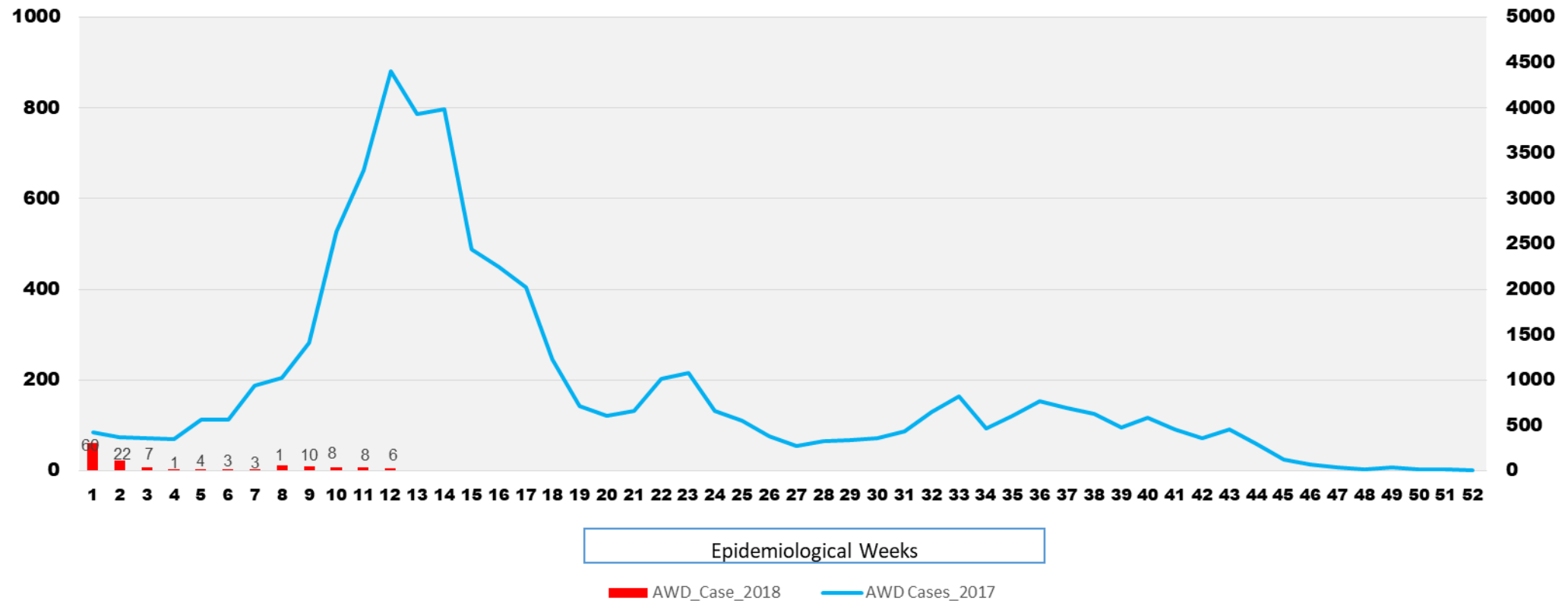
Indicator	2017	January, 2018
# of New South Sudanese arrival in 2017	87,678	4,218
# of <15 Childern	55,265	2,253
# of 6mth. - <15Yrs Childern	50,152	2,146
# of <15 Childern Obtained arrival Polio Vaccination	59,121	2,257
# of 6mth to 15Yrs childern Obtained Measles Vaccination	53,231	2,149

South Sudanese refugees, Gambella Sites, January 31, 2018



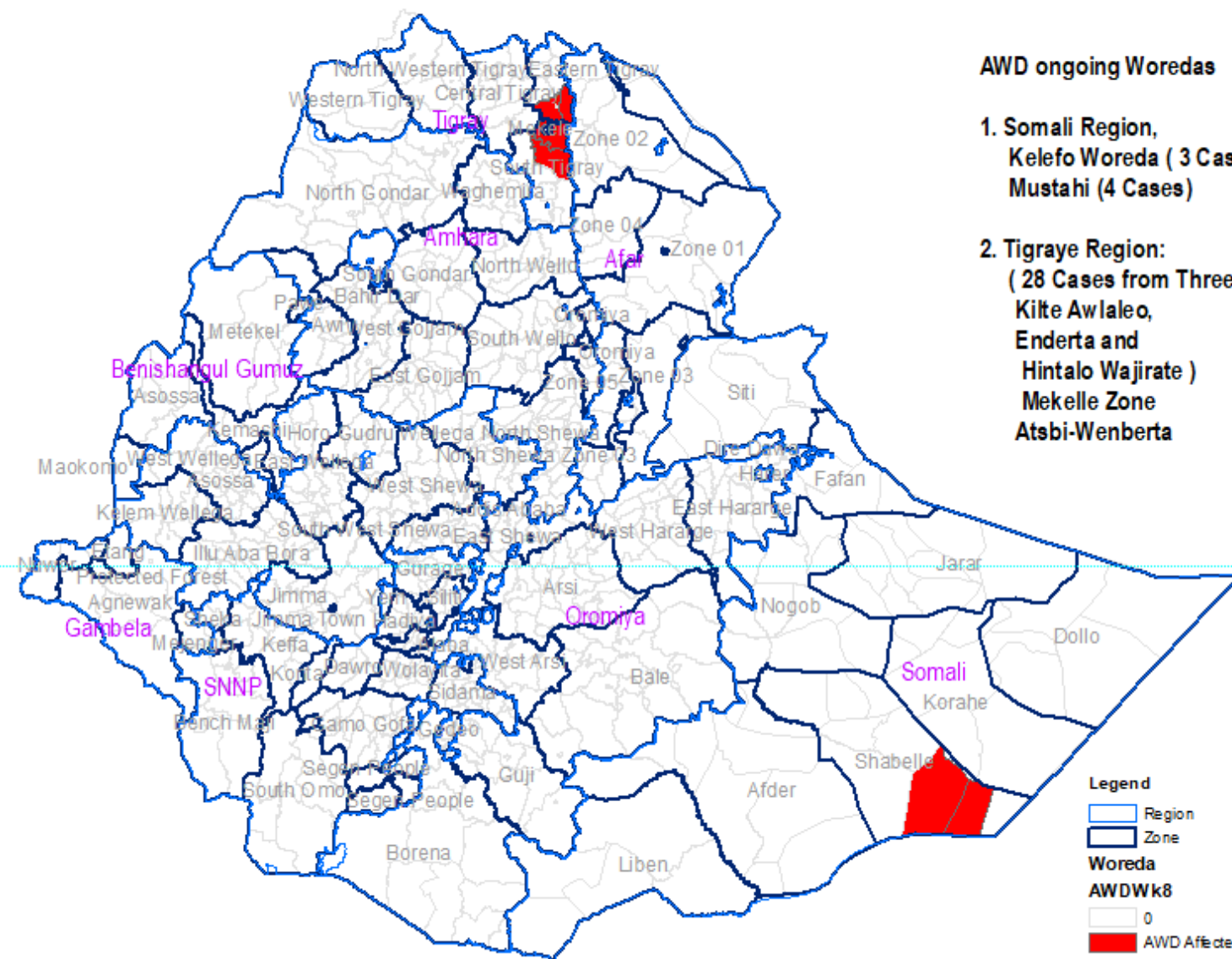
Data Source : Gambella UNICEF Field Office

National AWD distribution, Jan 1 to March 28 2018, Ethiopia



In 2018 total AWD Caseload became = 143. From Tigraye (37) and Ethio-Somali region (106)

Current AWD Active Transmission Woredas as of March 28, 2018 Ethiopia



AWD Hot spot woredas-

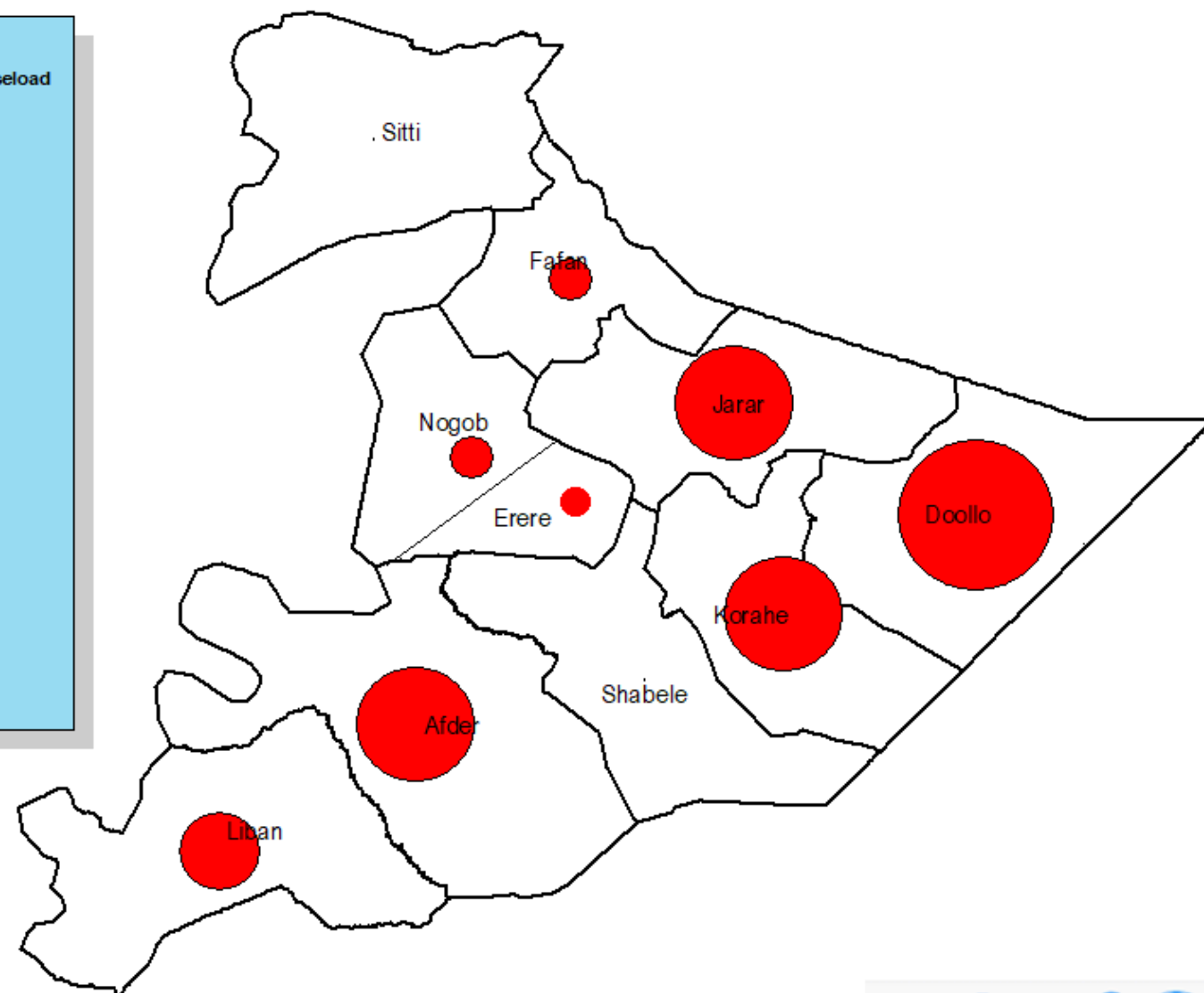
1. Tigray - Current AWD hot spot Wordas are four : ***Kilte-Awalelo, Hawelti, Semen, Adihaki, Quiha,***
2. Ethio-Somali - Current AWD hot spot Woredas are : Mustahi woreda was reported four (4) AWD confirmed cases yesterday (March 27, 2018)

Reported Measles Cases in Ethio-Somali Region from January - March, 2018, Ethiopia

Legend

Ethio_Somali Region Measles Caseload

Measles Caseload



Data Source:// Ethio-Somali Region -
Regional Health Bureau (PHEM Team)
Date of Production: March 15, 2018
Mapped by : UNICEF - Health Emergency Team
AddisAbaba, Ethiopia

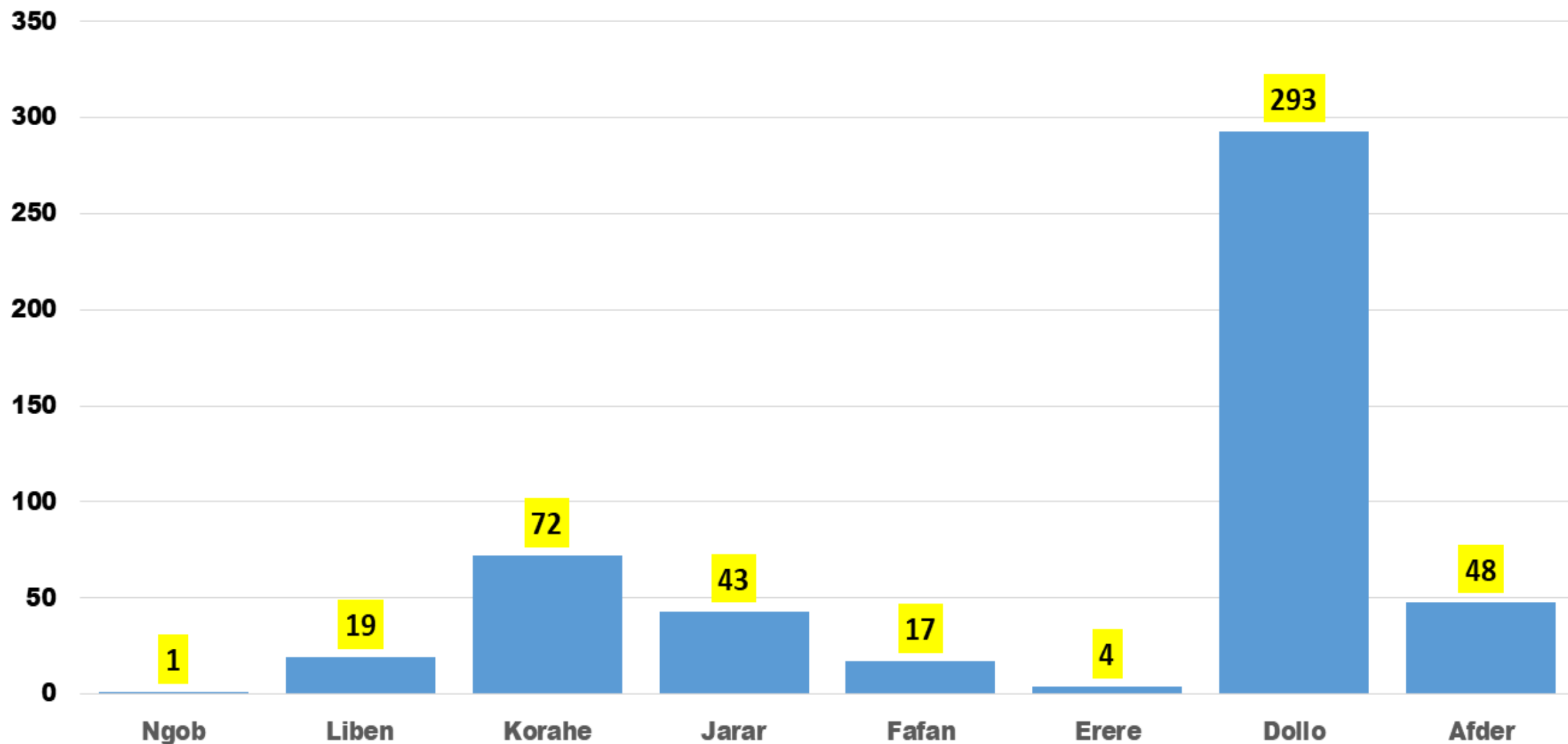
Ethiopian Somali Regional
Health Bureau

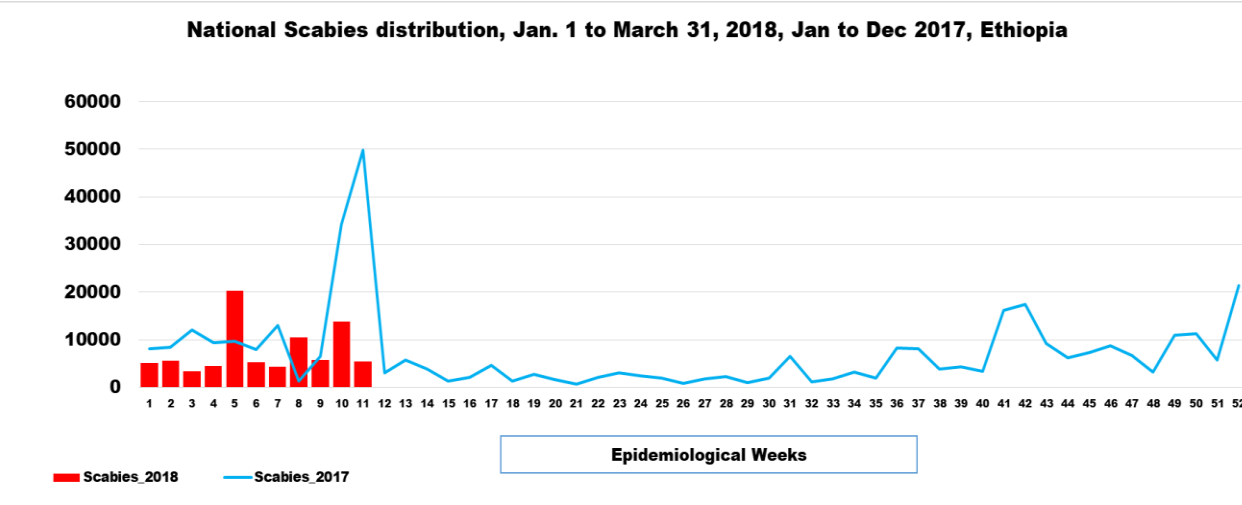
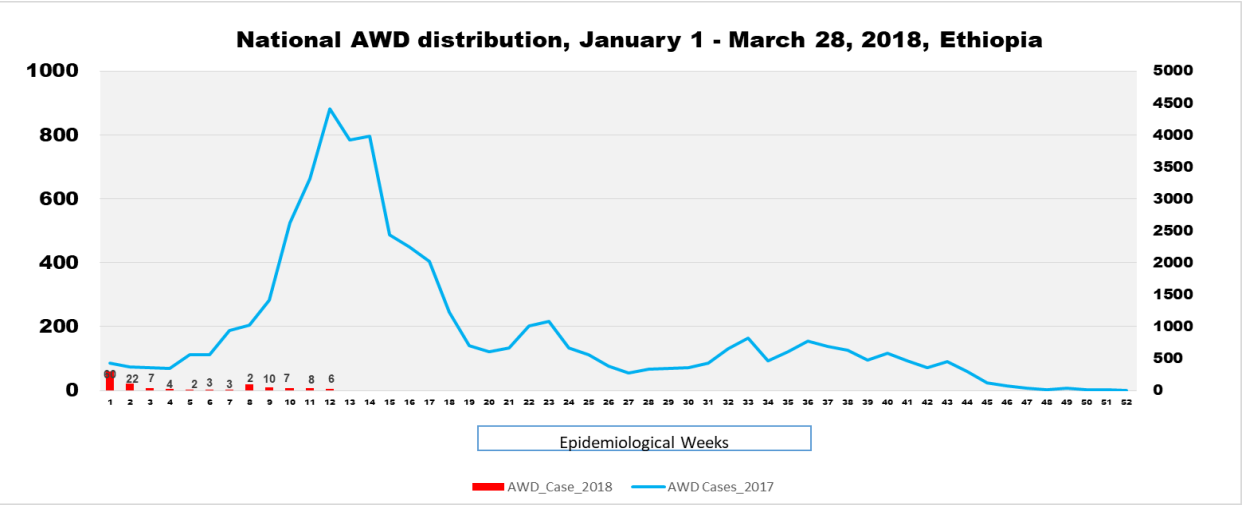


Public Health Emergency
Management core process



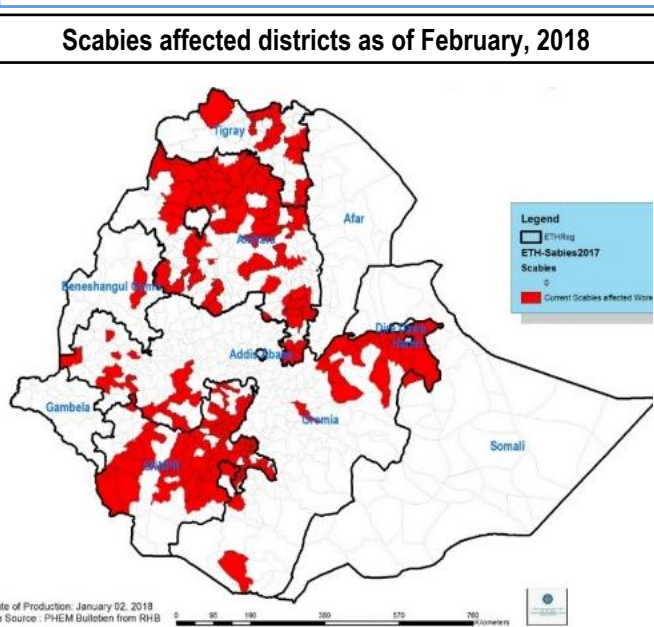
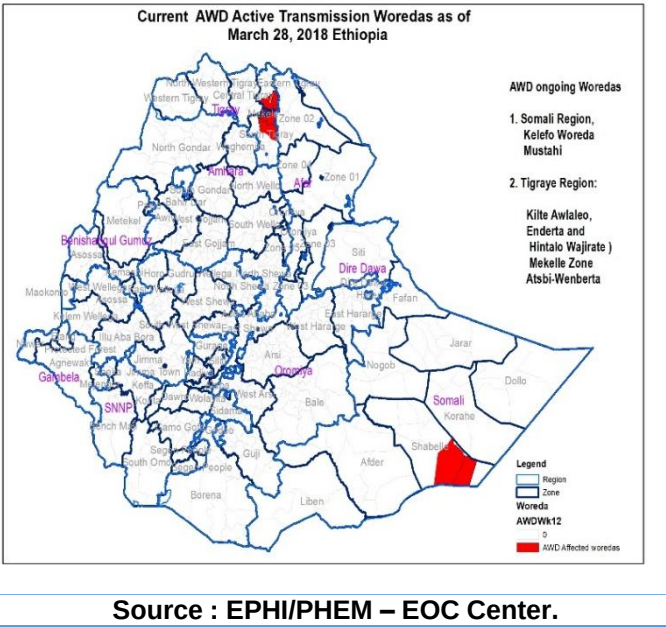
Distribution of Reported Measles Cases in Ethio-Somali Region from January - March, 2018, Ethiopia n=497 Cases)





Source : Regional Health Bureaus

Data Source : Regional Health Bureau



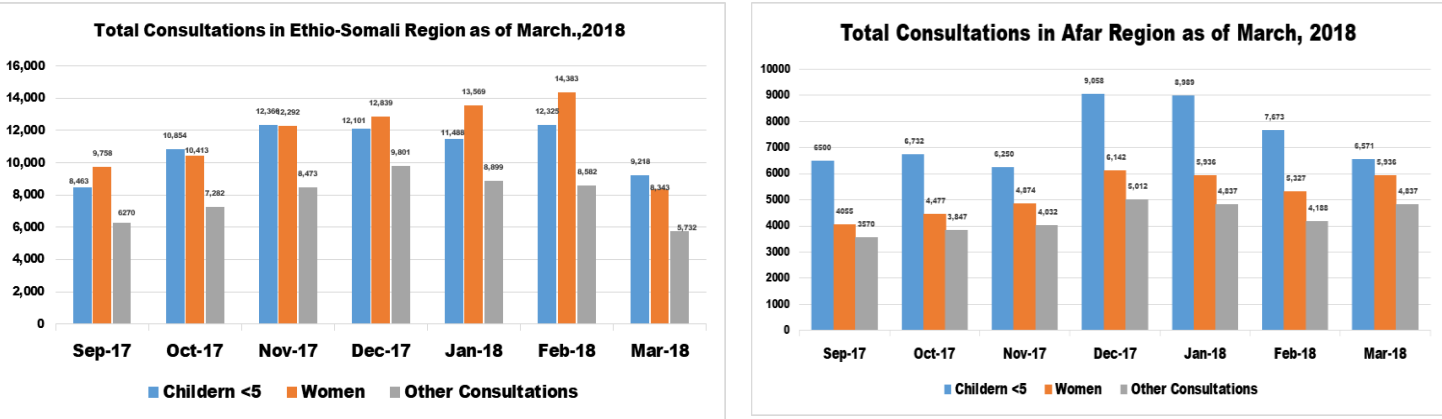
From March 1 – 31, 2018, a total number of 31 AWD Cases with zero death were reported from Somali (8) and Tigraye (23) regions. This is significantly lower compared to the same period in 2017. Total Cumulative AWD cases from January 1 to March 31, 2018 is 152.

On epidemiological Week 12, 2018 four (4) cases were reported from Ethio-Somali and to (3) from Tigray Regions.

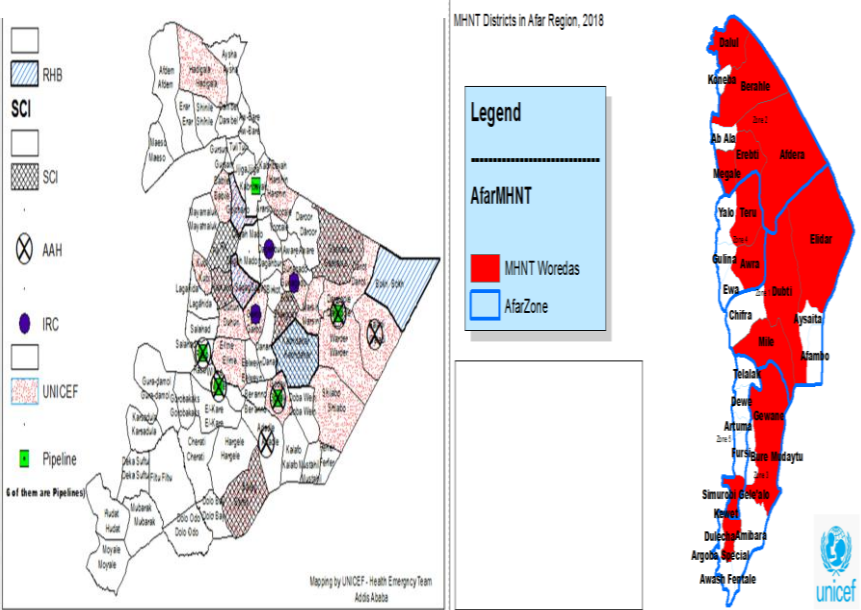
From March 1 – 31, 2018, a total number of 25,215 new Scabies Cases were reported from five regions. Oromia region reported highest number of cases. Please find case distribution in the table below. Comparing to the same period in 2017, Scabies cases were reduced by 73%.

Regions	No. of Scabies Cases from March 1 - 31, 2018	Cumulative Scabies Cases from Jan. 1 - March 31, 2018	Scabies Cases 2017
Oromia	14,335	20,055	29,876
Amhara	6,154	50,719	100,301
SNNP	4,210	10,377	225,150
Tigraye	500	2,078	12,962
B_Gumuz	16	1,516	3445
Total	25,215	84,745	371,734

Mobile Health and Nutrition Teams, Total Consultations, Afar and Somali Regions, Ethiopia as of March, 31, 2018



Locations of MHNT Team, Afar and Somali region, Ethiopia



Data Source :Regional Health Bureau

From March 1–31, 2018 a cumulative total of 40,637 medical consultations conducted in hard to reach areas of Afar and Somali regions, of which, 15,789 (39%) under five children and 14,279 (35%) were Women.

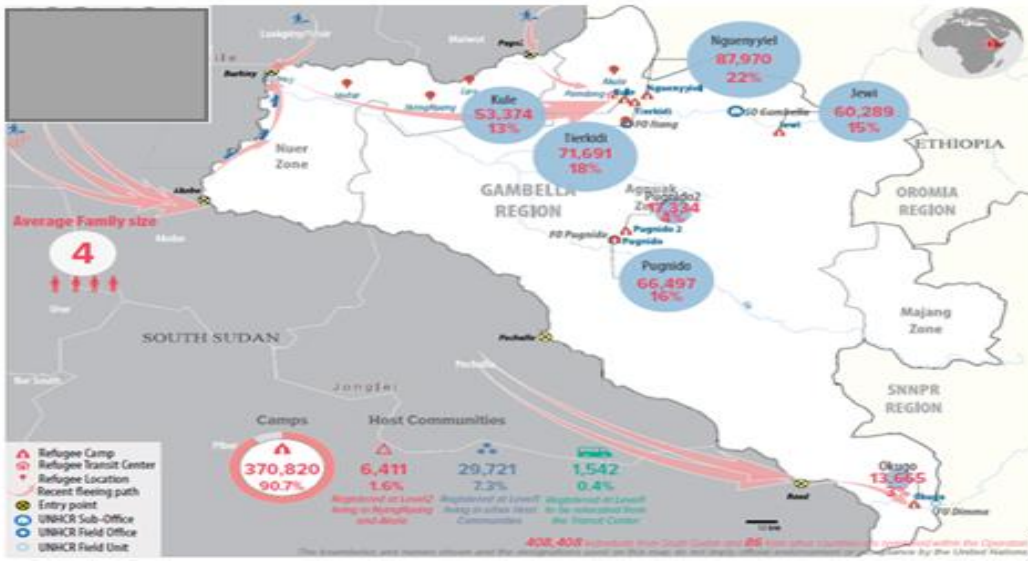
From January 1–March 31, 2018, a total cumulative 146,833 MHNT consultations were made (54,294 from Afar and 92,539 from Ethio-Somali region). Of which 56,264 (38%) Under five children, 53,494 (36%) Women and 37,075 (25%) were from Other adults.

The medical Consultation has been performed by 49 UNICEF supported Mobile Health and Nutrition Teams plus 11 International NGOs and 6 RHBs in Somali in region ; Of these consultations, 17,561 were from Somali (46 teams) and 12,507 from Afar with 20 Teams. In addition, from January to February 2018, a total 10,817 medical consultation has been conducted by 15 Sustainable Outreach Service Team (SOS). The consultation has been increasing in both regions.

South Sudanese Refugees Response, Jan. 1 – March 28, 2018, Ethiopia

Indicator	2017	March, 2018
# of New South Sudanese arrival in 2017	87,678	15,778
# of <15 Children	55,265	8,186
# of 6mth. - <15Yrs Children	50,152	7,782
# of <15 Children Obtained arrival Polio Vaccination	59,121	7,782
# of 6mth to 15Yrs children Obtained Measles Vaccination	53,231	7,782

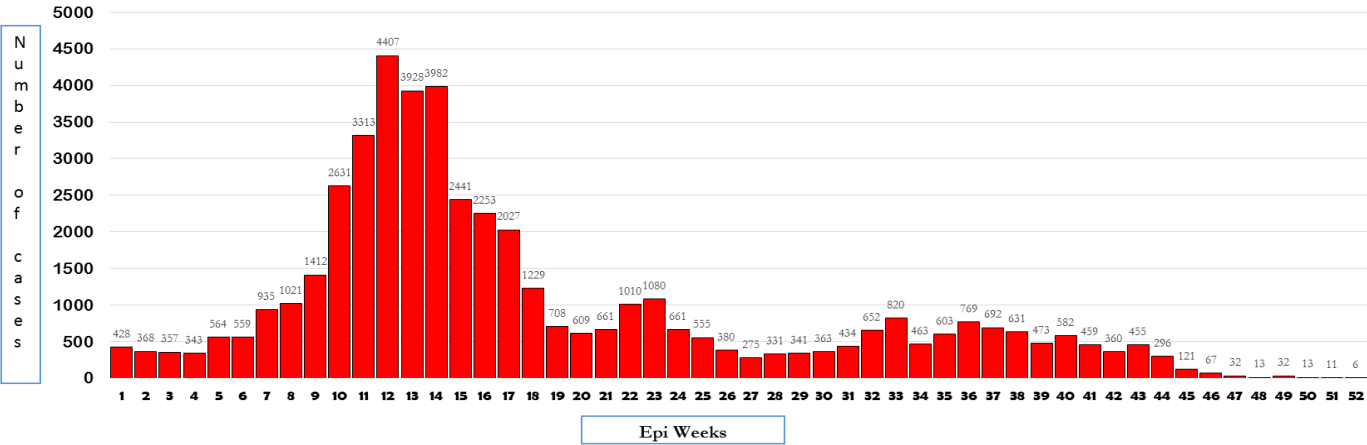
South Sudanese refugees, Gambella Sites, March, 2018



Data Source : Gambella UNHCR Field Office

AWD Epi Curve (January – December, 2017)

National AWD trend as of Week 52 (January - December), 2017, Ethiopia



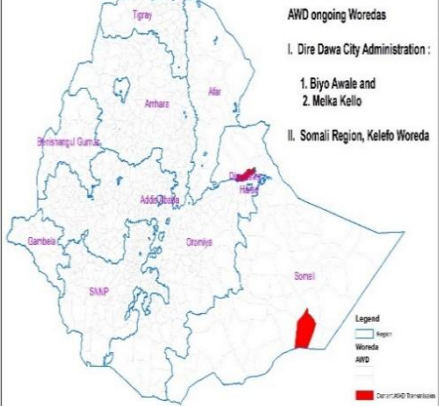
Source : RHBs and EPHI/PHEM

Current AWD transmission as of Week 52, 2017

In 2017 as of Week 52, from January – December 31, 2017, a cumulative of 48,638 AWD Cases and 859 AWD Deaths with CFR of 1.76% were reported from Seven regions in Ethiopia with one city Administrative Council. On week 51, a total of 11 AWD cases were reported from two clusters of Dire Dawa City Administrative Council.

The incidence rate of AWD continued to decrease significantly at national level as well as at regional level especially in Amhara, Oromia and Somali regions. The Only AWD active transmission Woredas is at Kelafo in Somali region and Biyo Awale and Melka Kello cluster from Dire Dawa as of Week 52.

Current AWD Transmission Woredas as of January 04, 2018, Ethiopia

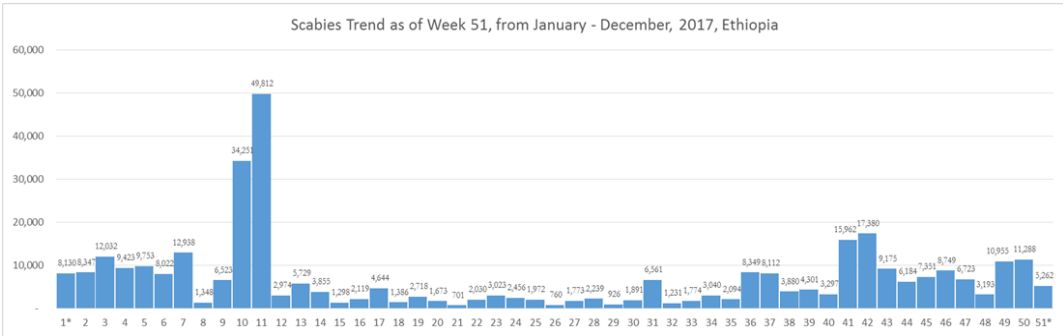


Current AWD transmission Sites as of Week 52 Source : EPHI/PHEM – EOC Center.

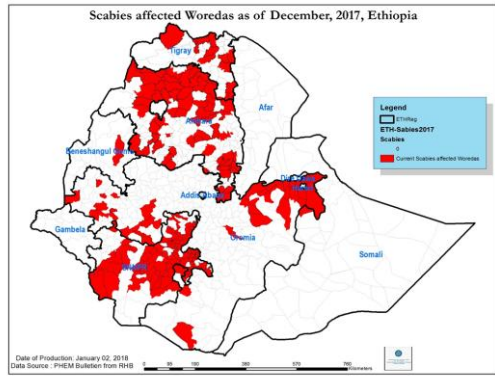
Scabies (From January – December, 2017

Source : Regional Health Bureaus – PHEM (RHBs)

Scabies Trend as of Week 51, from January - December, 2017, Ethiopia



Updated Scabies Case load, trend and Mapping , 2017, (Data Source RHBs/PHEM Bulletin,) (Jan. – December 31, 2017)



Scabies outbreak is ongoing in five regions from 237 districts from Amhara (62), Oromia (45), SNNPR (118), Beneshangul Gumuz (one woredas (13 Kebeles) and Tigray (11) districts have been affected with a total case of Scabies (n=349,607).

In December, 2017 from Week 47-52, a total case of 37,421 Scabies cases were reported from Oromia, Amhara, SNNP, Tigray and Beneshangul Gumuz. A new outbreak of Scabies cases emerged in Beneshangul Gumuz Since the first week of December, 2017. In 2017, 516,768 Permethrin drugs have been distributed including (SNNP=412,208), Amhara=57,793, Tigray=23,983) and Oromia=22,784 Permethrin lotions have been distributed in the affected districts in 237 districts.

Distribution of Scabies Cases with Premetir Lotion as of December, 2017, Ethiopia

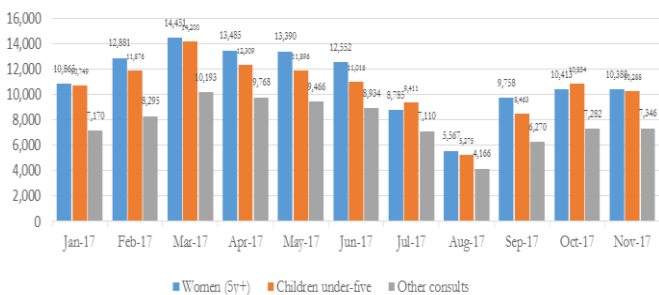
Region	December, 2017 (Week 47-52)	Total Scabies Cases by 2017	Number of Districts Currently affected by Scabies Cases	Peremetrin Distribuion, 2017		
				March, 2017	September, 2017	October, 2017
Oromia	5984	28,221	45	10184	12600	
Amhara	21,653	82,201	62	30600	27193	
SNNP	5,540	223,034	118	156768	125840	129600
Tigray	911	12,817	11	10000	6783	7200
B_Gumuz	3334	3334	1			
Total	37,421	349,607	237	166,768	173,407	176,593

Generally, the Scabies outbreak trend has been increased in the last 16 weeks in all regions which can be attributed to the schools opening last September, 2017.

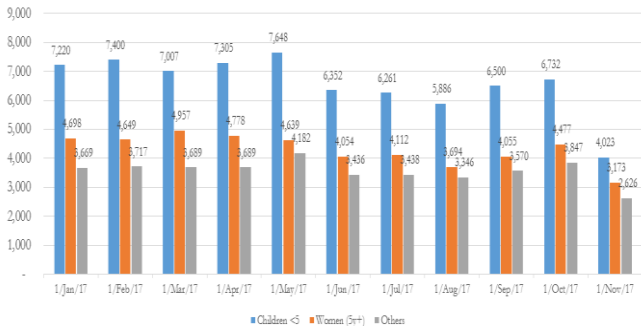
Ethiopia: Mobile Health and Nutrition (MHNT) and Refugees Influx (Situation as of 31 December 2017) : No.13

Mobile Health and Nutrition (MHNT), Total Consultations,
Afar and Somali Regions, Ethiopia as of December 31, 2017
(613,281 Consultations)

MHNT Consultation as of November, 2017, Ethio-Somali Region, Ethiopia (n=324,872)

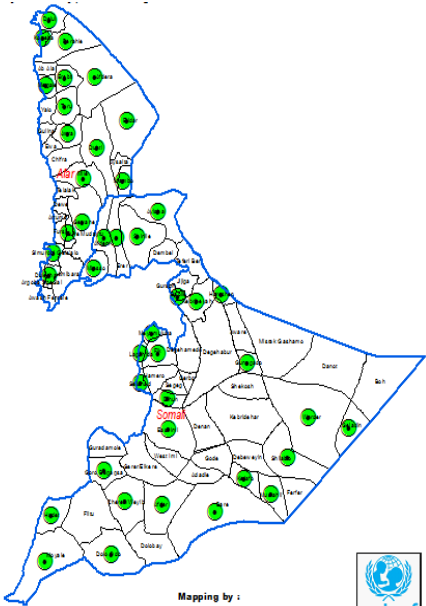
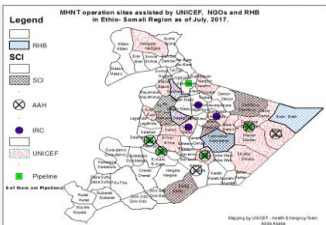


MHNT Consultation as of November, 2017, Afar Region, Ethiopia (n=158,829)



Data Source :RHBs

Location of MHNT Team, Afar and Somali region, Ethiopia



As of December 31, 2017, a cumulative total number of n=483,701 Consultations were made through n=49 Mobile Health and Nutrition Teams (MHNT program) in both Somali and Afar regions; of these, 158,829 from Afar region and 324,872 from Somali region. Out of the total consultation from both regions, 188,671 (39%) for children 5yrs+, 169,821 (35%) for Women and 125,209 (26%) for other groups.

Currently for Somali region, additional n=15 MHNT programs have been implemented through RHB (14) and 19 through NGOs

In Afar the reporting period of November, 2017 the completeness and timeliness of the reporting period coverage was 100%.

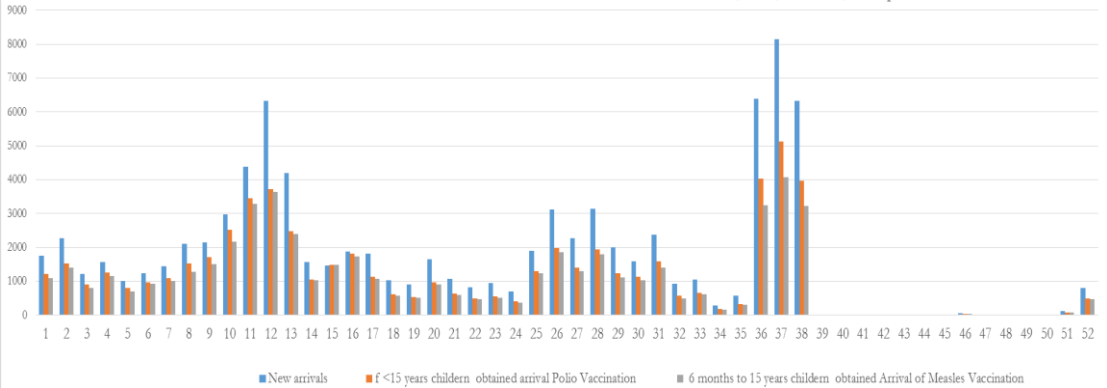


Influx of South Sudan Refugees in Gambella, Ethiopia , as of December 31, 2017

In 2017, a total of 87,678 new South Sudan refugees arrived in Lare and Akobo Woredas in Gambella Region. As of date week 52 (December 31, 2017), a total of 59,121 and 53,231 children less than 15 years received a dose of Polio and Measles Vaccines respectively at the entry points.

Source : Gambella UNICEF and UNHCR Field Offices)

Total new arrival of South Sudan Influx and their vaccination Status as of December, 2017, Gambella, Ethiopia



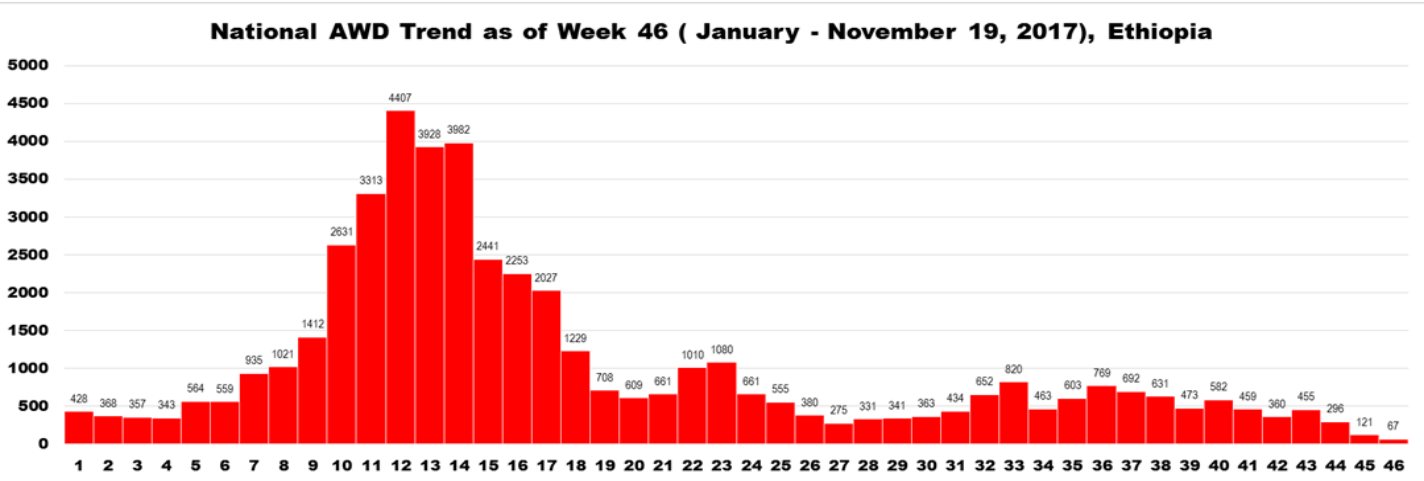
South Sudanese refugees new arrival in Gambella Sites at Gambella , December 31, 2017



Indicator	Number
# of New South Sudanese arrival in 2017	87,678
# of <15 Children	55,265
# of 6mth. - <15Yrs Children	50,152
# of <15 Children Obtained arrival Polio Vaccination	59,121
# of 6mth to 15Yrs children Obtained Measles Vaccination	53,231

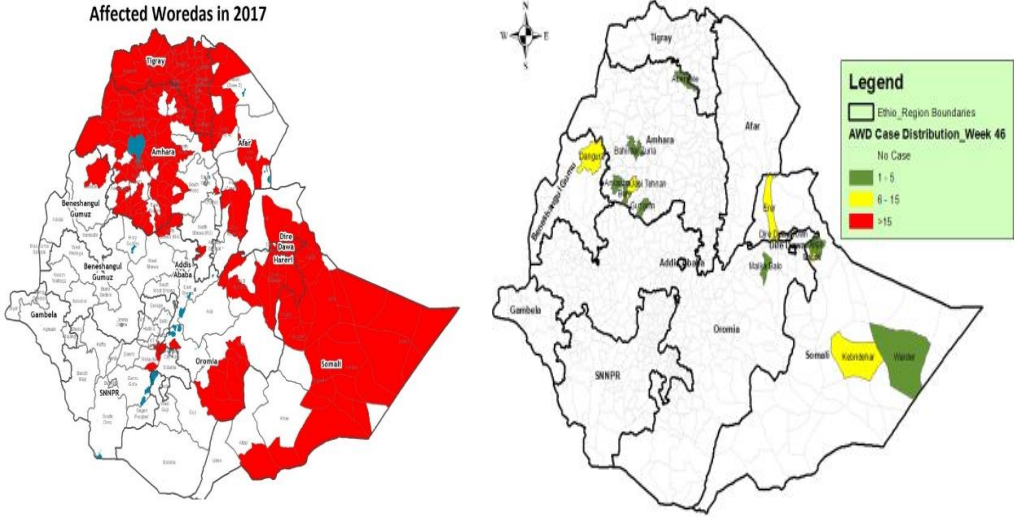
Data Source : UNICEF and UNHCR Gambella Field Offices

AWD Epi Curve (January – November 30, 2017)



Source : RHBs, PHEM and HCC

Current AWD transmission as of week 46, 2017



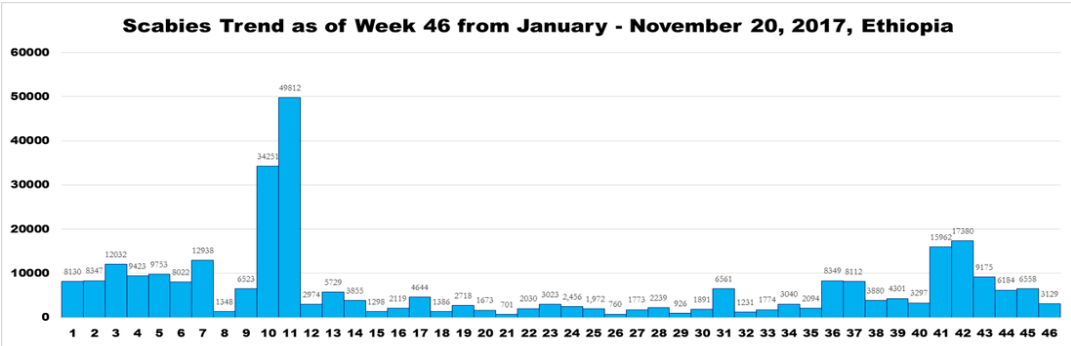
In 2017 as of Week 46, from January – November 19, 2017, a cumulative total of 48,638 AWD Cases and 859 AWD Deaths with CFR of 1.8% were reported from Seven regions in Ethiopia with one city Administrative Council. On week 46, a total of 67 AWD cases from 16 woredas were reported at national level; the distribution of the cases are: Oromia (4), Amhara (22), Somali (26), Dire Dawa (8), BGumuz (6), and Tigraye (1).

The incidence rate of AWD continued to decrease significantly at national level as well as at regional level especially in Amhara, Oromia and Somali regions.

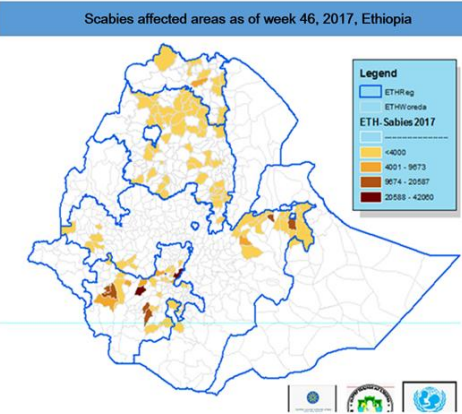
Current AWD hotspot with Caseload as of November 30, 2017 Source : EPHI/PHEM

Scabies (From January – November 30, 2017

Source : Regional Health Bureaus – PHEM (RHBs)



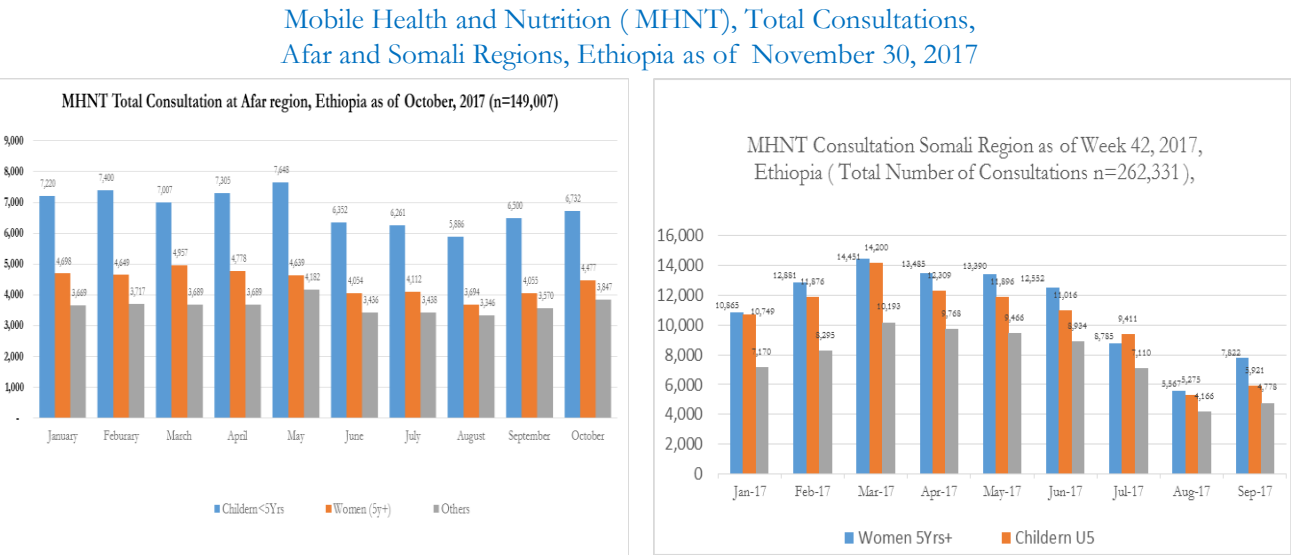
Updated Scabies Case load, trend and Mapping , 2017,
(Data Source RHBs/PHEM Bulletin,) (Jan. – November 30, 2017)



Scabies outbreak is ongoing in four regions from 74 districts from Amhara (33), Oromia (12), SNNPR (20), and Tigray (9) districts. A total of 42,426 new scabies cases were reported for the month November, 2017 (Epidemic Weeks 42-46). The total cumulative reported scabies cases in 2017 became 305,773 from the four regions: Amhara (56,358), Oromia (20,015), SNNP (217,494) and Tigray (11,906). Generally, the Scabies outbreak trend has increased in the last Six weeks in all regions which can be attributed to the schools opening last September.

Region	November, 2017 (Week 42 - 46)	2017
Amhara	11,832	56,358
Oromia	1509	20,015
SNNP	28,014	217,494
Tigray	1,071	11,906
Total	42,426	305,773

Response interventions are ongoing in the ground which including: active surveillance case finding, case management, social mobilization activities in the affected woredas and distribution of 350,000 Permethrin drugs (SNNP=255,440), Amhara=57,793), Tigraye=13,983) and Oromia=22,784 Permethrin lotions have been distributed in the affected districts.

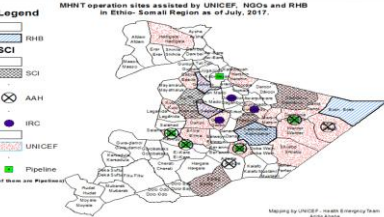


Data Source :RHBs

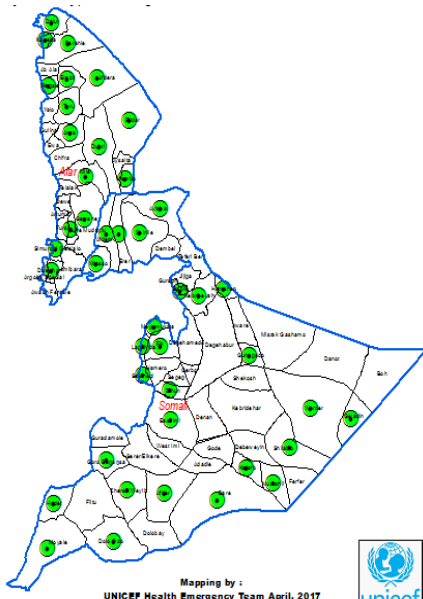
Location of MHNT Team, Afar and Somali region, Ethiopia

Legend

MHNT operation sites assisted by UNICEF, NGOs and RHB in Afar-Somali Region as of July, 2017.







Mapping by :
UNICEF Health Emergency Team April, 2017

In 2017, a cumulative total of n=411,338 Consultations were made through n=49 Mobile Health and Nutrition Teams (MHNT program) in both Somali and Afar regions; of these, 149,007 from Afar and 262,331 from Somali. Out of the total consultation from both regions, 143,911 (35%) were for Women, 160,964 5yrs+ Children under 5 years and 106,463 (26%) for other groups.

Currently for Somali region, additional n=15 MHNT programs have been implemented through RHB (5), NGOs (10) while 9 additional teams from the NGOs are in the pipeline.

Please take note that the October report on the consultations is still incomplete and will be updated during the next reporting period including the timeliness and completeness.

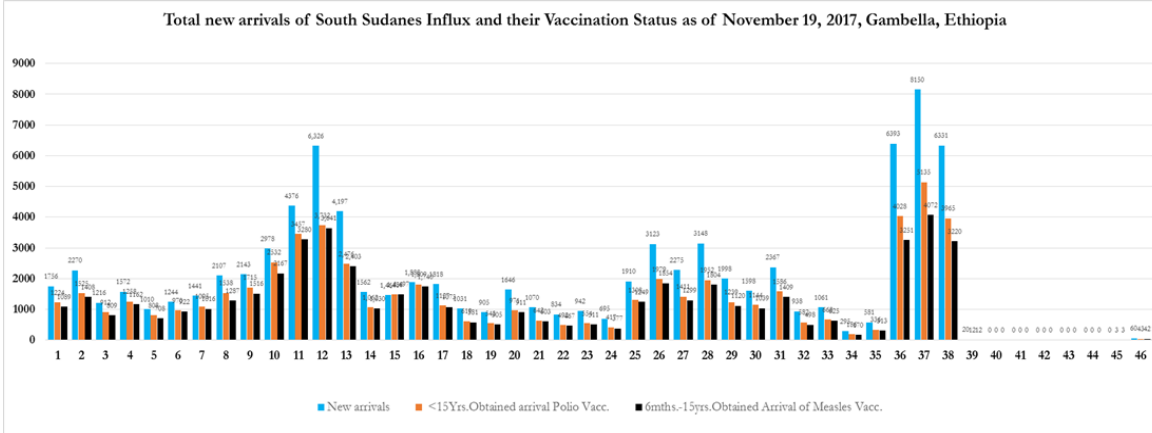


Influx of South Sudan Refugees in Gambella, Ethiopia , as of November 30, 2017

In 2017, a total of 86,739 new South Sudan refugees arrived in Lare and Akobo Woredas in Gambella Region. As of date week 46 (November 19, 2017), a total of 58,495 and 52,683 children less than 15 years received a dose of Polio and Measles Vaccines respectively at the entry points.

Source : Gambella UNICEF and UNHCR Field Offices)

Total new arrivals of South Sudanese Influx and their Vaccination Status as of November 19, 2017, Gambella, Ethiopia



Week	New arrivals	<15yrs.Obtained arrival Polio Vacc.	6mths.-15yrs.Obtained Arrival of Measles Vacc.
1	1736	1256	1256
2	2270	1515	1515
3	1372	1469	1469
4	2252	1515	1515
5	1244	1084	1084
6	2441	1084	1084
7	2207	1515	1515
8	2543	1515	1515
9	2578	1515	1515
10	4376	1515	1515
11	6326	1515	1515
12	4397	1515	1515
13	1542	1515	1515
14	1444	1515	1515
15	1444	1515	1515
16	1444	1515	1515
17	1444	1515	1515
18	1444	1515	1515
19	1444	1515	1515
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37	1444	1515	1515
38	1444	1515	1515
39	1444	1515	1515
40	1444	1515	1515
41	1444	1515	1515
42	1444	1515	1515
43	1444	1515	1515
44	1444	1515	1515
45	1444	1515	1515
46	1444	1515	1515

South Sudanese refugees new arrival in Gambella Sites at Gambella , November 30, 2017



Indicator	Number
# of New South Sudanese arrival in 2017	86,739
# of <15 Children	54,687
# of 6mth. - <15Yrs Children	46,391
# of <15 Children Obtained arrival Polio Vaccination	58,495
# of 6mth to 15Yrs children Obtained Measles Vaccination	52,683

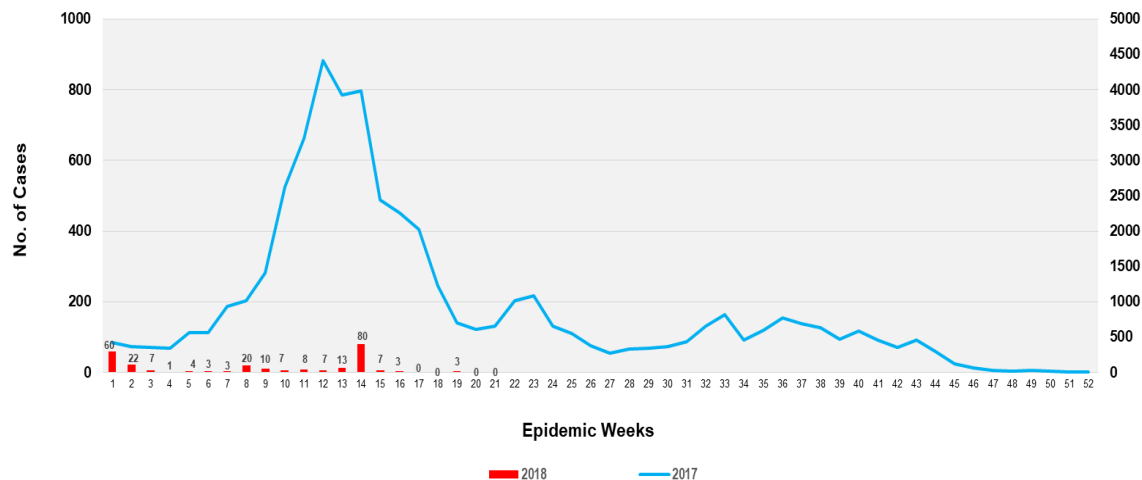
Data Source : UNICEF and UNHCR Gambella Field Offices

UNICEF - Health Section – Ethiopia (Monthly Health Emergency Info-graphics – for Internal Use Only)

For Further Information

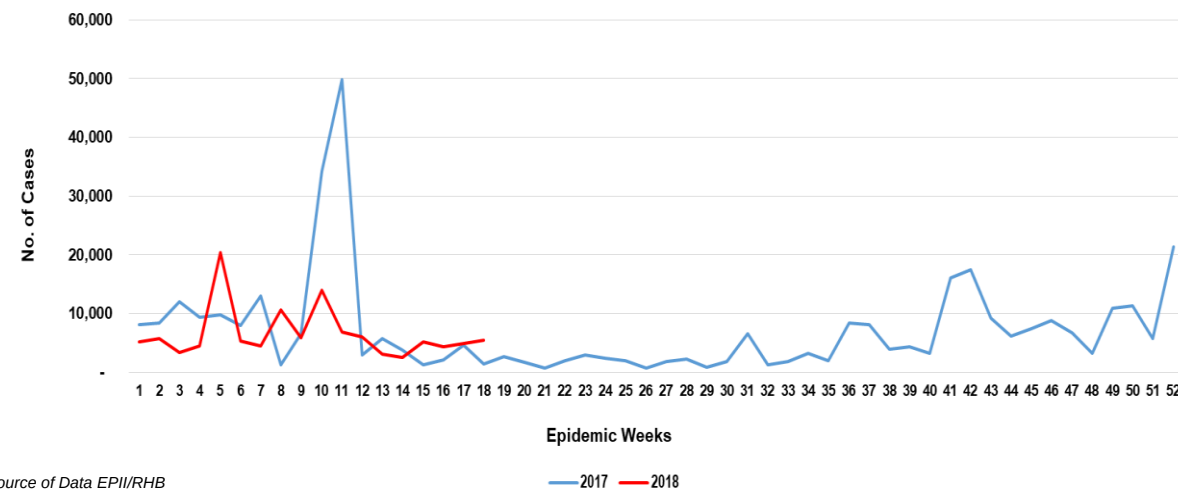
Dr. Macoura Oulare : moulare@unicef.org , Dr. Marisa Ricardo mricardo@unicef.org, Dr. Mohamed Diaeldin Omer, mdomer@unicef.org, Getachew Haile, gehaile@unicef.org , Negusu Worku (Consultant, IM & GIS) negusuworku@gmail.com

National trend of reported AWD Cases, 2017/2018, Ethiopia



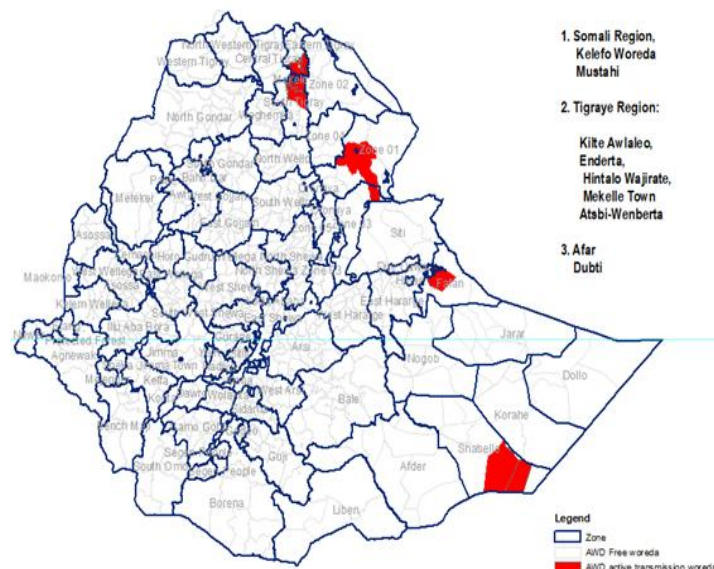
Source of Data EPII/RHB

National trend of reported Scabies cases 2017/2018, Ethiopia



Source of Data EPII/RHB

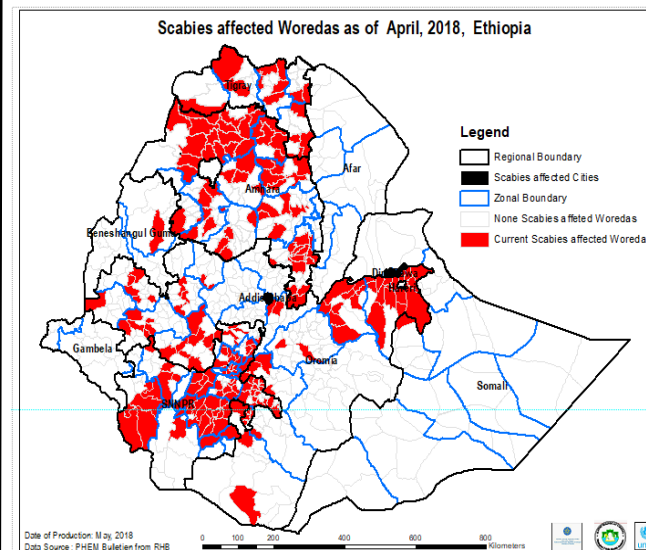
AWD active woredas as of April 30, 2018, Ethiopia



From April 1 – 30, 2018, a total number of 91 AWD Cases with zero death were reported from Somali (26), Tigraye (1) and Afar (64) regions. This is significantly lower compared to the same period in 2017. Total Cumulative AWD cases from January 1 to April 30, 2018 is 255.

On April 04, 2018, a new AWD outbreak occurred in Afar region. On epidemiological Week 17, 2018 Zero AWD cases were reported from the affected regions of Afar, Tigray and Ethio-Somali regions.

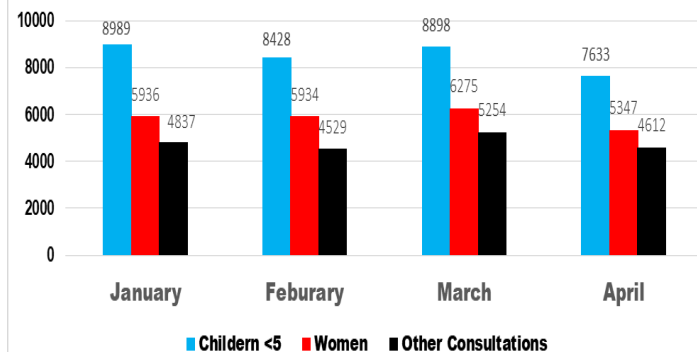
Scabies active woredas 2018, Ethiopia



From April 1 – 30, 2018, a total number of 20,365 new Scabies Cases were reported from Seven regions. Of which Dire Dawa and Addis Ababa regions became a newly Scabies affected city administrative regions. Amhara, SNNP and Oromia regions reported highest number of cases. Please find case distribution in the table below. Comparing to the same period in 2017, Scabies cases were reduced by 65%.

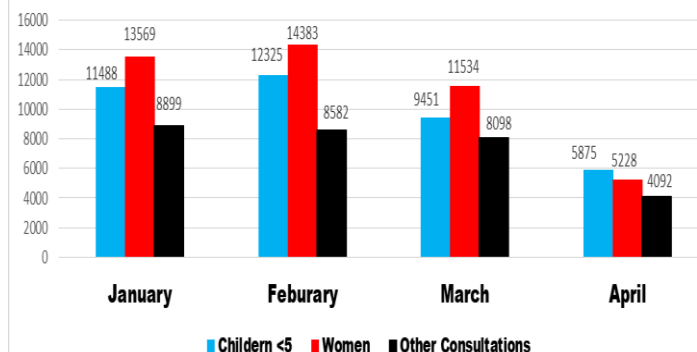
Regions	April, 2018	Cummulative 2018	Cummulative 2017
Oromia	6,070	28,075	29,876
Amhara	6,405	61,295	100,301
SNNP	6,044	17,076	225,150
Tigraye	856	3,384	12,962
B_Gumuz	8	1,528	3445
Addis Ababa	515	1,009	0
Dire Dawa	467	467	0
Total	20,365	112,834	371,734

MHNT : Medical Consultation in Afar Region, April, 2018



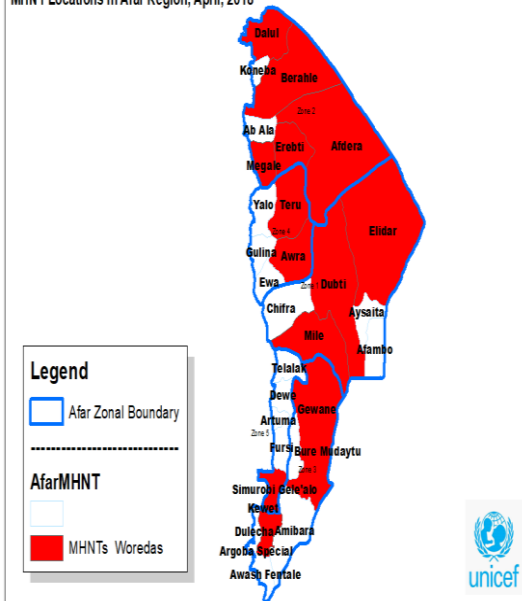
Source: Regional Health Bureaus

MHNT : Medical Consultation in Ethio-Somali Region, 2018.

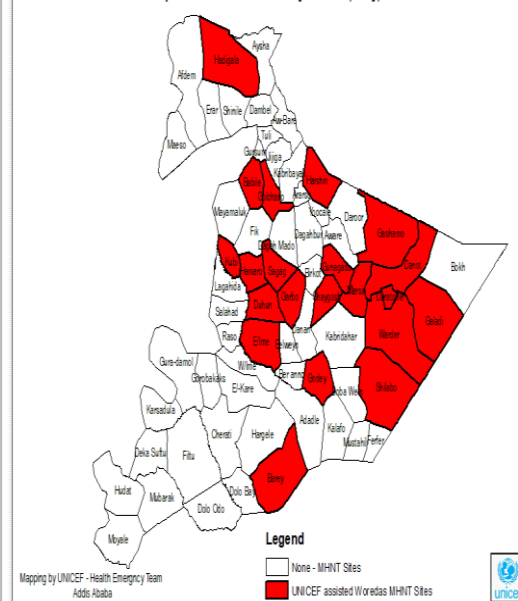


Locations of MHNTs, Afar and Ethio-Somali Regions, April, 2018.

MHNT Locations in Afar Region, April, 2018



MHNT operation sites assisted by UNICEF, May, 2018



From April 1–30, 2018 a cumulative total of 32,787 medical consultations conducted in hard to reach areas of Afar and Somali regions, of which, 13,508 (41%) under five children and 10,575 (32%) were Women.

From January 1–April 30, 2018, a total cumulative 190,196 MHNT consultations were made 76,672 from Afar and 113,524 from Ethio-Somali region. Of which 73,087 (38%) Under five children, 68,206 (36%) Women and 48,903 (26%) were from Other adults.

The medical Consultation has been performed by 49 UNICEF supported Mobile Health and Nutrition Teams plus 11 International NGOs and 6 RHBS in Somali in region.

Moreover, total MHNT beneficiaries from both regions became nearly 2.8 million. Of which 1,689,651 from Ethio-Somali and the rest 1,083,044 from Afar region.

South Sudanese Refugees: Selective Screening Vaccination, 2017/2018

Indicators	April 2018	Cummulative 2018	Cummulative 2017
Total No. of new arrivals	2777	19,178	87,678
No. of Childern Under 15 years	1742	10,333	55,265
No children 6 months - 14 years	1498	9,629	50,152
No. of children received polio vaccination	1711	10,344	59,121
No of children received Measles Vaccination	1521	9,684	53,231

Source: UNICEF and UNHCR Gambella Field Offices

South Sudanese Refugees in Gambella Sites, April, 2018

